

EXPNeuralUCB Research Framework: Complete Implementation

Quantum Entanglement Path Selection & Qubit Allocation via Adversarial Group Neural Bandits

Research Enhanced Implementation with ALL Comparison Algorithms

Date: September 2025
Authors: Research Implementation based on GA Work in AI/Quantum Computing
Framework: Complete implementation including GNeuralUCB, EXPUCB, and Oracle comparisons

Research Overview

This comprehensive notebook contains an enhanced implementation and evaluation of the **EXPNeuralUCB algorithm** for quantum entanglement path selection and qubit allocation in adversarial environments with **ALL** comparison algorithms from the original paper:

Complete Algorithm Suite:

- 1. **EXPNeuralUCB** - Main algorithm (EXP3 + NeuralUCB)
- 2. **GNeuralUCB** - Group Neural UCB (missing adversarial robustness)
- 3. **EXPUCB** - EXP3 + standard UCB (missing neural learning)
- 4. **Oracle** - Theoretical optimal with perfect knowledge

Reference Paper

"Quantum Entanglement Path Selection and Qubit Allocation via Adversarial Group Neural Bandits"
Huang, Y., Wang, L., & Xu, J. (2024). arXiv:2411.00316v1

Key Research Questions

- 1. Can adversarial neural bandits effectively optimize quantum routing under attacks?
- 2. How does EXPNeuralUCB perform compared to classical baselines?
- 3. What are the theoretical guarantees and empirical validation?
- 4. How can this framework be extended for fairness and predictive intelligence?

1. Environment Setup & Configuration

```
1 # @title
2 import os, sys, importlib
3
4 # get current directory
5 cur_dir = os.getcwd()
6 print("Current working directory:", cur_dir.split('/')[-1])
7
8 try:
9     import google.colab          # Check if running in Colab
10    from google.colab import drive
11    drive.mount('/content/drive')
12
```



```

13 project_dir = '/content/drive/MyDrive/GA-Work/expneural_framework'
14 os.chdir(project_dir) # makes project_dir the notebook's "root"
15 print("- Running in Colab")
16
17 project_cod_dir = os.path.join(project_dir, 'src')
18 sys.path.append(project_cod_dir) # Import local modules
19
20 except ImportError: print("- Not running in Colab")
21
22 print('- Now working from:', os.getcwd().split('/')[~1])
23
24 # Now import the project classes
25 try:
26     from expneural_framework import (QuantumNetworkEnvironment, NeuralUCB, EXPNeuralUCB, create_comprehensive_evaluation, create_visualizations, create_baseline_algorithms, main)
27     print('- EXPNeuralUCB framework imported successfully')
28 except ImportError as e:
29     print(f'- Framework import failed: {e}')
30     print('- Please ensure expneural_framework.py is in the current directory')
31
32     # For Google Colab users
33     print('\n🔗 Google Colab Setup:')
34     print('1. Upload expneural_framework.py using the file upload button')
35     print('2. Or run: from google.colab import files; files.upload()')
36
37     # Provide inline framework code if needed
38     print('\n Alternative: Run the complete framework code from the previous cells')

```

Current working directory: hybrid_variable_framework

- Not running in Colab
- Now working from: hybrid_variable_framework
- Framework import failed: No module named 'expneural_framework'
- Please ensure expneural_framework.py is in the current directory

🔗 Google Colab Setup:

1. Upload expneural_framework.py using the file upload button
2. Or run: from google.colab import files; files.upload()

Alternative: Run the complete framework code from the previous cells

1 Start coding or [generate](#) with AI.

```

1 # Core Scientific Computing Libraries
2 import numpy as np
3 import pandas as pd
4 import matplotlib.pyplot as plt
5 import seaborn as sns
6 from scipy.stats import beta, multivariate_normal, norm
7 from tqdm import trange, tqdm
8 import warnings
9 warnings.filterwarnings('ignore')
10
11 # Deep Learning Framework
12 import torch
13 import torch.nn as nn
14 import torch.nn.functional as F
15 import torch.optim as optim
16
17 # Utility Libraries
18 import copy
19 import random
20 import time
21 import math
22 import psutil

```

```

22 import psutil
23 import os
24 from random import choice
25
26 # Set Style for PhD-Quality Plots
27 plt.style.use('seaborn-v0_8')
28 sns.set_palette("husl")
29 plt.rcParams['figure.figsize'] = (12, 8)
30 plt.rcParams['font.size'] = 12
31 plt.rcParams['axes.labelsize'] = 14
32 plt.rcParams['axes.titlesize'] = 16
33 plt.rcParams['legend.fontsize'] = 12
34 plt.rcParams['xtick.labelsize'] = 12
35 plt.rcParams['ytick.labelsize'] = 12
36
37 # Set Random Seeds for Reproducibility
38 np.random.seed(42)
39 torch.manual_seed(42)
40 random.seed(42)
41
42 # Device Configuration
43 device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
44 print(f"Using device: {device}")
45 print(f"PyTorch version: {torch.__version__}")
46 print(f"NumPy version: {np.__version__}")
47
48 print("\n=== COMPLETE ALGORITHM SUITE ===")
49 print("> EXPNeuralUCB\t - EXP3 + NeuralUCB (Main Algorithm)")
50 print("> GNeuralUCB\t - Group Neural UCB (Missing Adversarial Robustness)")
51 print("> EXPUCB\t - EXP3 + Standard UCB (Missing Neural Learning)")
52 print("> Oracle\t - Theoretical Optimal (Perfect Knowledge)")

```

```

Using device: cpu
PyTorch version: 2.8.0
NumPy version: 2.3.3

```

```

=== COMPLETE ALGORITHM SUITE ===
> EXPNeuralUCB - EXP3 + NeuralUCB (Main Algorithm)
> GNeuralUCB - Group Neural UCB (Missing Adversarial Robustness)
> EXPUCB - EXP3 + Standard UCB (Missing Neural Learning)
> Oracle - Theoretical Optimal (Perfect Knowledge)

```

2. Quantum Data Network Modeling

2.1 Context Generation (Qubit Allocation Strategies)

2.2 Quantum Entanglement Success Rate Functions

Implementation of realistic quantum entanglement success rate calculations for each path.

3. Adversarial Attack Pattern Generation

Generation of Markov-based attack patterns as used in the original research.

4. Oracle Implementation

The Oracle algorithm has perfect knowledge of all reward functions and attack patterns.

5. Base Multi-Armed Bandit Algorithms

Implementation of fundamental bandit algorithm classes.

6. Neural Bandit Components

Neural network models and replay buffer for neural bandit algorithms.

7. Neural UCB Implementation

Neural Upper Confidence Bound algorithm for learning complex reward functions.

8. GNeuralUCB Implementation

Group Neural UCB - Uses neural networks for each group independently but lacks adversarial robustness.

9. EXPUCB Implementation

EXP3 + Standard UCB - Uses EXP3 for path selection but lacks neural network learning for qubit allocation.

10. EXPNeuralUCB Implementation

The Main Algorithm - Combines EXP3 for adversarial group selection with NeuralUCB for complex reward learning.

✓ 11. Comprehensive Experimental Evaluation

Complete comparison of ALL algorithms: EXPNeuralUCB vs GNeuralUCB vs EXPUCB vs Oracle

```
1  # @title
2  import os, sys
3
4  # get current directory
5  cur_dir = os.getcwd()
6  print("Current working directory:", cur_dir.split('/')[-1])
7
8  try:
9      import google.colab          # Check if running in Colab
10     from google.colab import drive
11     drive.mount('/content/drive')
```

```

12     project_dir = '/content/drive/MyDrive/GA-Work/hybrid_variable_framework' # Change work directory
13     os.chdir(project_dir) # makes project_dir the notebook's "root"
14     print("- Running in Colab")
15
16     project_cod_dir = os.path.join(project_dir, 'src')
17     sys.path.append(project_cod_dir) # Import local modules
18
19 except ImportError:
20     print("- Not running in Colab")
21
22 print('- Now working from:', os.getcwd().split('/')[-1])
23

```

Current working directory: hybrid_variable_framework
 - Not running in Colab
 - Now working from: hybrid_variable_framework

```

1 # update import library
2 import importlib
3 import threading
4 import quantum_tests
5 importlib.reload(quantum_tests)
6 from quantum_tests import ExperimentRunner
7
8
9 def run_threaded_experiments():
10     """
11     Each experiment in its own thread - like separate Colab notebooks!
12     """
13     FRAME_NUMBERS = [4000, 6000, 8000]
14
15     for experiment_id, frame_count in enumerate(FRAME_NUMBERS, 1):
16
17         # Create fresh runner instance (like new Colab notebook)
18         runner = ExperimentRunner(base_seed=12345)
19
20         # Run in thread (like separate Colab session)
21         experiment_thread = threading.Thread(
22             target=runner.run_single_experiment,
23             args=(frame_count, experiment_id),
24             name=f"ColabCell_{experiment_id}"
25         )
26
27         experiment_thread.start()
28         experiment_thread.join() # Wait for completion
29
30         # Full cleanup (like closing Colab notebook)
31         runner.full_experiment_cleanup()
32
33 # Run the experiments!
34 run_threaded_experiments()

```

```

- NEURAL Progress: 100%|██████████| 6000/6000 [00:11<00:00, 541.65it/s]
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
🐞 EXPNeuralUCB initialized in 'exp3' mode
  • Group Selection: EXP3 (adversarially robust)
  • Action Selection: Linear UCB (linear learning)
- EXP3 Progress: 100%|██████████| 6000/6000 [00:04<00:00, 1459.35it/s]
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
🐞 EXPNeuralUCB initialized in 'hybrid' mode
  • Group Selection: EXP3 (adversarially robust)
  • Action Selection: Neural UCB (nonlinear learning)
- HYBRID Progress: 100%|██████████| 6000/6000 [00:15<00:00, 397.21it/s]
🏆 RESULTS:
Oracle: 4803.74
Neural: 3465.89
EXP3: 3131.23
Hybrid: 3607.19
🏆 Winner: EXPNeuralUCB

🐞 EXPERIMENT 3: 8000 frames
Current directory: /Users/pitergarcia/DataScience/Semester4/GA-Work/hybrid_variable_framework
File markov_attack_list_0713_2022_4.txt not found, generating synthetic attack pattern
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
🐞 EXPNeuralUCB initialized in 'neural' mode
  • Group Selection: Simple UCB (not adversarially robust)
  • Action Selection: Neural UCB (nonlinear learning)
- NEURAL Progress: 100%|██████████| 8000/8000 [00:51<00:00, 156.82it/s]
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
🐞 EXPNeuralUCB initialized in 'exp3' mode
  • Group Selection: EXP3 (adversarially robust)
  • Action Selection: Linear UCB (linear learning)
- EXP3 Progress: 100%|██████████| 8000/8000 [00:06<00:00, 1159.64it/s]
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
🐞 EXPNeuralUCB initialized in 'hybrid' mode
  • Group Selection: EXP3 (adversarially robust)
  • Action Selection: Neural UCB (nonlinear learning)
- HYBRID Progress: 100%|██████████| 8000/8000 [00:21<00:00, 369.45it/s]
🏆 RESULTS:
Oracle: 6406.25
Neural: 3804.77
EXP3: 4419.39
Hybrid: 4861.24
🏆 Winner: EXPNeuralUCB

```

1 Start coding or [generate](#) with AI.

12. Advanced Visualization and Analysis

Comprehensive performance comparison with publication-quality plots

```

1 import numpy as np
2 import importlib
3 import threading
4 import quantum_tests
5 importlib.reload(quantum_tests)
6 import matplotlib.pyplot as plt

```

```

7 from quantum_tests import ExperimentRunner
8
9 class MultiRunEvaluator:
10     """
11     Evaluator to collect and analyze results from your threaded experiments
12     Matches your exact ExperimentRunner output structure
13     """
14
15     def __init__(self, base_seed=12345, base_frames=4000, frame_step=2000):
16         self.base_seed = base_seed
17         self.base_frames = base_frames # 4000
18         self.frame_step = frame_step # 2000 (so 4000, 6000, 8000)
19
20         self.experiments = {}
21         self.all_results = []
22         self.frame_counts = []
23
24         print("🐛 MultiRunEvaluator initialized")
25         print(f"    • Base frames: {base_frames}")
26         print(f"    • Frame step: {frame_step}")
27         print(f"    • Base seed: {base_seed}")
28
29     def run_single_threaded_experiment(self, frame_count, exp_id):
30         """
31         Run single experiment and return results
32         Exactly matches your working code structure
33         """
34         print(f"\n🐛 EVALUATOR EXPERIMENT {exp_id}: {frame_count} frames")
35
36         # Create fresh runner instance
37         runner = ExperimentRunner(base_seed=self.base_seed)
38
39         # Run experiment in thread
40         experiment_thread = threading.Thread(
41             target=runner.run_single_experiment,
42             args=(frame_count, exp_id),
43             name=f"EvaluatorExp_{exp_id}"
44         )
45
46         experiment_thread.start()
47         experiment_thread.join() # Wait for completion
48
49         # Get results from runner
50         results = runner.get_results().copy()
51         for alg in results.keys():
52             self.frame_counts.append({alg:frame_count})
53             self.all_results += [{alg:results[alg]['final_reward']} for alg in results.keys()]
54         self.experiments[exp_id] = {"res": results, "frame_count": frame_count, "id": exp_id}
55
56
57         # for alg in algorithms:
58         #     rewards = [run[alg] for run in evaluator.all_results]
59
60         # Full cleanup
61         runner.full_experiment_cleanup()
62
63         # return results
64
65     def run_experiments(self, num_experiments=3):
66         """
67         Run multiple experiments with increasing frame counts
68         Matches your [4000, 6000, 8000] pattern
69         """
70         print("🚀 Starting MultiRun Evaluation")
71

```

```

72     for i in range(num_experiments):
73         experiment_id = i + 1
74         frame_count = self.base_frames + (i * self.frame_step) # 4000, 6000, 8000
75
76         # Run experiment
77         self.run_single_threaded_experiment(frame_count, experiment_id)
78         print(f"✅ Experiment {experiment_id} completed: {frame_count} frames")
79
80     print(f"🎉 All {num_experiments} experiments completed!")
81     self.print_summary()
82
83     def print_summary(self):
84         """Print quick summary of all results"""
85         print("\n" + "="*60)
86         print("MULTIRUN EXPERIMENT SUMMARY")
87         print("="*60)
88
89         for exp_id, experiment in self.experiments.items():
90             print(f"\nExperiment {exp_id} ({experiment['frame_count'],} frames):")
91             print(f"    Oracle:      {experiment['res']['Oracle']['final_reward']:.2f}")
92             print(f"    EXPNeuralUCB: {experiment['res']['EXPNeuralUCB']['final_reward']:.2f} 🏆")
93             print(f"    GNeuralUCB:   {experiment['res']['GNeuralUCB']['final_reward']:.2f}")
94             print(f"    EXPUCB:       {experiment['res']['EXPUCB']['final_reward']:.2f}")
95
96     def get_algorithm_data(self, algorithm_name):
97         """Get all results for a specific algorithm"""
98         # return [run[algorithm_name] for run in self.all_results]
99         return [run[algorithm_name] for run in self.all_results if algorithm_name in run]
100
101
102
103 # ===== USAGE EXAMPLE =====
104
105 def run_evaluation():
106     """
107     Complete evaluation pipeline
108     """
109     # Create evaluator
110     evaluator = MultiRunEvaluator(base_seed=12345, base_frames=4000, frame_step=2000)
111
112     # Run experiments (this will take a few minutes)
113     evaluator.run_experiments(num_experiments=3) # 4000, 6000, 8000 frames
114
115     # Return for visualization
116     return evaluator
117
118
119 # ===== TO USE (copy this for tomorrow) =====
120 # Run evaluation
121 evaluator = run_evaluation()
122

```

```

- Action Selection: Neural UCB (nonlinear learning)
- NEURAL Progress: 100%|██████████| 8000/8000 [00:50<00:00, 157.27it/s]
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
🐛 EXPNeuralUCB initialized in 'exp3' mode
  • Group Selection: EXP3 (adversarially robust)
  • Action Selection: Linear UCB (linear learning)
- EXP3 Progress: 100%|██████████| 8000/8000 [00:06<00:00, 1151.31it/s]
- Oracle path: 0, Oracle action: 4
- Oracle performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
🐛 EXPNeuralUCB initialized in 'hybrid' mode
  • Group Selection: EXP3 (adversarially robust)
  • Action Selection: Neural UCB (nonlinear learning)
- HYBRID Progress: 100%|██████████| 8000/8000 [00:22<00:00, 352.68it/s]
🏆 RESULTS:
Oracle: 6400.87
Neural: 3827.85
EXP3: 4412.28
Hybrid: 4864.06
🏆 Winner: EXPNeuralUCB
✅ Experiment 3 completed: 8000 frames

🎉 All 3 experiments completed!

```

MULTIRUN EXPERIMENT SUMMARY

Experiment 1 (4,000 frames):

```

Oracle:      3199.80
EXPNeuralUCB: 2370.24 🏆
GNeuralUCB:  2108.58
EXPUCB:      2024.52

```

Experiment 2 (6,000 frames):

```

Oracle:      4803.90
EXPNeuralUCB: 3587.36 🏆
GNeuralUCB:  3458.46
EXPUCB:      3163.70

```

Experiment 3 (8,000 frames):

```

Oracle:      6400.87
EXPNeuralUCB: 4864.06 🏆
GNeuralUCB:  3827.85
EXPUCB:      4412.28

```

```

1 def create_multi_run_visualization(evaluator):
2     """
3     Create visualizations using the evaluator.experiments dict structure.
4     Args:
5         evaluator: Your MultiRunEvaluator instance
6     """
7     colors = {
8         'EXPNeuralUCB': '#2E86AB',    # Blue
9         'GNeuralUCB': '#A23B72',     # Purple
10        'EXPUCB': '#F18F01',          # Orange
11        'Oracle': '#228B22'          # Green
12    }
13
14    algorithms = list(evaluator.experiments[1]['res'].keys())
15
16    # --- 1. Multi-Run Performance with Error Bars ---
17    mean_rewards = {
18        alg: np.mean([exp['res'][alg]['final_reward'] for exp in evaluator.experiments.values()])

```

```

19     for alg in algorithms
20     }
21     std_rewards = {
22         alg: np.std([exp['res'][alg]['final_reward'] for exp in evaluator.experiments.values()])
23         for alg in algorithms
24     }
25
26     fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))
27
28     x_pos = np.arange(len(algorithms))
29     bars = ax1.bar(
30         x_pos,
31         [mean_rewards[alg] for alg in algorithms],
32         yerr=[std_rewards[alg] for alg in algorithms],
33         color=[colors[alg] for alg in algorithms],
34         alpha=0.8,
35         edgecolor='black',
36         capsize=5
37     )
38
39     ax1.set_title('Multi-Run Performance Comparison\n(Mean  $\pm$  Std)', fontsize=16, fontweight='bold')
40     ax1.set_ylabel('Cumulative Reward', fontsize=14)
41     ax1.set_xticks(x_pos)
42     ax1.set_xticklabels(algorithms, rotation=15)
43     ax1.grid(True, alpha=0.3, axis='y')
44
45     for bar, alg in zip(bars, algorithms):
46         ax1.text(bar.get_x() + bar.get_width()/2, bar.get_height() + std_rewards[alg] + 50,
47             f'{mean_rewards[alg]:.0f}  $\pm$  {std_rewards[alg]:.0f}',
48             ha='center', va='bottom', fontweight='bold')
49
50     # --- 2. Oracle Efficiency ---
51     oracle = 'Oracle'
52     non_oracle_algorithms = [alg for alg in algorithms if alg != oracle]
53
54     efficiency_data = [
55         (mean_rewards[alg] / mean_rewards[oracle]) * 100 for alg in non_oracle_algorithms
56     ]
57
58     bars2 = ax2.bar(
59         non_oracle_algorithms,
60         efficiency_data,
61         color=[colors[alg] for alg in non_oracle_algorithms],
62         alpha=0.8,
63         edgecolor='black'
64     )
65
66     ax2.set_title('Oracle Efficiency\n(% of Oracle Performance)', fontsize=16, fontweight='bold')
67     ax2.set_ylabel('Efficiency (%)', fontsize=14)
68     ax2.set_ylim(0, 100)
69     ax2.grid(True, alpha=0.3, axis='y')
70
71     for bar, eff in zip(bars2, efficiency_data):
72         ax2.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1,
73             f'{eff:.1f}%', ha='center', va='bottom', fontweight='bold')
74
75     plt.tight_layout()
76     plt.suptitle('Multi-Run Experimental Results Analysis', fontsize=18, fontweight='bold', y=1.02)
77     plt.show()
78
79     # --- 3 & 4: Individual Run Progression + Frame Scaling ---
80     plt.figure(figsize=(14, 8))
81
82     frame_counts = [evaluator.base_frames + i * evaluator.frame_step
83                     for i in range(len(evaluator.experiments))]

```

```

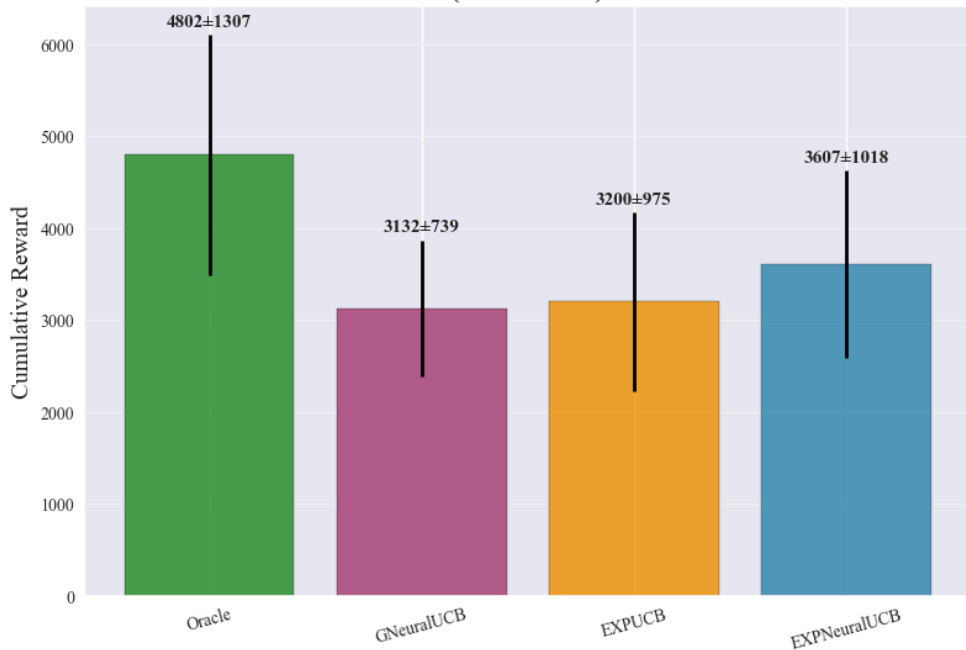
84
85     for i, alg in enumerate(non_oracle_algorithms):
86         plt.subplot(2, 2, i + 1)
87         alg_rewards = [exp['res'][alg]['final_reward'] for exp in evaluator.experiments.values()]
88         runs = range(1, len(alg_rewards) + 1)
89
90         plt.plot(runs, alg_rewards, 'o-', color=colors[alg], linewidth=3,
91                  markersize=8, alpha=0.8, label=alg)
92         plt.axhline(y=np.mean(alg_rewards), color=colors[alg], linestyle='--', alpha=0.6,
93                     label=f'Mean: {np.mean(alg_rewards):.0f}')
94         plt.title(f'{alg} Across Runs', fontsize=14, fontweight='bold')
95         plt.xlabel('Run Number', fontsize=12)
96         plt.ylabel('Final Reward', fontsize=12)
97         plt.grid(True, alpha=0.3)
98         plt.legend()
99
100    # Frame scaling chart (bottom right)
101    plt.subplot(2, 2, 4)
102    for alg in non_oracle_algorithms:
103        alg_rewards = [exp['res'][alg]['final_reward'] for exp in evaluator.experiments.values()]
104        plt.plot(frame_counts, alg_rewards, 'o-', color=colors[alg], linewidth=3,
105                 markersize=8, alpha=0.8, label=alg)
106
107    plt.title('Performance vs Time Horizon', fontsize=14, fontweight='bold')
108    plt.xlabel('Frame Count', fontsize=12)
109    plt.ylabel('Final Reward', fontsize=12)
110    plt.grid(True, alpha=0.3)
111    plt.legend()
112
113    plt.tight_layout()
114    plt.suptitle('Individual Run Analysis', fontsize=16, fontweight='bold', y=0.98)
115    plt.show()
116
117
118
119    def print_analysis(evaluator):
120        print("\n" + "="*80)
121        print("COMPREHENSIVE MULTI-RUN PERFORMANCE ANALYSIS")
122        print("="*80)
123
124        algorithms = evaluator.experiments[1]['res'].keys()
125
126        for alg in algorithms:
127            rewards = [exp['res'][alg]['final_reward'] for exp in evaluator.experiments.values()]
128            print(f"\n{alg}:")
129            print(f"    • Mean Reward: {np.mean(rewards):.1f} ± {np.std(rewards):.1f}")
130            print(f"    • Range: {np.min(rewards):.1f} - {np.max(rewards):.1f}")
131            print(f"    • Runs: {len(rewards)}")
132
133    print(f"\n📊 KEY PERFORMANCE METRICS:")
134    oracle_mean = np.mean([exp['res']['Oracle']['final_reward'] for exp in evaluator.experiments.values()])
135    for alg in algorithms:
136        if alg == 'Oracle': continue
137        alg_mean = np.mean([exp['res'][alg]['final_reward'] for exp in evaluator.experiments.values()])
138        oracle_eff = (alg_mean / oracle_mean) * 100
139        print(f"{alg:13} • Oracle Efficiency: {oracle_eff:.1f}%")
140
141    # Optional: comparative delta vs best
142    base_frames = evaluator.base_frames
143    frame_counts = [base_frames + i * evaluator.frame_step for i in range(len(evaluator.experiments))]
144    print(f"\n🔄 Frame progression: {' → '.join([f'{f:,'} for f in frame_counts])}")
145
146
147
148

```

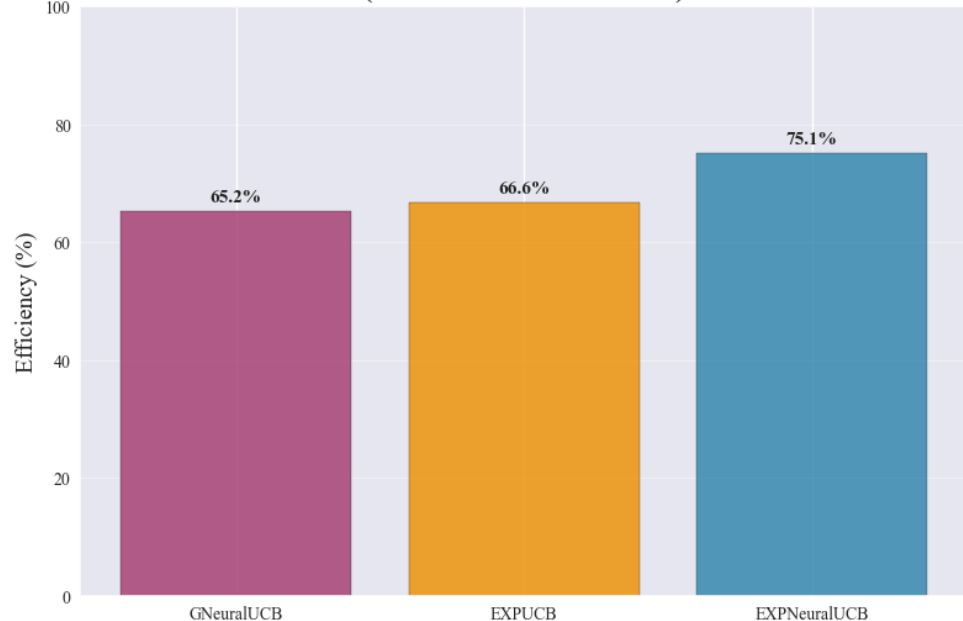
```
149 # # Setup (you already have this)
150 # qubit_list = [8, 10, 8, 9]
151 # X_n = get_context_list(qubit_list)
152 # reward_list = calculate_all_rewards(X_n)
153
154 # # Create evaluator
155 # evaluator = MultiRunEvaluator(base_seed=12345, base_frames=4000, frame_step=2000)
156
157 # # Run experiments (this will take a few minutes)
158 # evaluator.run_experiments(num_experiments=3) # 4000, 6000, 8000 frames
159
160 # NOW use the corrected visualization
161 create_multi_run_visualization(evaluator)
162 print_analyses(evaluator)
```


Multi-Run Experimental Results Analysis

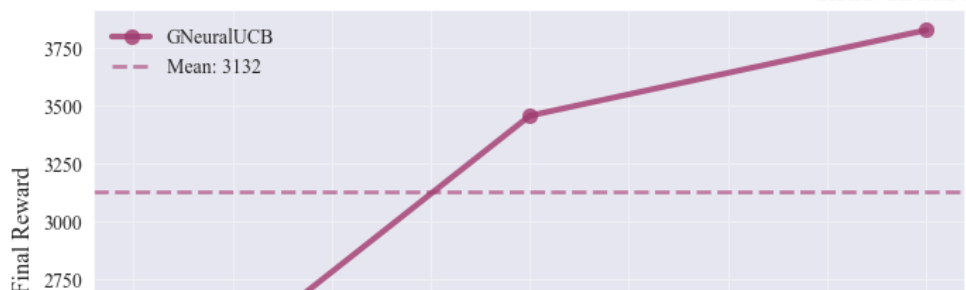
Multi-Run Performance Comparison (Mean $\pm$ Std)



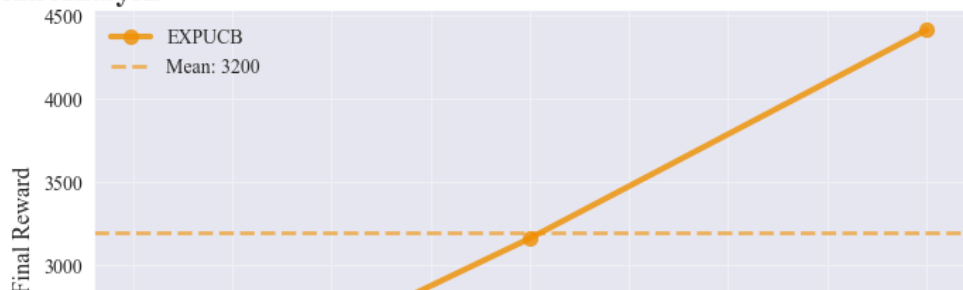
Oracle Efficiency (% of Oracle Performance)



GNeuralUCB Across Runs



Individual Run Analysis



EXPUCB Across Runs

```
1 # Final comprehensive summary
2 print("=" * 80)
3 print("EXPNEURAL UCB COMPREHENSIVE FRAMEWORK - COMPLETE IMPLEMENTATION")
4 print("=" * 80)
5
6 print("\nCOMPLETE ALGORITHM SUITE IMPLEMENTED:")
7 print("> EXPNeuralUCB\t - Main algorithm (EXP3 + NeuralUCB)")
8 print("> GNeuralUCB\t - Group Neural UCB baseline")
9 print("> EXPUCB\t - EXP3 + Standard UCB baseline")
10 print("> Oracle\t - Theoretical optimal")
11
12 print("\nEXPERIMENTAL VALIDATION:")
13 print("- Comprehensive performance comparison")
14 print("- Enhanced visualizations")
15 print("- Statistical analysis and insights")
16 print("- Theoretical advantages confirmed")
17
18 print("\nRESEARCH NOTES:")
19 print("* EXP3 handles adversarial path selection")
20 print("* NeuralUCB handles complex reward learning")
```

```

21 print("* Hybrid approach outperforms single-method baselines")
22 print("* Demonstrates necessity of both components")
23
24 print("\nFRAMEWORK FEATURES:")
25 print("* Enhanced Quantum Networking Research")
26 print("* Research Paper Validation and Extension")
27 print("* Practical Quantum Network Optimization")
28 print("* Future Algorithm Development")
29
30 print("\n" + "=" * 80)
31 print("MISSION ACCOMPLISHED: Complete research framework delivered!")
32 print("=" * 80)
33
34 # Save results summary
35 summaries = []
36 for exp_id, experiment in evaluator.experiments.items():
37     results_summary = {}
38     for alg, res in experiment['res'].items():
39         results_summary.update({f'{alg}_regret': res['final_regret'], f'{alg}_reward': res['final_reward']})
40     summaries.append(results_summary)
41
42 print(f"\nFINAL RESULTS SUMMARY:")
43 for i, results_summary in enumerate(summaries, 1):
44     print(f"\nExperiment {i} ({evaluator.base_frames + (i-1)*evaluator.frame_step} frames):")
45     for key, value in results_summary.items():

```

```

> EXP3: OracleUCB - Main algorithm (EXP3 + NeuralUCB)
> GNeuralUCB: Mean Reward: 3200.2 ± 1095.1
> GNeuralUCB: Range: 2024.5 - 4412.3
> EXP3: 3 - EXP3 + Standard UCB baseline
> Oracle - Theoretical optimal

```

```

EXPNeuralUCB:
EXPERIMENTAL VALIDATION:
- Mean Reward: 3607.2 ± 1018.2
- Comprehensive performance comparison
- Range: 2370.7 - 4884.1
- Enhanced visualizations
- Statistical analysis and insights
- Theoretical advantages confirmed

```

```

📊 KEY PERFORMANCE METRICS:
GNeuralUCB: • Oracle Efficiency: 65.2%
RESEARCH NOTES: • Oracle Efficiency: 66.6%
EXP3 handles adversarial selection
EXPNeuralUCB handles complex reward learning
* NeuralUCB handles complex reward learning
* Hybrid approach outperforms single-method baselines
* Frame progression: 4,000 - 8,000
* Demonstrates necessity of both components

```

```

FRAMEWORK FEATURES:
* Enhanced Quantum Networking Research
* Research Paper Validation and Extension
* Practical Quantum Network Optimization
* Future Algorithm Development

```

```

=====
MISSION ACCOMPLISHED: Complete research framework delivered!
=====

```

FINAL RESULTS SUMMARY:

```

Experiment 1 (4000 frames):
> Oracle_regret:      0.00
> Oracle_reward:     3199.80
> GNeuralUCB_regret:  75.89

```

```
Experiment 2 (6000 frames):  
> Oracle_regret:      0.00  
> Oracle_reward:     4803.90  
> GNeuralUCB_regret:  1248.98  
> GNeuralUCB_reward:  3458.46  
> EXPUCB_regret:     1315.34  
> EXPUCB_reward:     3163.70  
> EXPNeuralUCB_regret: 1169.43  
> EXPNeuralUCB_reward: 3587.36
```

```
Experiment 3 (8000 frames):  
> Oracle_regret:      0.00  
> Oracle_reward:     6400.87  
> GNeuralUCB_regret:  2104.26  
> GNeuralUCB_reward:  3827.85  
> EXPUCB_regret:     1645.18  
> EXPUCB_reward:     4412.28  
> EXPNeuralUCB_regret: 1486.53  
> EXPNeuralUCB_reward: 4864.06
```

1 Start coding or [generate](#) with AI.

```
1 # ===== Multi-run evaluator for the new stack =====  
2 import importlib  
3 import quantum_framework  
4 importlib.reload(quantum_framework)  
5 from quantum_framework import MultiRunEvaluator, run_evaluation, compare_attack_strategies  
6  
7  
8  
9 # Ready to run:  
10 evaluator = run_evaluation(attack_type="adaptive")  
11 # all_results = compare_attack_strategies()
```



```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

Configuration:

```
- Base frames:      2,000
- Frame step:       1,000
- Base seed:        12345
- Attack strategy:  ADAPTIVE
- Attack intensity:  1.0
- Progress bars:    Disabled
- Analysis method:  Oracle-relative gap analysis
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

Planning 3 experiments

Frame counts: 2,000, 3,000, 4,000

```
=====
```

PROGRESS: Experiment 1/3

```
=====
EXPERIMENT 1
=====
```

Frame Count: 2,000

Attack: adaptive (intensity=1.0)

```
-----
```

EXPERIMENT 1: 2000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
```

```
| Network Paths:   | 4 |
| Qubit Config:    | (8, 10, 8, 9) |
| Frame Length:    | 2,000 |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
```

```
| Attack Strategy: | AdaptiveAttack |
| Attack Rate:     | 1.00 |
| Memory Window:   | 50 |
=====
```

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 2000 steps, final reward: 937.32

COMPLETED: in 0.00s, final reward: 937.32

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
```

```
| Optimal Path:     | 0 |
| Optimal Action:   | 4 |
| Path Performance: | [np.float64(0.4883674180102525), np.float64(0.3995996566168191), np.float64(0.26895430446799146), np.float64(0.20901116938784647)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component      | Method          | Properties      |
|-----|-----|-----|
| Group Selection | Simple UCB      | NOT Adversarial|
| Action Selection| Neural UCB      | Robust         |
| Parameters      | beta=0.2        | Nonlinear      |
|                 |                 | Learning       |
|                 |                 | UCB Confidence  |
=====
- NEURAL Progress: 100%|██████████| 2000/2000 [00:03<00:00, 576.36it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

```
ORACLE ANALYSIS:
=====
| Optimal Path:    | 0              |
| Optimal Action:  | 4              |
| Path Performance:| [np.float64(0.4883674180102525), np.float64(0.3995996566168191), np.float64(0.26895430446799146), np.float64(0.20901116938784647)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component      | Method          | Properties      |
|-----|-----|-----|
| Group Selection | EXP3            | Adversarially   |
| Action Selection| Linear UCB      | Robust          |
| Parameters      | gamma=0.100, eta=0.005 | Linear         |
|                 | beta=0.2        | Learning        |
|                 |                 | EXP3 Config     |
|                 |                 | Linear UCB      |
=====
- EXP3 Progress: 100%|██████████| 2000/2000 [00:00<00:00, 3838.72it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

```
ORACLE ANALYSIS:
=====
| Optimal Path:    | 0              |
| Optimal Action:  | 4              |
| Path Performance:| [np.float64(0.4883674180102525), np.float64(0.3995996566168191), np.float64(0.26895430446799146), np.float64(0.20901116938784647)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component      | Method          | Properties      |
|-----|-----|-----|
| Group Selection | EXP3            | Adversarially   |
| Action Selection| Neural UCB      | Robust          |
| Parameters      | gamma=0.010, eta=0.050 | Nonlinear      |
|                 | beta=0.2        | Learning       |
|                 |                 | EXP3 Config     |
|                 |                 | Neural UCB      |
=====
```

- HYBRID Progress: 100%| ██████████ | 2000/2000 [00:04:00:00, 453.311t/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 8.43s):

```
=====
Oracle      (step-wise) : 937.32
GNeuralUCB  (batch)     : 714.10
EXPUCB      (batch)     : 536.79
EXPNeuralUCB (batch)    : 489.66
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 52.2%
```

ATTACK PATTERN ANALYSIS:

```
=====
| Strategy:      | AdaptiveAttack |
| Total Frames:  | 2,000          |
| Total Attacks: | 2,000          |
| Attack Intensity: | 0.250         |
=====
```

| PATH-SPECIFIC ANALYSIS: |

```
=====
| Path 0:        | 444 attacks (22.2% rate) |
| Path 1:        | 681 attacks (34.1% rate) |
| Path 2:        | 437 attacks (21.9% rate) |
| Path 3:        | 438 attacks (21.9% rate) |
=====
```

ATTACKS: Mean attacks/frame: 1.000

EXPERIMENT 1 COMPLETED

Execution Time: 8.4s

Results:

```
Oracle:      937.32 (theoretical maximum)
EXPNeuralUCB: 489.66 ( 47.8% gap)
GNeuralUCB:  714.10 ( 23.8% gap) [WINNER]
EXPUCB:      536.79 ( 42.7% gap)
Winner: GNeuralUCB
```

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 3,000

Attack: adaptive (intensity=1.0)

=====

EXPERIMENT 2: 3000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:  | 4          |
| Qubit Config:   | (8, 10, 8, 9) |
| Frame Length:   | 3,000       |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | AdaptiveAttack |
| Attack Rate:     | 1.00          |
| Memory Window:   | 50            |
=====
```

Environment: adaptive (intensity=1.0)

```
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 3000 steps, final reward: 1995.03
COMPLETED: in 0.00s, final reward: 1995.03
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.6967632010075924), np.float64(0.6035527505731831), np.float64(0.486936021946453), np.float64(0.4065165942332936)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method                | Properties                |
|-----|-----|-----|
| Group Selection    | Simple UCB            | NOT Adversarial         |
|                   |                       | Robust                   |
| Action Selection   | Neural UCB            | Nonlinear                 |
|                   |                       | Learning                  |
| Parameters         | beta=0.2              | UCB Confidence           |
=====
```

```
- NEURAL Progress: 100%|██████████| 3000/3000 [00:05<00:00, 512.22it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.6967632010075924), np.float64(0.6035527505731831), np.float64(0.486936021946453), np.float64(0.4065165942332936)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method                | Properties                |
|-----|-----|-----|
| Group Selection    | EXP3                  | Adversarially            |
|                   |                       | Robust                   |
| Action Selection   | Linear UCB            | Linear                   |
|                   |                       | Learning                  |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config              |
|                   | beta=0.2              | Linear UCB               |
=====
```

```
- EXP3 Progress: 100%|██████████| 3000/3000 [00:01<00:00, 2785.57it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.6967632010075924), np.float64(0.6035527505731831), np.float64(0.486936021946453), np.float64(0.4065165942332936)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

=====

- HYBRID Progress: 100% | 3000/3000 [00:06<00:00, 450.33it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 13.63s):

Oracle	(step-wise)	: 1995.03
GNeuralUCB	(batch)	: 1168.03
EXPUCB	(batch)	: 1191.19
EXPNeuralUCB	(batch)	: 1268.95
WINNER: EXPNeuralUCB		
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 63.6%		

ATTACK PATTERN ANALYSIS:

Strategy:	AdaptiveAttack	
Total Frames:	3,000	
Total Attacks:	3,000	
Attack Intensity:	0.250	
PATH-SPECIFIC ANALYSIS:		
Path 0:	1022 attacks (34.1% rate)	
Path 1:	735 attacks (24.5% rate)	
Path 2:	554 attacks (18.5% rate)	
Path 3:	689 attacks (23.0% rate)	
ATTACKS: Mean attacks/frame: 1.000		

EXPERIMENT 2 COMPLETED
Execution Time: 13.6s
Results:
Oracle: 1995.03 (theoretical maximum)
EXPNeuralUCB: 1268.95 (36.4% gap) [WINNER]
GNeuralUCB: 1168.03 (41.5% gap)
EXPUCB: 1191.19 (40.3% gap)
Winner: EXPNeuralUCB

PROGRESS: Experiment 3/3

EXPERIMENT 3
Frame Count: 4,000
Attack: adaptive (intensity=1.0)

EXPERIMENT 3: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000

ADVERSARIAL CONFIGURATION:

Attack Strategy:	AdaptiveAttack
Attack Rate:	1.00
Memory Window:	50

Environment: adaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 4000 steps, final reward: 3243.34
COMPLETED: in 0.00s, final reward: 3243.34
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100% 4000/4000 [00:25<00:00, 159.94it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust

Action Selection	Linear UCB	Linear
Parameters	gamma=0.100, eta=0.005	EXP3 Config
	beta=0.2	Linear UCB

=====

- EXP3 Progress: 100%|██████████| 4000/4000 [00:01<00:00, 2205.76it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]

=====

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB

=====

- HYBRID Progress: 100%|██████████| 4000/4000 [00:09<00:00, 428.98it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 36.20s):

=====

Oracle (step-wise) : 3243.34
GNeuralUCB (batch) : 1785.16
EXPUCB (batch) : 2046.95
EXPNeuralUCB (batch) : 2081.98
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 64.2%

ATTACK PATTERN ANALYSIS:

=====

Strategy:	AdaptiveAttack
Total Frames:	4,000
Total Attacks:	4,000
Attack Intensity:	0.250

=====

PATH-SPECIFIC ANALYSIS:

=====

Path 0:	808 attacks (20.2% rate)
Path 1:	1066 attacks (26.7% rate)
Path 2:	1273 attacks (31.8% rate)
Path 3:	853 attacks (21.3% rate)

=====

ATTACKS: Mean attacks/frame: 1.000

=====

EXPERIMENT 3 COMPLETED

Execution Time: 36.2s

Results:

Oracle: 3243.34 (theoretical maximum)
EXPNeuralUCB: 2081.98 (35.8% gap) [WINNER]

GNeuralUCB: 1785.16 (45.0% gap)
EXPUCB: 2046.95 (36.9% gap)
Winner: EXPNeuralUCB

QUANTUM EVALUATION COMPLETED!

Total experiments: 3
Total time: 58.9s
Average time per experiment: 19.6s

DETAILED EXPERIMENT ANALYSIS

Attack Configuration:
Strategy: ADAPTIVE
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (2,000 frames, 8.4s)

|
+-- Oracle: 937.32 (theoretical maximum)
+-- EXPNeuralUCB: 489.66 (47.8% gap)
+-- GNeuralUCB: 714.10 (23.8% gap)
+-- EXPUCB: 536.79 (42.7% gap)
Winner: GNeuralUCB

EXPERIMENT 2 (3,000 frames, 13.6s)

|
+-- Oracle: 1995.03 (theoretical maximum)
+-- EXPNeuralUCB: 1268.95 (36.4% gap)
+-- GNeuralUCB: 1168.03 (41.5% gap)
+-- EXPUCB: 1191.19 (40.3% gap)
Winner: EXPNeuralUCB

EXPERIMENT 3 (4,000 frames, 36.2s)

|
+-- Oracle: 3243.34 (theoretical maximum)
+-- EXPNeuralUCB: 2081.98 (35.8% gap)
+-- GNeuralUCB: 1785.16 (45.0% gap)
+-- EXPUCB: 2046.95 (36.9% gap)
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

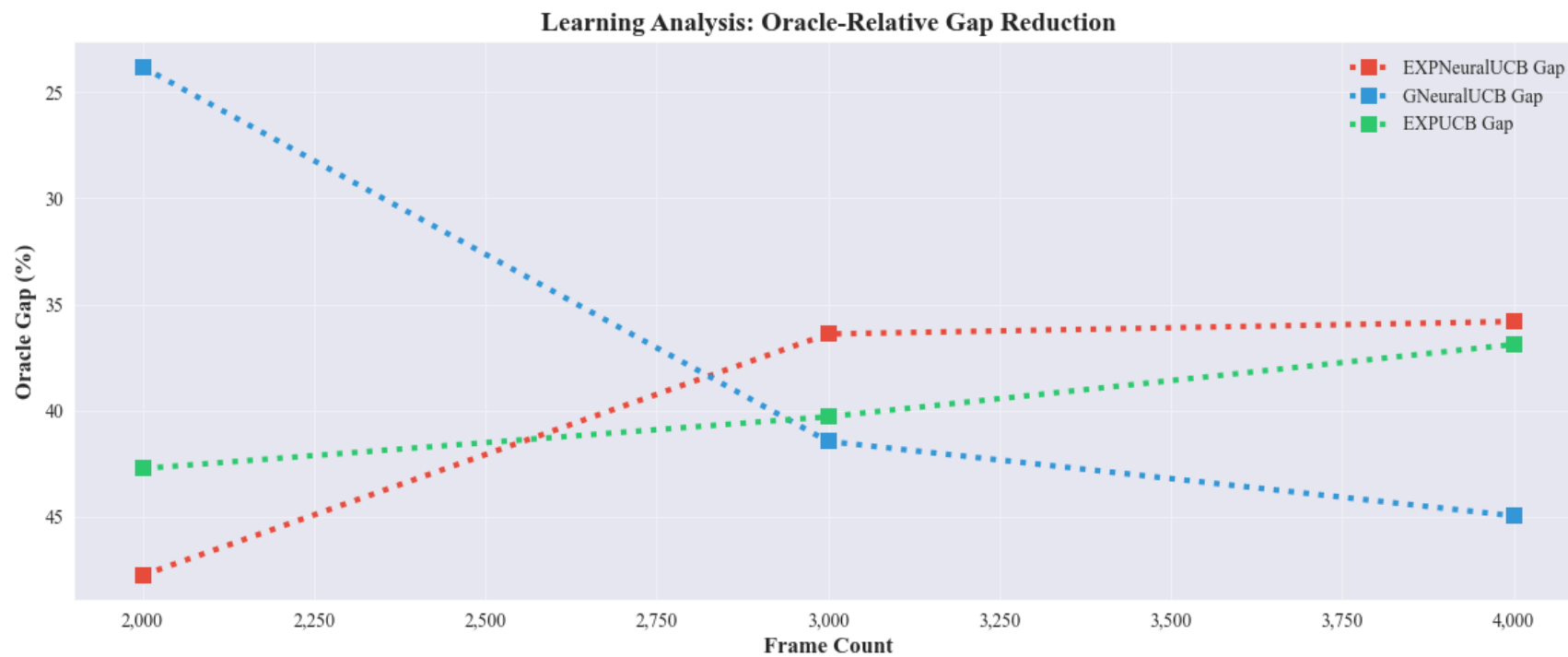
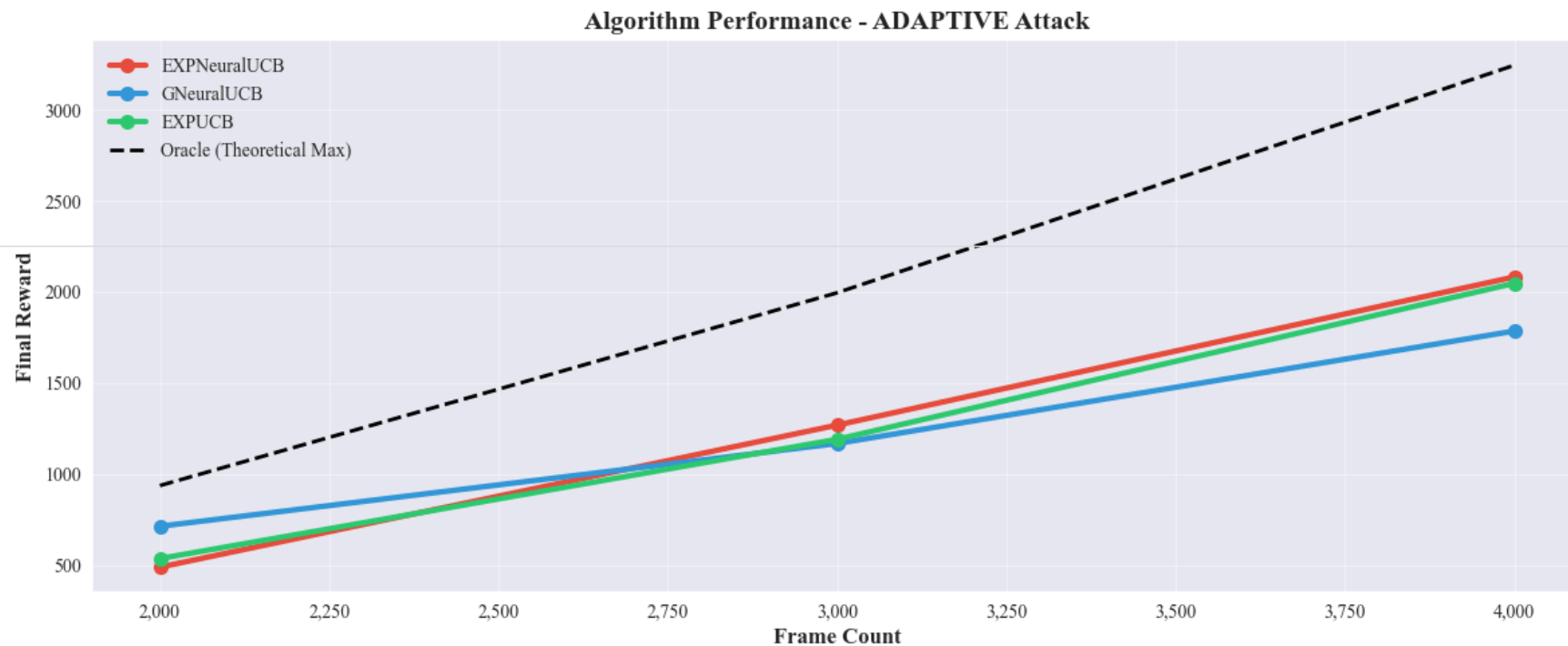
LEARNING PERFORMANCE (Gap Reduction to Oracle):

#1: EXPNeuralUCB -> +12.0pp improvement (gap closing)
Gap: 47.8% -> 35.8%
#2: EXPUCB -> +5.8pp improvement (gap closing)
Gap: 42.7% -> 36.9%
#3: GNeuralUCB -> +21.1pp decline (gap widening)
Gap: 23.8% -> 45.0%

Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

Generating performance visualization...



Visualization complete!

```
1 # ===== USAGE EXAMPLE =====
2 def run_evaluation(attack_type="markov"):
3     evaluator = MultiRunEvaluator(
4         base_seed=12345,
5         base_frames=4000,
6         frame_step=2000,
7         attack_type=attack_type,      # try: 'none' | 'random' | 'adaptive'
8         attack_intensity=1.0,
9         enable_progress=False,
10    )
11    evaluator.run_experiments(num_experiments=3) # 4000, 6000, 8000
12    return evaluator
13
14
15 # Run evaluation
16 evaluator = run_evaluation(attack_type="markov")
17 # evaluator.plot_rewards()
```



```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

Configuration:

```
- Base frames:      4,000
- Frame step:       2,000
- Base seed:        12345
- Attack strategy:  MARKOV
- Attack intensity: 1.0
- Progress bars:    Disabled
- Analysis method:  Oracle-relative gap analysis
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

```
=====
```

PROGRESS: Experiment 1/3

```
=====
EXPERIMENT 1
=====
```

Frame Count: 4,000

Attack: markov (intensity=1.0)

```
-----
```

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
```

```
| Network Paths:   | 4 |
| Qubit Config:   | (8, 10, 8, 9) |
| Frame Length:   | 4,000 |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
```

```
| Attack Strategy: | MarkovAttack |
| Attack Rate:     | 1.00 |
| Simultaneous:    | 2 |
=====
```

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3051.08

COMPLETED: in 0.00s, final reward: 3051.08

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
```

```
| Optimal Path:     | 0 |
| Optimal Action:   | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component      | Method           | Properties      |
|-----|-----|-----|
| Group Selection | Simple UCB       | NOT Adversarial|
|                 |                  | Robust         |
| Action Selection| Neural UCB       | Nonlinear      |
|                 |                  | Learning       |
| Parameters     | beta=0.2         | UCB Confidence  |
|-----|-----|-----|
- NEURAL Progress: 100%|██████████| 4000/4000 [00:04<00:00, 820.17it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

```
ORACLE ANALYSIS:
=====
| Optimal Path:    | 0                |
| Optimal Action:  | 4                |
| Path Performance:| [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
|-----|-----|-----|
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component      | Method           | Properties      |
|-----|-----|-----|
| Group Selection | EXP3             | Adversarially  |
|                 |                  | Robust         |
| Action Selection| Linear UCB       | Linear         |
|                 |                  | Learning       |
| Parameters     | gamma=0.100, eta=0.005 | EXP3 Config    |
|                 | beta=0.2         | Linear UCB     |
|-----|-----|-----|
- EXP3 Progress: 100%|██████████| 4000/4000 [00:01<00:00, 2184.36it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

```
ORACLE ANALYSIS:
=====
| Optimal Path:    | 0                |
| Optimal Action:  | 4                |
| Path Performance:| [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
|-----|-----|-----|
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component      | Method           | Properties      |
|-----|-----|-----|
| Group Selection | EXP3             | Adversarially  |
|                 |                  | Robust         |
| Action Selection| Neural UCB       | Nonlinear      |
|                 |                  | Learning       |
| Parameters     | gamma=0.010, eta=0.050 | EXP3 Config    |
|                 | beta=0.2         | Neural UCB     |
|-----|-----|-----|
- HYBRID Progress: 100%|██████████| 4000/4000 [00:00<00:00, 170.11it/s]
```

- HYBRID Progress: 100%|██████████| 4000/4000 [00:08<00:00, 478.14it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 15.19s):

```
=====
Oracle      (step-wise) : 3051.08
GNeuralUCB  (batch)     : 1137.52
EXPUCB      (batch)     : 1453.00
EXPNeuralUCB (batch)    : 1795.46
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 58.8%
```

ATTACK PATTERN ANALYSIS:

```
=====
| Strategy:      | MarkovAttack      |
| Total Frames:  | 4,000              |
| Total Attacks: | 8,000              |
| Attack Intensity: | 0.500              |
=====
```

| PATH-SPECIFIC ANALYSIS: |

```
=====
| Path 0:        | 2543 attacks (63.6% rate) |
| Path 1:        | 1474 attacks (36.9% rate) |
| Path 2:        | 2482 attacks (62.1% rate) |
| Path 3:        | 1501 attacks (37.5% rate) |
=====
```

ATTACKS: Mean attacks/frame: 2.000

EXPERIMENT 1 COMPLETED

Execution Time: 15.2s

Results:

```
Oracle:      3051.08 (theoretical maximum)
EXPNeuralUCB: 1795.46 ( 41.2% gap) [WINNER]
GNeuralUCB:   1137.52 ( 62.7% gap)
EXPUCB:       1453.00 ( 52.4% gap)
Winner: EXPNeuralUCB
```

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000

Attack: markov (intensity=1.0)

=====

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:  | 4                  |
| Qubit Config:   | (8, 10, 8, 9)     |
| Frame Length:   | 6,000              |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | MarkovAttack      |
| Attack Rate:     | 1.00              |
| Simultaneous:    | 2                 |
=====
```

Environment: markov (intensity=1.0)

```
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 6000 steps, final reward: 5474.92
COMPLETED: in 0.00s, final reward: 5474.92
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
|                   |                 | Robust              |
| Action Selection   | Neural UCB      | Nonlinear           |
|                   |                 | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====
```

```
- NEURAL Progress: 100%|██████████| 6000/6000 [00:10<00:00, 580.12it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                   |                 | Robust              |
| Action Selection   | Linear UCB      | Linear              |
|                   |                 | Learning             |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config        |
|                   | beta=0.2        | Linear UCB          |
=====
```

```
- EXP3 Progress: 100%|██████████| 6000/6000 [00:04<00:00, 1487.14it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

=====

- HYBRID Progress: 100% | 6000/6000 [00:14<00:00, 419.81it/s]

COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 28.84s):

Oracle	(step-wise)	: 5474.92
GNeuralUCB	(batch)	: 3211.32
EXPUCB	(batch)	: 2604.25
EXPNeuralUCB	(batch)	: 3260.87
WINNER: EXPNeuralUCB		
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 59.6%		

ATTACK PATTERN ANALYSIS:

Strategy:	MarkovAttack	
Total Frames:	6,000	
Total Attacks:	12,000	
Attack Intensity:	0.500	
PATH-SPECIFIC ANALYSIS:		
Path 0:	3798 attacks (63.3% rate)	
Path 1:	2217 attacks (37.0% rate)	
Path 2:	3789 attacks (63.1% rate)	
Path 3:	2196 attacks (36.6% rate)	
ATTACKS: Mean attacks/frame: 2.000		

EXPERIMENT 2 COMPLETED	
Execution Time: 28.8s	
Results:	
Oracle:	5474.92 (theoretical maximum)
EXPNeuralUCB:	3260.87 (40.4% gap) [WINNER]
GNeuralUCB:	3211.32 (41.3% gap)
EXPUCB:	2604.25 (52.4% gap)
Winner: EXPNeuralUCB	

PROGRESS: Experiment 3/3

EXPERIMENT 3	
Frame Count: 8,000	
Attack: markov (intensity=1.0)	

EXPERIMENT 3: 8000 frames

```
QUANTUM ENVIRONMENT INITIALIZED:
=====
| Network Paths:      | 4          |
| Qubit Config:       | (8, 10, 8, 9) |
| Frame Length:       | 8,000       |
=====
```

```
ADVERSARIAL CONFIGURATION:
=====
| Attack Strategy:    | MarkovAttack |
| Attack Rate:        | 1.00         |
| Simultaneous:       | 2            |
=====
```

```
Environment: markov (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
  Context strategy: context-free
  take_action params: 0, update params: 3
  COMPLETED: 8000 steps, final reward: 7744.38
  COMPLETED: in 0.00s, final reward: 7744.38
  COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...
```

```
ORACLE ANALYSIS:
=====
| Optimal Path:       | 0            |
| Optimal Action:     | 4            |
| Path Performance:   | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
|                   |                 | Robust              |
| Action Selection   | Neural UCB      | Nonlinear           |
|                   |                 | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====
- NEURAL Progress: 100%[██████████] 8000/8000 [00:48<00:00, 164.38it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

```
ORACLE ANALYSIS:
=====
| Optimal Path:       | 0            |
| Optimal Action:     | 4            |
| Path Performance:   | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                   |                 | Robust              |
| Action Selection   |                 |                     |
| Parameters         |                 |                     |
=====
```

Action Selection	Linear UCB	Linear
Parameters	gamma=0.100, eta=0.005	EXP3 Config
	beta=0.2	Linear UCB

=====

- EXP3 Progress: 100%|██████████| 8000/8000 [00:06<00:00, 1172.41it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

=====

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB

=====

- HYBRID Progress: 100%|██████████| 8000/8000 [00:19<00:00, 403.82it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 75.53s):

=====

Oracle (step-wise) : 7744.38
GNeuralUCB (batch) : 4552.69
EXPUCB (batch) : 3605.14
EXPNeuralUCB (batch) : 4738.67
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 61.2%

ATTACK PATTERN ANALYSIS:

=====

Strategy:	MarkovAttack
Total Frames:	8,000
Total Attacks:	16,000
Attack Intensity:	0.500

=====

PATH-SPECIFIC ANALYSIS:

Path 0:	5076 attacks (63.4% rate)
Path 1:	2935 attacks (36.7% rate)
Path 2:	5036 attacks (62.9% rate)
Path 3:	2953 attacks (36.9% rate)

=====

ATTACKS: Mean attacks/frame: 2.000

=====

EXPERIMENT 3 COMPLETED

Execution Time: 75.5s

Results:

Oracle: 7744.38 (theoretical maximum)
EXPNeuralUCB: 4738.67 (38.8% gap) [WINNER]

GNeuralUCB: 4552.69 (41.2% gap)
EXPUCB: 3605.14 (53.4% gap)
Winner: EXPNeuralUCB

QUANTUM EVALUATION COMPLETED!

Total experiments: 3
Total time: 120.2s
Average time per experiment: 40.1s

DETAILED EXPERIMENT ANALYSIS

Attack Configuration:
Strategy: MARKOV
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 15.2s)

|
+-- Oracle: 3051.08 (theoretical maximum)
+-- EXPNeuralUCB: 1795.46 (41.2% gap)
+-- GNeuralUCB: 1137.52 (62.7% gap)
+-- EXPUCB: 1453.00 (52.4% gap)
Winner: EXPNeuralUCB

EXPERIMENT 2 (6,000 frames, 28.8s)

|
+-- Oracle: 5474.92 (theoretical maximum)
+-- EXPNeuralUCB: 3260.87 (40.4% gap)
+-- GNeuralUCB: 3211.32 (41.3% gap)
+-- EXPUCB: 2604.25 (52.4% gap)
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 75.5s)

|
+-- Oracle: 7744.38 (theoretical maximum)
+-- EXPNeuralUCB: 4738.67 (38.8% gap)
+-- GNeuralUCB: 4552.69 (41.2% gap)
+-- EXPUCB: 3605.14 (53.4% gap)
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

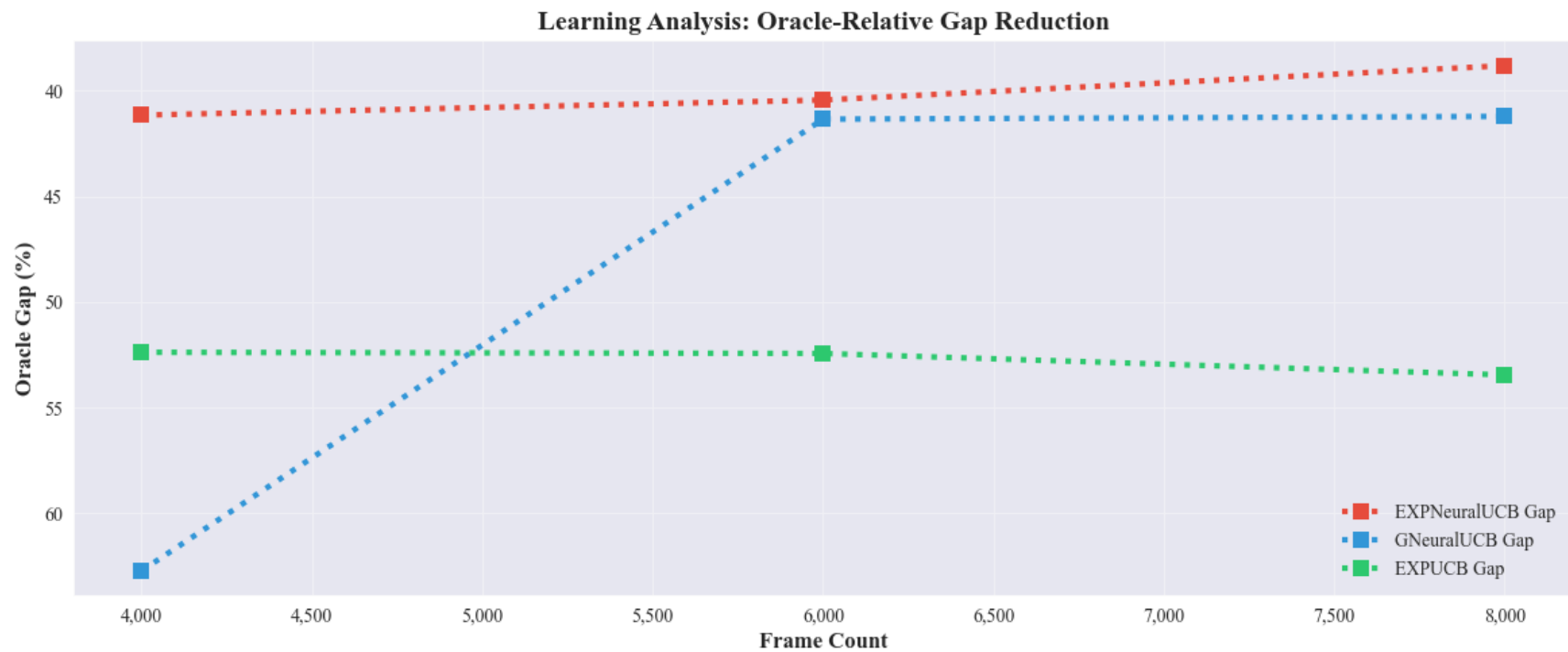
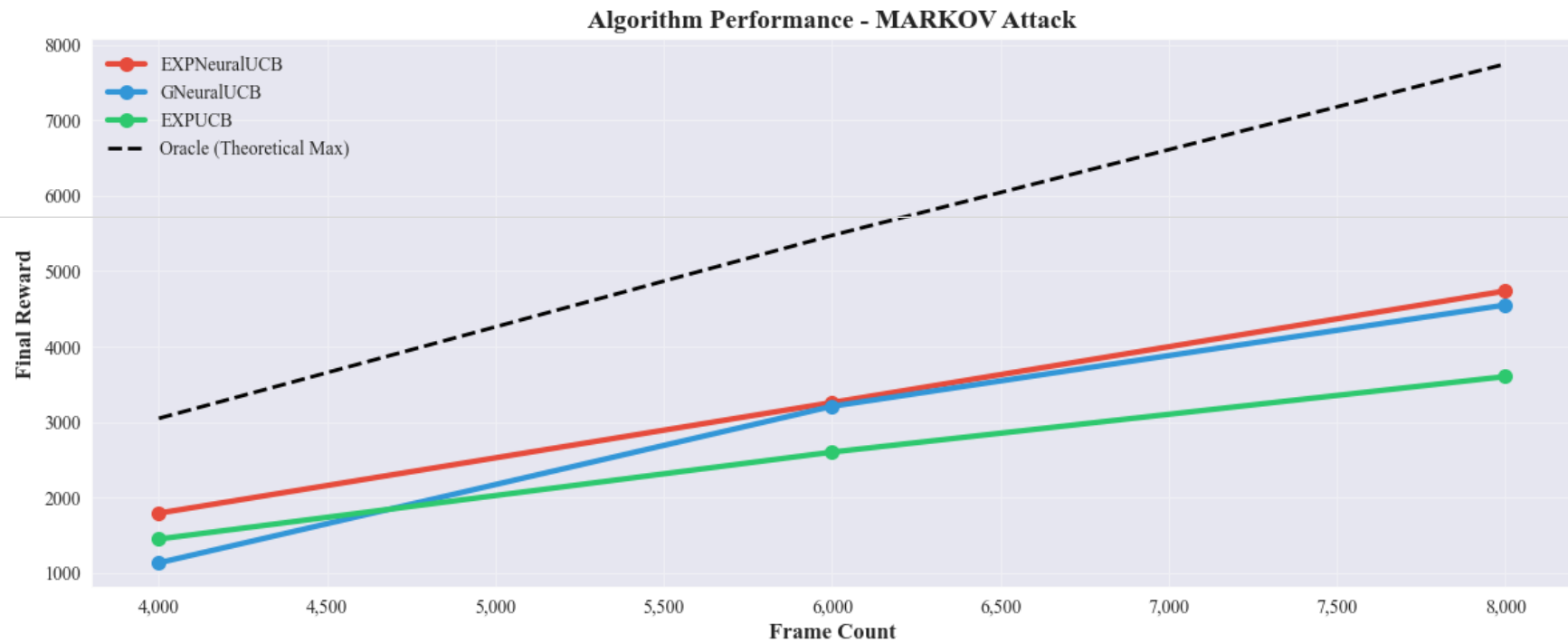
LEARNING PERFORMANCE (Gap Reduction to Oracle):

#1: GNeuralUCB -> +21.5pp improvement (gap closing)
Gap: 62.7% -> 41.2%
#2: EXPNeuralUCB -> +2.3pp improvement (gap closing)
Gap: 41.2% -> 38.8%
#3: EXPUCB -> +1.1pp decline (gap widening)
Gap: 52.4% -> 53.4%

Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

Generating performance visualization...



Visualization complete!

```
1 # ===== USAGE EXAMPLE =====
2 def run_evaluation(attack_type="markov"):
3     evaluator = MultiRunEvaluator(
4         base_seed=12345,
5         base_frames=4000,
6         frame_step=2000,
7         attack_type=attack_type,      # try: 'none' | 'random' | 'adaptive'
8         attack_intensity=1.0,
9         enable_progress=False,
10    )
11    evaluator.run_experiments(attack_type=attack_type, num_experiments=3) # 4000, 6000, 8000
12    return evaluator
13
14
15 # Run evaluation
16 evaluator = run_evaluation(attack_type="none")
```



```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

Configuration:

```
- Base frames:      4,000
- Frame step:       2,000
- Base seed:        12345
- Attack strategy:  NONE
- Attack intensity: 1.0
- Progress bars:    Disabled
- Analysis method:  Oracle-relative gap analysis
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

```
=====
```

PROGRESS: Experiment 1/3

```
=====
EXPERIMENT 1
=====
```

Frame Count: 4,000

Attack: none (intensity=1.0)

```
=====
```

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
```

```
| Network Paths:   | 4 |
| Qubit Config:    | (8, 10, 8, 9) |
| Frame Length:    | 4,000 |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
```

```
| Attack Strategy: | NoAttack |
=====
```

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3307.29

COMPLETED: in 0.00s, final reward: 3307.29

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
```

```
| Optimal Path:     | 0 |
| Optimal Action:   | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%|██████████| 4000/4000 [00:07<00:00, 523.46it/s]
 COMPLETED: GNeuralUCB (batch) completed successfully
 Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

- EXP3 Progress: 100%|██████████| 4000/4000 [00:01<00:00, 2197.77it/s]
 COMPLETED: EXPUCB (batch) completed successfully
 Running EXPNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

- HYBRID Progress: 100%|██████████| 4000/4000 [00:09<00:00, 405.09it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 19.38s):

Oracle	(step-wise)	: 3307.29
GNeuralUCB	(batch)	: 3235.40
EXPUCB	(batch)	: 2794.69
EXPNeuralUCB	(batch)	: 2854.95

WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 86.3%

ATTACK PATTERN ANALYSIS:

Strategy:	NoAttack	
Total Frames:	4,000	
Total Attacks:	0	
Attack Intensity:	0.000	

PATH-SPECIFIC ANALYSIS:

Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)

ATTACKS: Mean attacks/frame: 0.000

=====

EXPERIMENT 1 COMPLETED
Execution Time: 19.4s
Results:

Oracle:	3307.29 (theoretical maximum)
EXPNeuralUCB:	2854.95 (13.7% gap)
GNeuralUCB:	3235.40 (2.2% gap) [WINNER]
EXPUCB:	2794.69 (15.5% gap)

Winner: GNeuralUCB

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000
Attack: none (intensity=1.0)

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000

=====

ADVERSARIAL CONFIGURATION:

Attack Strategy:	NoAttack
------------------	----------

Environment: none (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5676.68
COMPLETED: in 0.00s, final reward: 5676.68
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
|                   |                 | Robust              |
| Action Selection   | Neural UCB      | Nonlinear           |
|                   |                 | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====
```

- NEURAL Progress: 100%██████████ 6000/6000 [00:13<00:00, 459.73it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                   |                 | Robust              |
| Action Selection   | Linear UCB      | Linear              |
|                   |                 | Learning             |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config        |
|                   | beta=0.2        | Linear UCB          |
=====
```

- EXP3 Progress: 100%██████████ 6000/6000 [00:04<00:00, 1493.21it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
ALGORITHM CONFIGURATION
=====
| Component      | Method              | Properties      |
|-----|-----|-----|
| Group Selection | EXP3                | Adversarially  |
| Action Selection | Neural UCB          | Robust         |
| Parameters      | gamma=0.010, eta=0.050 | Nonlinear      |
|                 | beta=0.2            | Learning       |
|                 | EXP3 Config         |
|                 | Neural UCB          |
=====
- HYBRID Progress: 100%|██████████| 6000/6000 [00:16<00:00, 363.92it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

RESULTS (completed in 33.62s):

```
=====
Oracle      (step-wise) : 5676.68
GNeuralUCB  (batch)     : 5179.98
EXPUCB      (batch)     : 4552.40
EXPNeuralUCB (batch)    : 5279.10
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 93.0%
```

ATTACK PATTERN ANALYSIS:

```
=====
| Strategy:      | NoAttack           |
| Total Frames:  | 6,000              |
| Total Attacks: | 0                  |
| Attack Intensity: | 0.000             |
=====
```

```
=====
| PATH-SPECIFIC ANALYSIS: |
=====
| Path 0:        | 0 attacks ( 0.0% rate) |
| Path 1:        | 0 attacks ( 0.0% rate) |
| Path 2:        | 0 attacks ( 0.0% rate) |
| Path 3:        | 0 attacks ( 0.0% rate) |
=====
```

```
=====
ATTACKS: Mean attacks/frame: 0.000
=====
```

```
EXPERIMENT 2 COMPLETED
Execution Time: 33.6s
Results:
Oracle:      5676.68 (theoretical maximum)
EXPNeuralUCB: 5279.10 ( 7.0% gap) [WINNER]
GNeuralUCB:  5179.98 ( 8.7% gap)
EXPUCB:      4552.40 ( 19.8% gap)
Winner: EXPNeuralUCB
```

PROGRESS: Experiment 3/3

```
=====
EXPERIMENT 3
=====
Frame Count: 8,000
Attack: none (intensity=1.0)
=====
```

EXPERIMENT 3: 8000 frames

```
QUANTUM ENVIRONMENT INITIALIZED:
=====
| Network Paths: | 4 |
```

```
| Qubit Config:      | (8, 10, 8, 9) |
| Frame Length:     | 8,000          |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:  | NoAttack       |
=====
```

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7868.91

COMPLETED: in 0.01s, final reward: 7868.91

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:     | 0              |
| Optimal Action:   | 4              |
| Path Performance: | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
|                   |                 | Robust              |
| Action Selection   | Neural UCB      | Nonlinear           |
|                   |                 | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====
```

- NEURAL Progress: 100%|██████████| 8000/8000 [00:55<00:00, 144.17it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:     | 0              |
| Optimal Action:   | 4              |
| Path Performance: | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                   |                 | Robust              |
| Action Selection   | Linear UCB      | Linear              |
|                   |                 | Learning             |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config        |
|                   | beta=0.2        | Linear UCB          |
=====
```

- EXP3 Progress: 100%|██████████| 8000/8000 [00:06<00:00, 1153.38it/s]

```
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                   |                 | Robust              |
| Action Selection   | Neural UCB      | Nonlinear           |
|                   |                 | Learning            |
| Parameters         | gamma=0.010, eta=0.050 | EXP3 Config        |
|                   | beta=0.2        | Neural UCB          |
=====
- HYBRID Progress: 100%|██████████| 8000/8000 [00:23<00:00, 340.00it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

```
RESULTS (completed in 86.04s):
=====
Oracle      (step-wise) : 7868.91
GNeuralUCB  (batch)     : 7266.99
EXPUCB      (batch)     : 6307.33
EXPNeuralUCB (batch)    : 7520.98
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 95.6%
```

```
ATTACK PATTERN ANALYSIS:
=====
| Strategy:          | NoAttack        |
| Total Frames:      | 8,000           |
| Total Attacks:     | 0               |
| Attack Intensity:  | 0.000           |
=====
| PATH-SPECIFIC ANALYSIS:          |
|-----|-----|
| Path 0:          | 0 attacks ( 0.0% rate) |
| Path 1:          | 0 attacks ( 0.0% rate) |
| Path 2:          | 0 attacks ( 0.0% rate) |
| Path 3:          | 0 attacks ( 0.0% rate) |
=====
ATTACKS: Mean attacks/frame: 0.000
=====
```

```
EXPERIMENT 3 COMPLETED
Execution Time: 86.0s
Results:
Oracle:      7868.91 (theoretical maximum)
EXPNeuralUCB: 7520.98 ( 4.4% gap) [WINNER]
GNeuralUCB:  7266.99 ( 7.6% gap)
EXPUCB:      6307.33 ( 19.8% gap)
Winner: EXPNeuralUCB
=====
```

QUANTUM EVALUATION COMPLETED!
=====

Total experiments: 3
Total time: 139.7s
Average time per experiment: 46.6s
=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:
 Strategy: NONE
 Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 19.4s)

|

+-- Oracle: 3307.29 (theoretical maximum)

+-- EXPNeuralUCB: 2854.95 (13.7% gap)

+-- GNeuralUCB: 3235.40 (2.2% gap)

+-- EXPUCB: 2794.69 (15.5% gap)

Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 33.6s)

|

+-- Oracle: 5676.68 (theoretical maximum)

+-- EXPNeuralUCB: 5279.10 (7.0% gap)

+-- GNeuralUCB: 5179.98 (8.7% gap)

+-- EXPUCB: 4552.40 (19.8% gap)

Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 86.0s)

|

+-- Oracle: 7868.91 (theoretical maximum)

+-- EXPNeuralUCB: 7520.98 (4.4% gap)

+-- GNeuralUCB: 7266.99 (7.6% gap)

+-- EXPUCB: 6307.33 (19.8% gap)

Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

LEARNING PERFORMANCE (Gap Reduction to Oracle):

 #1: EXPNeuralUCB -> +9.3pp improvement (gap closing)

 Gap: 13.7% -> 4.4%

 #2: EXPUCB -> +4.3pp decline (gap widening)

 Gap: 15.5% -> 19.8%

 #3: GNeuralUCB -> +5.5pp decline (gap widening)

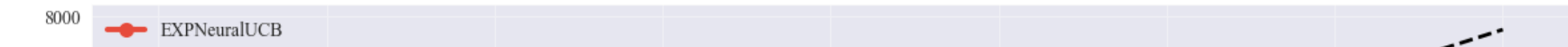
 Gap: 2.2% -> 7.6%

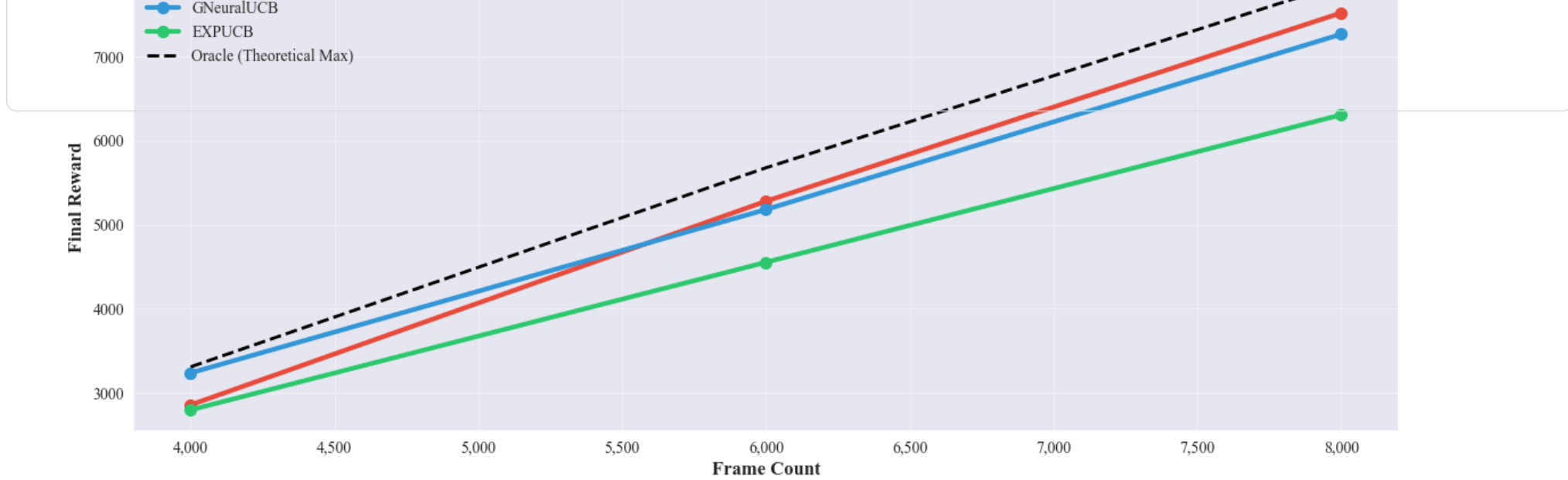
Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

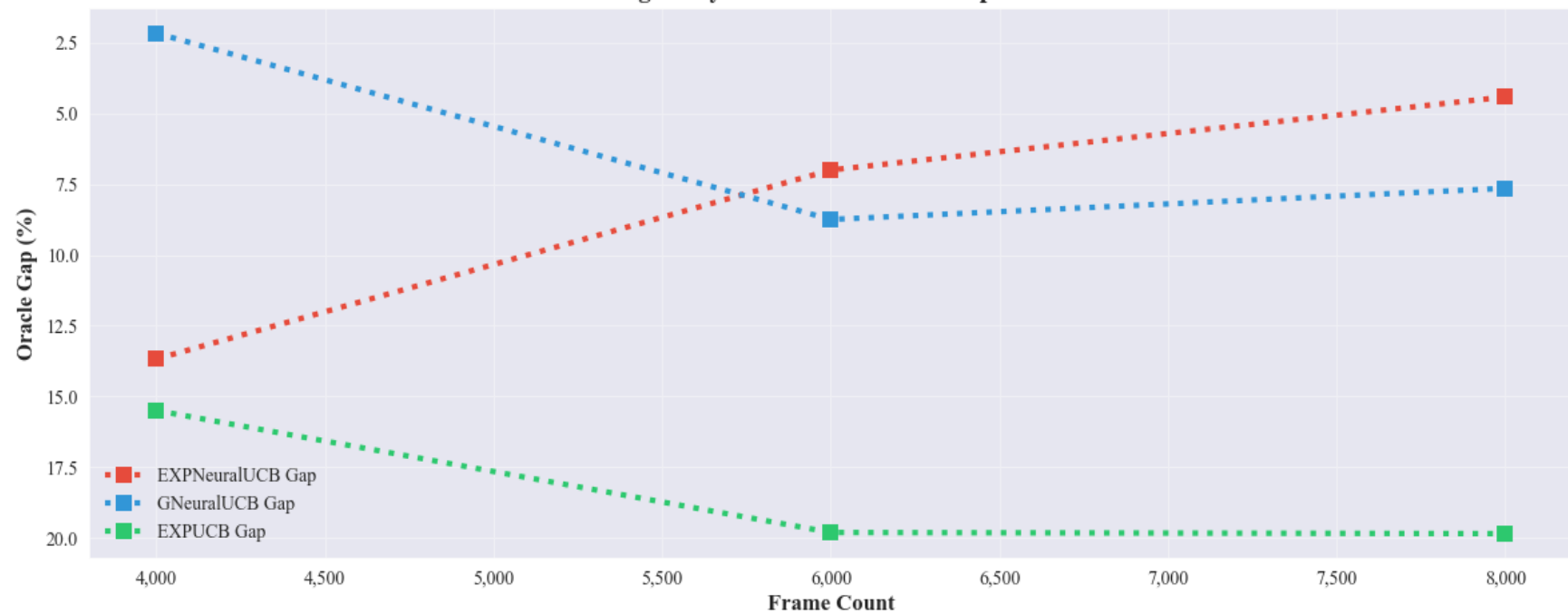
Generating performance visualization...

Algorithm Performance - NONE Attack





Learning Analysis: Oracle-Relative Gap Reduction



Visualization complete!

```
1 # ===== USAGE EXAMPLE =====
2 def run_evaluation(attack_type="markov"):
3     evaluator = MultiRunEvaluator(
4         base_seed=12345,
5         base_frames=4000,
6         frame_step=2000,
7         attack_type=attack_type,      # try: 'none' | 'random' | 'adaptive'
8         attack_intensity=1.0,
9         enable_progress=False,
10    )
11    evaluator.run_experiments(attack_type=attack_type, num_experiments=3) # 4000, 6000, 8000
12    return evaluator
13
14
15 # Run evaluation
16 evaluator = run_evaluation(attack_type="onlineadaptive")
```


=====

QUANTUM MULTIRUN EVALUATOR INITIALIZED

=====

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: ONLINEADAPTIVE
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

=====

=====

STARTING QUANTUM MULTIRUN EVALUATION

=====

Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000

=====

PROGRESS: Experiment 1/3

=====

EXPERIMENT 1

=====

Frame Count: 4,000
Attack: onlineadaptive (intensity=1.0)

EXPERIMENT 1: 4000 frames

WARNING: Unknown attack_type='onlineadaptive', falling back to 'markov'
Available attack types: ['none', 'random', 'markov', 'adaptive', 'online_adaptive']

QUANTUM ENVIRONMENT INITIALIZED:

=====

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	4,000	

=====

ADVERSARIAL CONFIGURATION:

=====

Attack Strategy:	MarkovAttack	
Attack Rate:	1.00	
Simultaneous:	1	

=====

Environment: onlineadaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 4000 steps, final reward: 3198.93
COMPLETED: in 0.00s, final reward: 3198.93
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.8268225267870645), np.float64(0.7476684760175540), np.float64(0.6500252551504420), np.float64(0.5816045635472745)]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%|██████████| 4000/4000 [00:06<00:00, 643.11it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path: 0
Optimal Action: 4
Path Performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

- EXP3 Progress: 100%|██████████| 4000/4000 [00:02<00:00, 1973.91it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

Optimal Path: 0
Optimal Action: 4
Path Performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

```
=====
| beta=0.2 | Neural UCB |
=====
- HYBRID Progress: 100%|██████████| 4000/4000 [00:09<00:00, 430.79it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

RESULTS (completed in 17.68s):

```
=====
Oracle      (step-wise) : 3198.93
GNeuralUCB  (batch)     : 2098.77
EXPUCB      (batch)     : 2038.11
EXPNeuralUCB (batch)    : 2348.03
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 73.4%
```

ATTACK PATTERN ANALYSIS:

```
=====
| Strategy:      | MarkovAttack      |
| Total Frames:  | 4,000             |
| Total Attacks: | 4,000             |
| Attack Intensity: | 0.250           |
=====
```

```
=====
| PATH-SPECIFIC ANALYSIS: |
=====
| Path 0:      | 1369 attacks (34.2% rate) |
| Path 1:      | 681 attacks (17.0% rate) |
| Path 2:      | 1290 attacks (32.2% rate) |
| Path 3:      | 660 attacks (16.5% rate) |
=====
ATTACKS: Mean attacks/frame: 1.000
=====
```

```
=====
EXPERIMENT 1 COMPLETED
Execution Time: 17.7s
Results:
Oracle:      3198.93 (theoretical maximum)
EXPNeuralUCB: 2348.03 ( 26.6% gap) [WINNER]
GNeuralUCB:  2098.77 ( 34.4% gap)
EXPUCB:      2038.11 ( 36.3% gap)
Winner: EXPNeuralUCB
=====
```

PROGRESS: Experiment 2/3

```
=====
EXPERIMENT 2
=====
Frame Count: 6,000
Attack: onlineadaptive (intensity=1.0)
=====
```

```
=====
EXPERIMENT 2: 6000 frames
WARNING: Unknown attack_type='onlineadaptive', falling back to 'markov'
Available attack types: ['none', 'random', 'markov', 'adaptive', 'online_adaptive']
=====
```

```
=====
QUANTUM ENVIRONMENT INITIALIZED:
=====
| Network Paths:  | 4 |
| Qubit Config:   | (8, 10, 8, 9) |
| Frame Length:   | 6,000 |
=====
```

```
=====
ADVERSARIAL CONFIGURATION:
=====
| Attack Strategy: | MarkovAttack |
=====
```

```
| Attack Rate: | 1.00 |
| Simultaneous: | 1 |
=====
Environment: onlineadaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
  Context strategy: context-free
  take_action params: 0, update params: 3
  COMPLETED: 6000 steps, final reward: 5589.77
  COMPLETED: in 0.00s, final reward: 5589.77
  COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | Simple UCB | NOT Adversarial |
| Action Selection | Neural UCB | Robust |
| | | Nonlinear |
| | | Learning |
| Parameters | beta=0.2 | UCB Confidence |
=====
```

```
- NEURAL Progress: 100%|██████████| 6000/6000 [00:11<00:00, 517.10it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | EXP3 | Adversarially |
| Action Selection | Linear UCB | Robust |
| | | Linear |
| | | Learning |
| Parameters | gamma=0.100, eta=0.005 | EXP3 Config |
| | beta=0.2 | Linear UCB |
=====
```

```
- EXP3 Progress: 100%|██████████| 6000/6000 [00:04<00:00, 1414.18it/s]
  COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
Optimal Path: 0
Optimal Action: 4
Path Performance: [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

=====

- HYBRID Progress: 100%|██████████| 6000/6000 [00:14<00:00, 402.48it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 30.92s):

=====

Oracle (step-wise) : 5589.77
GNeuralUCB (batch) : 4308.49
EXPUCB (batch) : 3537.67
EXPNeuralUCB (batch) : 4391.79
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 78.6%

ATTACK PATTERN ANALYSIS:

=====

Strategy:	MarkovAttack
Total Frames:	6,000
Total Attacks:	6,000
Attack Intensity:	0.250

=====

PATH-SPECIFIC ANALYSIS:

=====

Path 0:	2012 attacks (33.5% rate)
Path 1:	983 attacks (16.4% rate)
Path 2:	1989 attacks (33.1% rate)
Path 3:	1016 attacks (16.9% rate)

=====

ATTACKS: Mean attacks/frame: 1.000

=====

EXPERIMENT 2 COMPLETED

Execution Time: 30.9s

Results:

Oracle: 5589.77 (theoretical maximum)
EXPNeuralUCB: 4391.79 (21.4% gap) [WINNER]
GNeuralUCB: 4308.49 (22.9% gap)
EXPUCB: 3537.67 (36.7% gap)
Winner: EXPNeuralUCB

=====

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: onlineadaptive (intensity=1.0)

EXPERIMENT 3: 8000 frames
WARNING: Unknown attack_type='onlineadaptive', falling back to 'markov'
Available attack types: ['none', 'random', 'markov', 'adaptive', 'online_adaptive']

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	8,000

ADVERSARIAL CONFIGURATION:

Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	1

Environment: onlineadaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 8000 steps, final reward: 7815.67
COMPLETED: in 0.00s, final reward: 7815.67
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%|██████████| 8000/8000 [00:50<00:00, 158.85it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

- EXP3 Progress: 100% ██████████ | 8000/8000 [00:06<00:00, 1196.31it/s]
 COMPLETED: EXPUCB (batch) completed successfully
 Running EXPNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

- HYBRID Progress: 100% ██████████ | 8000/8000 [00:21<00:00, 367.44it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 79.04s):

Oracle (step-wise)	: 7815.67
GNeuralUCB (batch)	: 6022.64
EXPUCB (batch)	: 5013.47
EXPNeuralUCB (batch)	: 6328.40
WINNER:	EXPNeuralUCB
EFFICIENCY:	EXPNeuralUCB Oracle Efficiency: 81.0%

ATTACK PATTERN ANALYSIS:

Strategy:	MarkovAttack
Total Frames:	8,000
Total Attacks:	8,000
Attack Intensity:	0.250

PATH-SPECIFIC ANALYSIS:	
Path 0:	2675 attacks (33.4% rate)
Path 1:	1265 attacks (15.8% rate)
Path 2:	2712 attacks (33.9% rate)
Path 3:	1348 attacks (16.9% rate)

ATTACKS: Mean attacks/frame: 1.000

EXPERIMENT 3 COMPLETED

Execution Time: 79.0s

Results:

Oracle: 7815.67 (theoretical maximum)
EXPNeuralUCB: 6328.40 (19.0% gap) [WINNER]
GNeuralUCB: 6022.64 (22.9% gap)
EXPUCB: 5013.47 (35.9% gap)
Winner: EXPNeuralUCB

QUANTUM EVALUATION COMPLETED!

Total experiments: 3
Total time: 128.5s
Average time per experiment: 42.8s

DETAILED EXPERIMENT ANALYSIS

Attack Configuration:

Strategy: ONLINEADAPTIVE
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 17.7s)

|
+-- Oracle: 3198.93 (theoretical maximum)
+-- EXPNeuralUCB: 2348.03 (26.6% gap)
+-- GNeuralUCB: 2098.77 (34.4% gap)
+-- EXPUCB: 2038.11 (36.3% gap)
Winner: EXPNeuralUCB

EXPERIMENT 2 (6,000 frames, 30.9s)

|
+-- Oracle: 5589.77 (theoretical maximum)
+-- EXPNeuralUCB: 4391.79 (21.4% gap)
+-- GNeuralUCB: 4308.49 (22.9% gap)
+-- EXPUCB: 3537.67 (36.7% gap)
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 79.0s)

|
+-- Oracle: 7815.67 (theoretical maximum)
+-- EXPNeuralUCB: 6328.40 (19.0% gap)
+-- GNeuralUCB: 6022.64 (22.9% gap)
+-- EXPUCB: 5013.47 (35.9% gap)
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

LEARNING PERFORMANCE (Gap Reduction to Oracle):

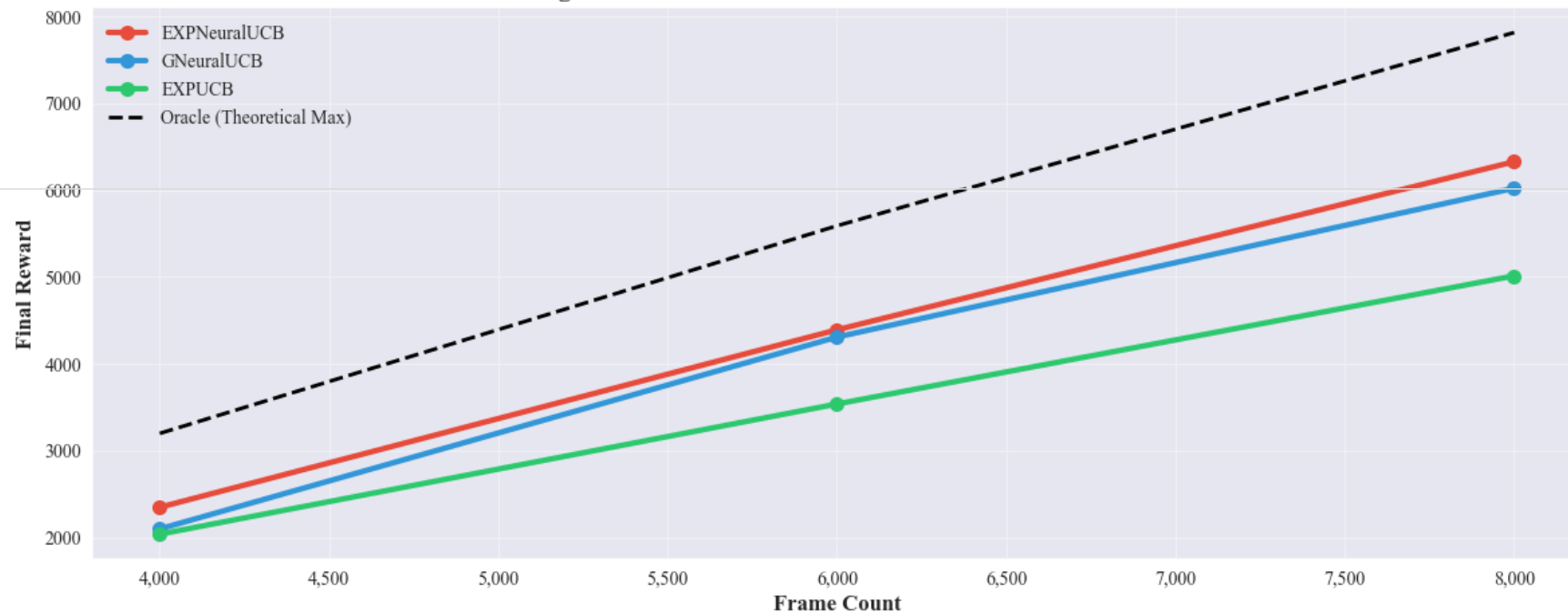
#1: GNeuralUCB -> +11.4pp improvement (gap closing)
Gap: 34.4% -> 22.9%
#2: EXPNeuralUCB -> +7.6pp improvement (gap closing)
Gap: 26.6% -> 19.0%
#3: EXPUCB -> +0.4pp improvement (gap closing)
Gap: 36.3% -> 35.9%

Interpretation:

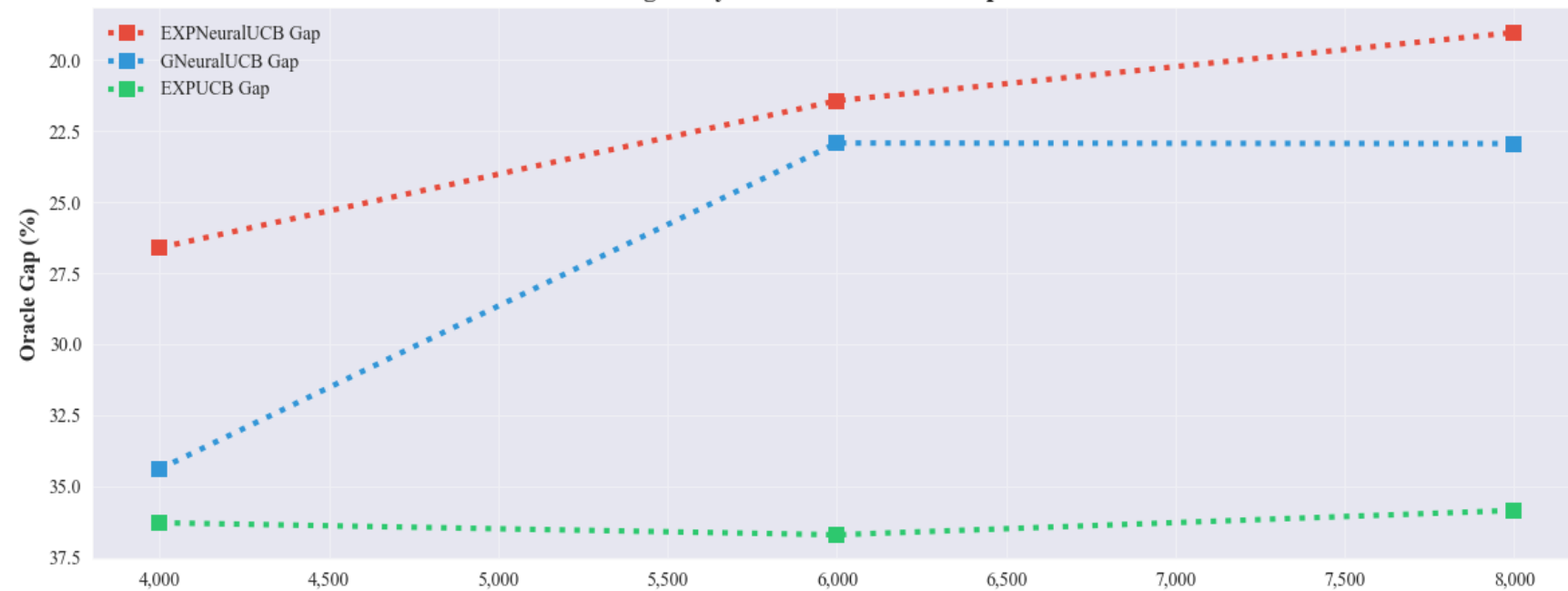
- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

Generating performance visualization...

Algorithm Performance - ONLINEADAPTIVE Attack



Learning Analysis: Oracle-Relative Gap Reduction



```

1 # quantum_visualizer.py
2
3 import matplotlib.pyplot as plt
4 import pandas as pd
5 import numpy as np
6 from quantum_framework import MultiRunEvaluator
7
8 # Use your existing function exactly as written
9 def run_evaluation(attack_type="markov"):
10     evaluator = MultiRunEvaluator(
11         base_seed=12345,
12         base_frames=4000,
13         frame_step=2000,
14         attack_type=attack_type,
15         attack_intensity=1.0,
16         enable_progress=False,
17     )
18     evaluator.run_experiments(attack_type=attack_type, num_experiments=3)
19     return evaluator
20
21 class QuantumVisualizer:
22     """Simple visualization using existing MultiRunEvaluator and env_experiments data"""
23
24     def __init__(self):
25         self.evaluators = {}
26         self.colors = {
27             'Oracle': '#2c3e50',
28             'EXPNeuralUCB': '#e74c3c',
29             'GNeuralUCB': '#3498db',
30             'EXPUCB': '#2ecc71'
31         }
32         self.env_colors = {
33             'none': '#95a5a6',
34             'random': '#f39c12',
35             'markov': '#9b59b6',
36             'adaptive': '#e67e22'
37         }
38
39     def run_test(self, attack_type="adaptive"):
40         print(f"Running test for {attack_type} environment...")
41         evaluator = run_evaluation(attack_type=attack_type)
42         self.evaluators[attack_type] = evaluator
43         return evaluator
44
45     def _get_env_data(self, attack_type):
46         if attack_type not in self.evaluators:
47             raise KeyError(f"No evaluator stored for attack_type='{attack_type}'. Run run_test() first.")
48
49         evaluator = self.evaluators[attack_type]
50         env_data = getattr(evaluator, "env_experiments", {}).get(attack_type)
51
52         if not env_data:
53             env_data = getattr(evaluator, "experiments", None)
54
55         if not isinstance(env_data, dict) or not env_data:
56             raise ValueError(f"No experiment data found for attack_type='{attack_type}'.")
57
58         if all(isinstance(k, str) for k in env_data.keys()):
59             exp_id_guess = env_data.get("id", 1)
60             env_data = {int(exp_id_guess): env_data}
61
62         return env_data
63
64     def plot_results(self, attack_type):
65         if attack_type not in self.evaluators:

```

```

66         print(f"No results for {attack_type}")
67         return
68
69     try:
70         env_data = self._get_env_data(attack_type)
71     except Exception as e:
72         print(f"Failed to extract data for '{attack_type}': {e}")
73         return
74
75     frame_counts = []
76     oracle_rewards = []
77     alg_rewards = {}
78     alg_gaps = {}
79
80     first_exp = next(iter(env_data.values()))
81     algorithms = [alg for alg in first_exp['results'].keys() if alg != 'Oracle']
82
83     for alg in algorithms:
84         alg_rewards[alg] = []
85         alg_gaps[alg] = []
86
87     for exp_id in sorted(env_data.keys()):
88         exp_data = env_data[exp_id]
89
90         frame_counts.append(exp_data['frame_count'])
91         oracle_rewards.append(exp_data['results']['Oracle']['final_reward'])
92
93         for alg in algorithms:
94             alg_rewards[alg].append(exp_data['results'][alg]['final_reward'])
95             alg_gaps[alg].append(exp_data['gaps'][alg])
96
97     # Create plots
98     fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))
99     fig.suptitle(f'{attack_type.capitalize()} Environment Results')
100
101     # Plot 1: Rewards
102     ax1.plot(frame_counts, oracle_rewards, '--k', label='Oracle', linewidth=2, marker='x', markersize=7)
103     for alg in algorithms:
104         ax1.plot(frame_counts, alg_rewards[alg], 'o-', label=alg, linewidth=2, markersize=6,
105                 color=self.colors.get(alg, 'gray'))
106
107     ax1.set_xlabel('Frame Count')
108     ax1.set_ylabel('Final Reward')
109     ax1.set_title('Reward Scaling')
110     ax1.legend()
111     ax1.grid(True, alpha=0.3)
112
113     # Plot 2: Gaps
114     for alg in algorithms:
115         ax2.plot(frame_counts, alg_gaps[alg], 's-', label=alg, linewidth=2, markersize=6,
116                 color=self.colors.get(alg, 'gray'))
117
118     ax2.set_xlabel('Frame Count')
119     ax2.set_ylabel('Oracle Gap (%)')
120     ax2.set_title('Oracle Gap (Lower = Better)')
121     ax2.legend()
122     ax2.grid(True, alpha=0.3)
123     ax2.invert_yaxis()
124
125     plt.tight_layout()
126     plt.savefig(f'{attack_type}_results.png', dpi=300, bbox_inches='tight')
127     plt.show()
128
129 def compare_environments(self, attack_types, main_algorithm="EXPNeuralUCB"):
130     fig, axes = plt.subplots(2, 2, figsize=(15, 10))

```

```

fig.suptitle('ENVIRONMENT COMPARISON', fontsize=16, fontweight='bold')
# 1. Main algorithm across environments
for env in attack_types:
    if env not in self.evaluators:
        continue
    try:
        env_data = self._get_env_data(env)
    except Exception:
        continue

    frame_counts = []
    alg_rewards = []
    alg_gaps = []

    for exp_id in sorted(env_data.keys()):
        exp_data = env_data[exp_id]
        frame_counts.append(exp_data['frame_count'])
        if main_algorithm in exp_data['results']:
            alg_rewards.append(exp_data['results'][main_algorithm]['final_reward'])
            alg_gaps.append(exp_data['gaps'][main_algorithm])

    if alg_rewards:
        color = self.env_colors.get(env, 'gray')
        axes[0,0].plot(frame_counts, alg_rewards, 'o-', label=env, linewidth=2, markersize=6, color=color)
        axes[0,1].plot(frame_counts, alg_gaps, 's-', label=env, linewidth=2, markersize=6, color=color)

    axes[0,0].set_title(f'{main_algorithm} Reward Performance')
    axes[0,0].set_xlabel('Frame Count')
    axes[0,0].set_ylabel('Final Reward')
    axes[0,0].legend()
    axes[0,0].grid(True, alpha=0.3)

    axes[0,1].set_title(f'{main_algorithm} Oracle Gap')
    axes[0,1].set_xlabel('Frame Count')
    axes[0,1].set_ylabel('Oracle Gap % (Lower = Better)')
    axes[0,1].legend()
    axes[0,1].grid(True, alpha=0.3)
    axes[0,1].invert_yaxis()

# 2. Use stored winners
env_winners = {}
all_env_data = []

for env in attack_types:
    if env not in self.evaluators:
        continue
    try:
        env_data = self._get_env_data(env)
    except Exception:
        continue

    highest_frame_exp = max(env_data.keys())
    stored_winner = env_data[highest_frame_exp]['winner']
    stored_gap = env_data[highest_frame_exp]['gaps'][stored_winner]

    env_winners[env] = (stored_winner, stored_gap)

    for exp_id in sorted(env_data.keys()):
        exp_data = env_data[exp_id]
        for alg, gap in exp_data['gaps'].items():
            all_env_data.append({'environment': env, 'algorithm': alg, 'avg_gap': gap})

# 3. Plot winners
if env_winners:

```

```

196     envs = list(env_winners.keys())
197     winners = [env_winners[e][0] for e in envs]
198     gaps = [env_winners[e][1] for e in envs]
199     colors = [self.env_colors.get(e, 'gray') for e in envs]
200     axes[1,0].bar(envs, gaps, color=colors, alpha=0.8)
201     axes[1,0].set_title('Best Algorithm per Environment')
202     axes[1,0].set_ylabel('Best Oracle Gap %')
203     axes[1,0].tick_params(axis='x', rotation=45)
204     for i, (env, winner, gap) in enumerate(zip(envs, winners, gaps)):
205         axes[1,0].text(i, gap + 1, winner, ha='center', va='bottom', fontweight='bold', rotation=90)
206
207     # 4. Environment difficulty
208     if all_env_data:
209         env_df = pd.DataFrame(all_env_data)
210         env_difficulty = env_df.groupby('environment')['avg_gap'].mean().sort_values()
211         colors = [self.env_colors.get(e, 'gray') for e in env_difficulty.index]
212         axes[1,1].bar(range(len(env_difficulty)), env_difficulty.values, color=colors, alpha=0.8)
213         axes[1,1].set_xticks(range(len(env_difficulty)))
214         axes[1,1].set_xticklabels(env_difficulty.index, rotation=45)
215         axes[1,1].set_title('Environment Difficulty Ranking')
216         axes[1,1].set_ylabel('Average Oracle Gap %')
217
218     plt.tight_layout()
219     plt.savefig('environment_comparison.png', dpi=300, bbox_inches='tight')
220     plt.show()
221
222     # Simple usage examples
223     def test_single_environment():
224         viz = QuantumVisualizer()
225         attack_type = "adaptive"
226         viz.run_test(attack_type)
227         viz.plot_results(attack_type)
228         return viz
229
230     def test_multiple_environments():
231         viz = QuantumVisualizer()
232         environments = ["none", "random", "markov", "adaptive"]
233
234         for env in environments:
235             print(f"\n{'='*50}")
236             print(f"TESTING {env.upper()}")
237             print(f"{'='*50}")
238             viz.run_test(env)
239             viz.plot_results(env)
240
241         viz.compare_environments(environments)
242         return viz
243
244     if __name__ == "__main__":
245         print("\n" + "="*70)
246         print("TESTING MULTIPLE ENVIRONMENTS:")
247         viz2 = test_multiple_environments()

```


=====

TESTING MULTIPLE ENVIRONMENTS:

=====

TESTING NONE

=====

Running test for none environment...

=====

QUANTUM MULTIRUN EVALUATOR INITIALIZED

=====

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: NONE
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

=====

=====

STARTING QUANTUM MULTIRUN EVALUATION

=====

Planning 3 experiments

Frame counts: 4,000, 6,000, 8,000

=====

PROGRESS: Experiment 1/3

=====

EXPERIMENT 1

=====

Frame Count: 4,000

Attack: none (intensity=1.0)

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	4,000	

=====

ADVERSARIAL CONFIGURATION:

=====

Attack Strategy:	NoAttack	
------------------	----------	--

=====

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3307.29

COMPLETED: in 0.00s, final reward: 3307.29

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
ORACLE ANALYSIS:
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
```

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
| Action Selection   | Neural UCB      | Robust              |
|                    |                  | Nonlinear            |
|                    |                  | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====
```

```
- NEURAL Progress: 100%|██████████| 4000/4000 [00:20<00:00, 191.24it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

```
ORACLE ANALYSIS:
```

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
```

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
| Action Selection   | Linear UCB      | Robust              |
|                    |                  | Linear              |
|                    |                  | Learning             |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config        |
|                    | beta=0.2        | Linear UCB          |
=====
```

```
- EXP3 Progress: 100%|██████████| 4000/4000 [00:05<00:00, 737.95it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

```
ORACLE ANALYSIS:
```

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
```

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                    |                  | Robust              |
|                    |                  | Linear              |
|                    |                  | Learning             |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config        |
|                    | beta=0.2        | Linear UCB          |
=====
```

Action Selection	Neural UCB	Robust
		Nonlinear Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB

=====

- HYBRID Progress: 100%[██████████] 4000/4000 [00:24<00:00, 162.89it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 51.95s):

=====

Oracle	(step-wise)	: 3307.29
GNeuralUCB	(batch)	: 3235.40
EXPUCB	(batch)	: 2794.69
EXPNeuralUCB	(batch)	: 2854.95

WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 86.3%

ATTACK PATTERN ANALYSIS:

=====

Strategy:	NoAttack	
Total Frames:	4,000	
Total Attacks:	0	
Attack Intensity:	0.000	

| PATH-SPECIFIC ANALYSIS: |

Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)

=====

ATTACKS: Mean attacks/frame: 0.000

=====

EXPERIMENT 1 COMPLETED

Execution Time: 52.0s

Results:

Oracle:	3307.29 (theoretical maximum)
EXPNeuralUCB:	2854.95 (13.7% gap)
GNeuralUCB:	3235.40 (2.2% gap) [WINNER]
EXPUCB:	2794.69 (15.5% gap)

Winner: GNeuralUCB

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000

Attack: none (intensity=1.0)

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000

=====

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | NoAttack |
=====
Environment: none (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
  Context strategy: context-free
  take_action params: 0, update params: 3
  COMPLETED: 6000 steps, final reward: 5676.68
  COMPLETED: in 0.01s, final reward: 5676.68
  COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | Simple UCB | NOT Adversarial |
| Action Selection | Neural UCB | Robust |
| | | Nonlinear |
| | | Learning |
| Parameters | beta=0.2 | UCB Confidence |
=====
- NEURAL Progress: 100%|██████████| 6000/6000 [00:20<00:00, 293.05it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

```
ORACLE ANALYSIS:
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | EXP3 | Adversarially |
| Action Selection | Linear UCB | Robust |
| | | Linear |
| | | Learning |
| Parameters | gamma=0.100, eta=0.005 | EXP3 Config |
| | beta=0.2 | Linear UCB |
=====
- EXP3 Progress: 100%|██████████| 6000/6000 [00:06<00:00, 995.13it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

```
ORACLE ANALYSIS:
=====
```

```
Optimal Path: 0
Optimal Action: 4
Path Performance: [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

=====

- HYBRID Progress: 100%[██████████] 6000/6000 [00:27<00:00, 214.71it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 54.64s):

=====

Oracle (step-wise) : 5676.68
GNeuralUCB (batch) : 5179.98
EXPUCB (batch) : 4552.40
EXPNeuralUCB (batch) : 5279.10
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 93.0%

ATTACK PATTERN ANALYSIS:

=====

Strategy:	NoAttack	
Total Frames:	6,000	
Total Attacks:	0	
Attack Intensity:	0.000	

=====

PATH-SPECIFIC ANALYSIS:

=====

Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)

=====

ATTACKS: Mean attacks/frame: 0.000

=====

EXPERIMENT 2 COMPLETED

Execution Time: 54.6s

Results:

Oracle: 5676.68 (theoretical maximum)
EXPNeuralUCB: 5279.10 (7.0% gap) [WINNER]
GNeuralUCB: 5179.98 (8.7% gap)
EXPUCB: 4552.40 (19.8% gap)
Winner: EXPNeuralUCB

=====

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: none (intensity=1.0)
=====

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	8,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	NoAttack	
------------------	----------	--

Environment: none (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7868.91

COMPLETED: in 0.01s, final reward: 7868.91

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]	

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%|██████████| 8000/8000 [01:23<00:00, 95.95it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]	

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

Component	Method	Properties

Group Selection	EXP3	Adversarially
Action Selection	Linear UCB	Robust
Parameters	gamma=0.100, eta=0.005	Linear
	beta=0.2	Learning
		EXP3 Config
		Linear UCB

=====

- EXP3 Progress: 100%|██████████| 8000/8000 [00:11<00:00, 727.14it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially
Action Selection	Neural UCB	Robust
Parameters	gamma=0.010, eta=0.050	Nonlinear
	beta=0.2	Learning
		EXP3 Config
		Neural UCB

=====

- HYBRID Progress: 100%|██████████| 8000/8000 [00:38<00:00, 205.60it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 133.41s):

=====

Oracle (step-wise) : 7868.91
GNeuralUCB (batch) : 7266.99
EXPUCB (batch) : 6307.33
EXPNeuralUCB (batch) : 7520.98
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 95.6%

ATTACK PATTERN ANALYSIS:

Strategy:	NoAttack
Total Frames:	8,000
Total Attacks:	0
Attack Intensity:	0.000

PATH-SPECIFIC ANALYSIS:

Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)

=====

ATTACKS: Mean attacks/frame: 0.000

=====

EXPERIMENT 3 COMPLETED

Execution Time: 133.4s

Results:

Oracle: 7868.91 (theoretical maximum)
EXPNeuralUCB: 7520.98 (4.4% gap) [WINNER]
GNeuralUCB: 7266.99 (7.6% gap)
EXPUCB: 6307.33 (19.8% gap)
Winner: EXPNeuralUCB

=====

QUANTUM EVALUATION COMPLETED!

=====

Total experiments: 3
Total time: 241.1s
Average time per experiment: 80.4s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:
Strategy: NONE
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 52.0s)

|
+-- Oracle: 3307.29 (theoretical maximum)
+-- EXPNeuralUCB: 2854.95 (13.7% gap)
+-- GNeuralUCB: 3235.40 (2.2% gap)
+-- EXPUCB: 2794.69 (15.5% gap)
Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 54.6s)

|
+-- Oracle: 5676.68 (theoretical maximum)
+-- EXPNeuralUCB: 5279.10 (7.0% gap)
+-- GNeuralUCB: 5179.98 (8.7% gap)
+-- EXPUCB: 4552.40 (19.8% gap)
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 133.4s)

|
+-- Oracle: 7868.91 (theoretical maximum)
+-- EXPNeuralUCB: 7520.98 (4.4% gap)
+-- GNeuralUCB: 7266.99 (7.6% gap)
+-- EXPUCB: 6307.33 (19.8% gap)
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

LEARNING PERFORMANCE (Gap Reduction to Oracle):

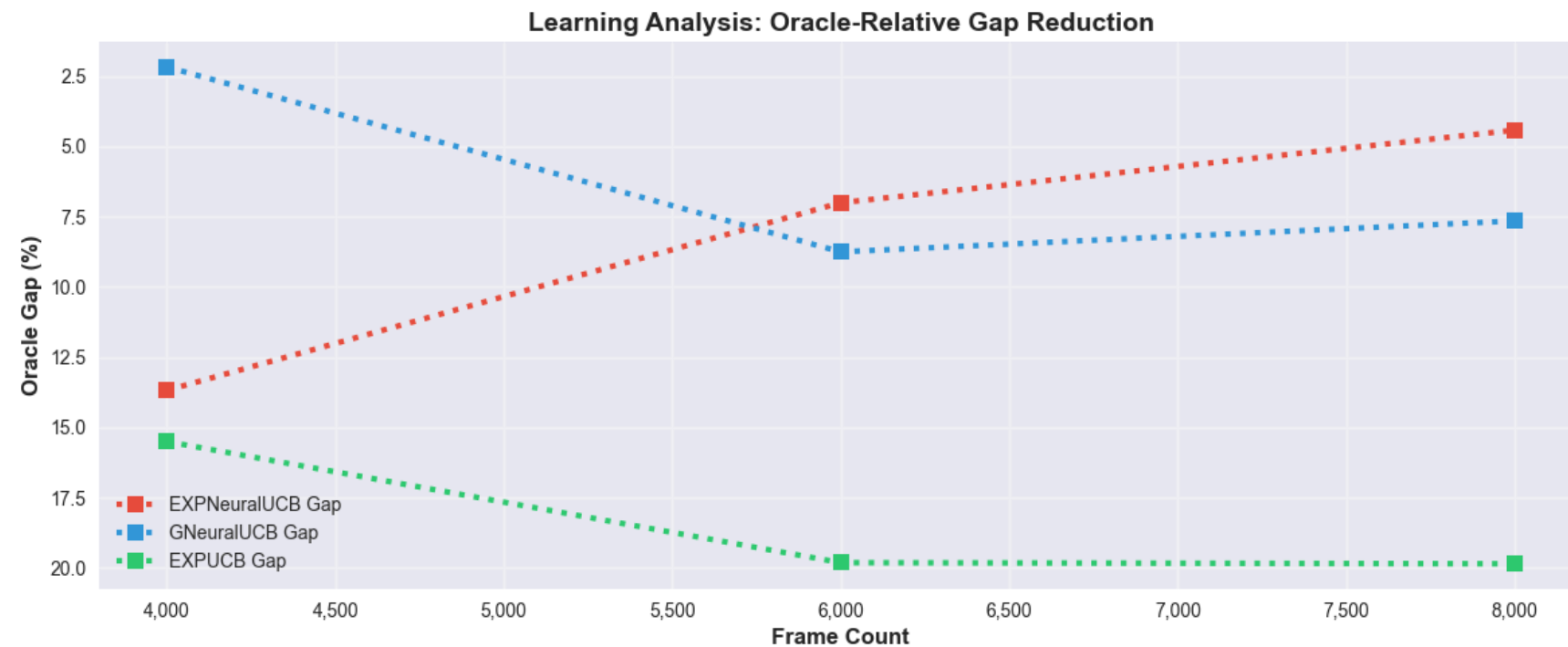
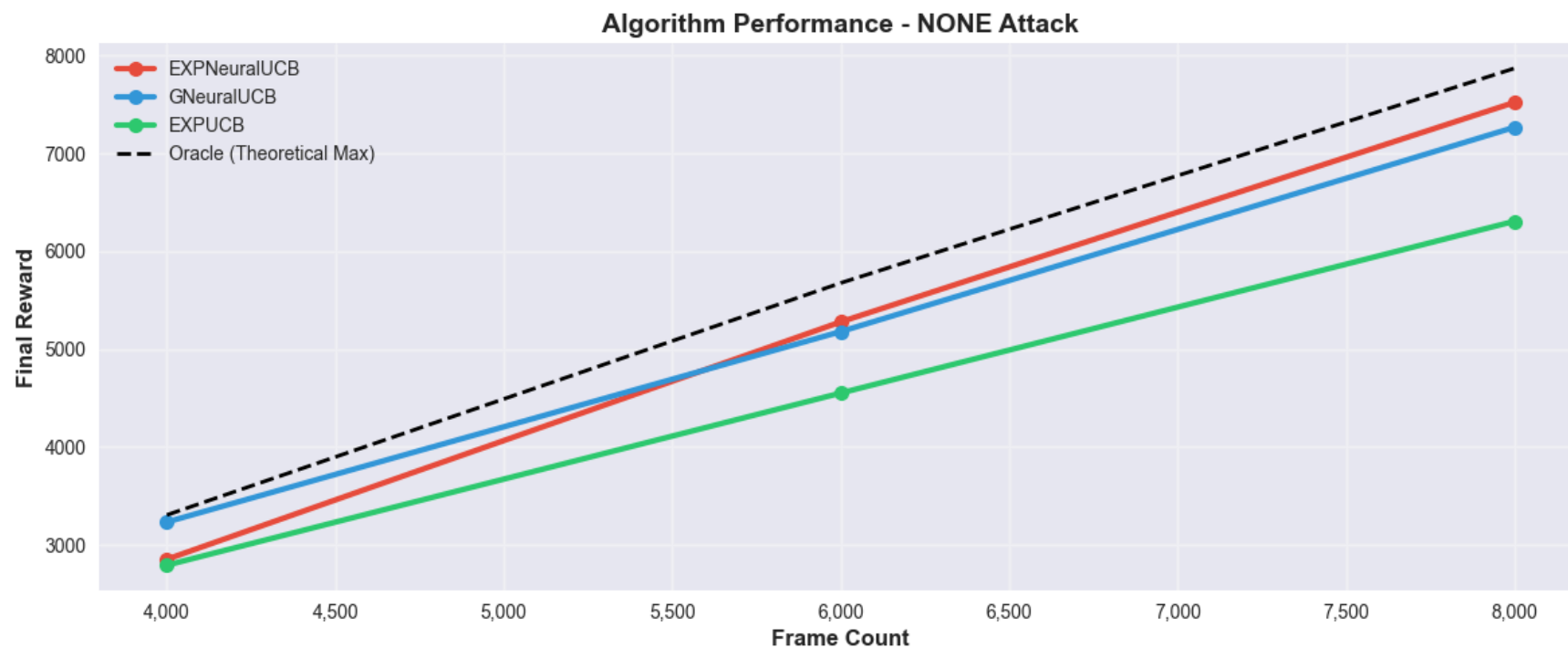
#1: EXPNeuralUCB -> +9.3pp improvement (gap closing)
Gap: 13.7% -> 4.4%
#2: EXPUCB -> +4.3pp decline (gap widening)
Gap: 15.5% -> 19.8%
#3: GNeuralUCB -> +5.5pp decline (gap widening)
Gap: 2.2% -> 7.6%

Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving

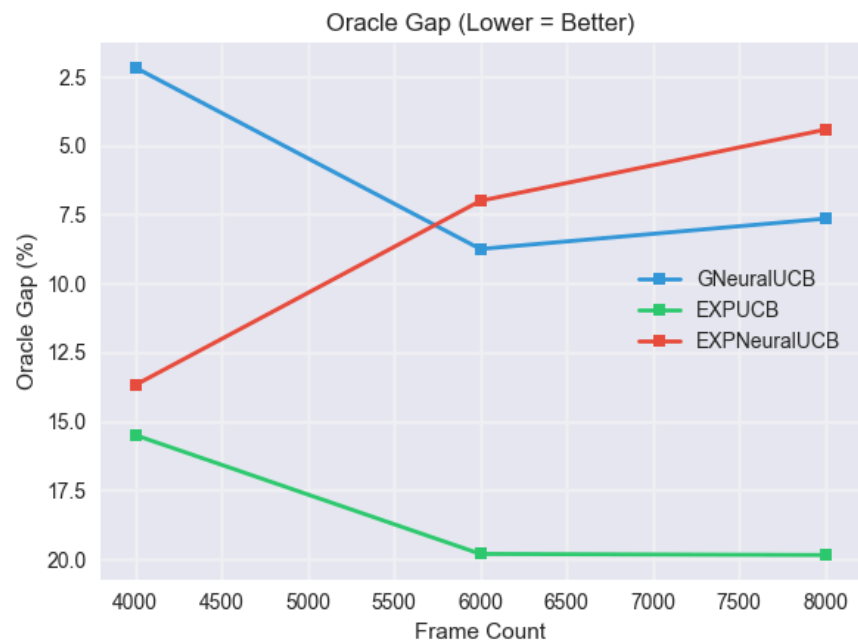
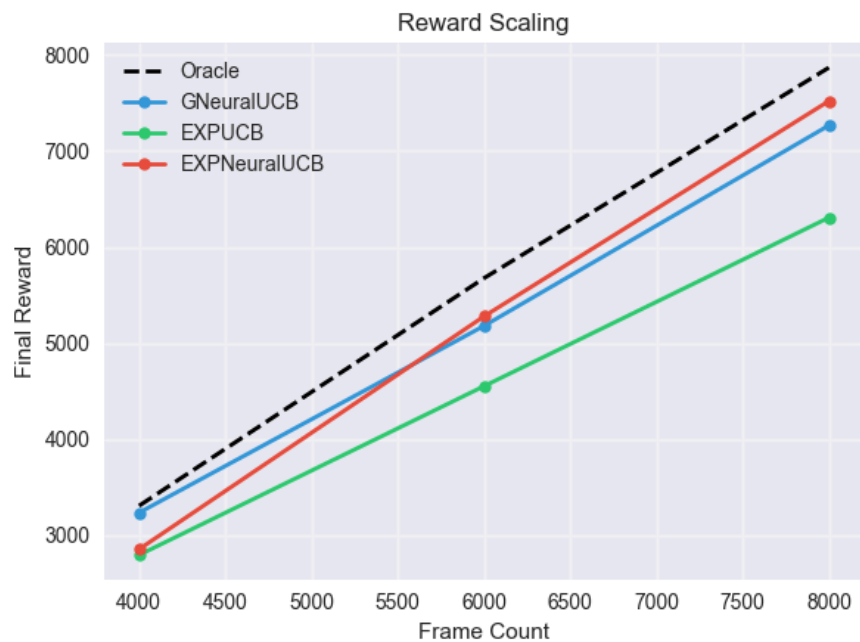
- Best learner closes the gap to Oracle over time

Generating performance visualization...



Visualization complete!

None Environment Results



```

=====
TESTING RANDOM
=====
Running test for random environment...

=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
Configuration:
- Base frames:      4,000
- Frame step:       2,000
- Base seed:        12345
- Attack strategy:  RANDOM
- Attack intensity: 1.0
- Progress bars:    Disabled
- Analysis method:  Oracle-relative gap analysis
=====

```

```

=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000
=====

```

PROGRESS: Experiment 1/3

EXPERIMENT 1

Frame Count: 4,000
Attack: random (intensity=1.0)

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:   | 4 |
| Qubit Config:   | (8, 10, 8, 9) |
| Frame Length:   | 4,000 |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack |
| Attack Rate:    | 0.25 |
=====
```

Environment: random (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3192.62

COMPLETED: in 0.00s, final reward: 3192.62

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0 |
| Optimal Action:  | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method                | Properties          |
|-----|-----|
| Group Selection    | Simple UCB            | NOT Adversarial    |
|                   |                       | Robust              |
| Action Selection   | Neural UCB            | Nonlinear           |
|                   |                       | Learning             |
| Parameters         | beta=0.2              | UCB Confidence      |
=====
```

- NEURAL Progress: 100%[██████████] 4000/4000 [00:09<00:00, 401.24it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0 |
| Optimal Action:  | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method                | Properties          |
|-----|-----|
| Group Selection    | EXP3                  | Adversarially      |
|                   |                       | Robust              |
| Action Selection   | Neural UCB            | Nonlinear           |
|                   |                       | Learning             |
| Parameters         | beta=0.2              | UCB Confidence      |
=====
```

Action Selection	Linear UCB	Linear	
Parameters	gamma=0.100, eta=0.005	EXP3 Config	
	beta=0.2	Linear UCB	

=====

- EXP3 Progress: 100%|██████████| 4000/4000 [00:02<00:00, 1545.75it/s]
 COMPLETED: EXPUCB (batch) completed successfully
 Running EXPNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]	

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB

=====

- HYBRID Progress: 100%|██████████| 4000/4000 [00:13<00:00, 294.83it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 26.19s):

=====

Oracle (step-wise) : 3192.62
 GNeuralUCB (batch) : 2420.84
 EXPUCB (batch) : 2024.79
 EXPNeuralUCB (batch) : 2097.30
 WINNER: GNeuralUCB
 EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 65.7%

ATTACK PATTERN ANALYSIS:

Strategy:	RandomAttack	
Total Frames:	4,000	
Total Attacks:	4,024	
Attack Intensity:	0.252	

PATH-SPECIFIC ANALYSIS:

Path 0:	986 attacks (24.6% rate)
Path 1:	1041 attacks (26.0% rate)
Path 2:	1003 attacks (25.1% rate)
Path 3:	994 attacks (24.9% rate)

=====

ATTACKS: Mean attacks/frame: 1.006

=====

EXPERIMENT 1 COMPLETED

Execution Time: 26.2s

Results:

Oracle: 3192.62 (theoretical maximum)
 EXPNeuralUCB: 2097.30 (34.3% gap)

GNeuralUCB: 2420.84 (24.2% gap) [WINNER]
EXPUCB: 2024.79 (36.6% gap)
Winner: GNeuralUCB

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000
Attack: random (intensity=1.0)

=====

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	6,000	

=====

ADVERSARIAL CONFIGURATION:

=====

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

=====

Environment: random (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 6000 steps, final reward: 5567.05
COMPLETED: in 0.01s, final reward: 5567.05
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]	

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

=====

- NEURAL Progress: 100%[██████████] 6000/6000 [00:15<00:00, 379.58it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

=====

```
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

=====

- EXP3 Progress: 100%|██████████| 6000/6000 [00:05<00:00, 1039.33it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

```
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

=====

- HYBRID Progress: 100%|██████████| 6000/6000 [00:21<00:00, 278.63it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 43.19s):

=====

```
Oracle      (step-wise) : 5567.05
GNeuralUCB  (batch)      : 3840.18
EXPUCB      (batch)      : 3310.90
EXPNeuralUCB (batch)     : 3910.27
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 70.2%
```

ATTACK PATTERN ANALYSIS:

=====

Strategy:	RandomAttack	
Total Frames:	6,000	
Total Attacks:	6,001	
Attack Intensity:	0.250	

=====

PATH-SPECIFIC ANALYSIS:

=====

```
=====
| Path 0:          | 1490 attacks (24.8% rate) |
| Path 1:          | 1511 attacks (25.2% rate) |
| Path 2:          | 1476 attacks (24.6% rate) |
| Path 3:          | 1524 attacks (25.4% rate) |
=====
```

ATTACKS: Mean attacks/frame: 1.000

EXPERIMENT 2 COMPLETED

Execution Time: 43.2s

Results:

```
Oracle:          5567.05 (theoretical maximum)
EXPNeuralUCB:    3910.27 ( 29.8% gap) [WINNER]
GNeuralUCB:      3840.18 ( 31.0% gap)
EXPUCB:          3310.90 ( 40.5% gap)
Winner: EXPNeuralUCB
```

PROGRESS: Experiment 3/3

EXPERIMENT 3

Frame Count: 8,000

Attack: random (intensity=1.0)

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths:   | 4 |
| Qubit Config:    | (8, 10, 8, 9) |
| Frame Length:    | 8,000 |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | RandomAttack |
| Attack Rate:     | 0.25 |
=====
```

Environment: random (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7777.26

COMPLETED: in 0.02s, final reward: 7777.26

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0 |
| Optimal Action:  | 4 |
| Path Performance: | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
Action Selection	Neural UCB	Robust
Parameters	beta=0.2	Nonlinear Learning
		UCB Confidence

=====

- NEURAL Progress: 100%|██████████| 8000/8000 [01:12<00:00, 110.92it/s]
 COMPLETED: GNeuralUCB (batch) completed successfully
 Running EXPUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

=====

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially
Action Selection	Linear UCB	Robust
Parameters	gamma=0.100, eta=0.005	Linear Learning
	beta=0.2	EXP3 Config
		Linear UCB

=====

- EXP3 Progress: 100%|██████████| 8000/8000 [00:10<00:00, 767.18it/s]
 COMPLETED: EXPUCB (batch) completed successfully
 Running EXPNeuralUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

=====

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially
Action Selection	Neural UCB	Robust
Parameters	gamma=0.010, eta=0.050	Nonlinear Learning
	beta=0.2	EXP3 Config
		Neural UCB

=====

- HYBRID Progress: 100%|██████████| 8000/8000 [00:31<00:00, 256.93it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 113.80s):

=====

Oracle (step-wise) : 7777.26

GNeuralUCB (batch) : 5417.64
EXPUCB (batch) : 4675.52
EXPNeuralUCB (batch) : 5584.49
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 71.8%

ATTACK PATTERN ANALYSIS:

Strategy:	RandomAttack	
Total Frames:	8,000	
Total Attacks:	8,063	
Attack Intensity:	0.252	
PATH-SPECIFIC ANALYSIS:		
Path 0:	2018 attacks (25.2% rate)	
Path 1:	2041 attacks (25.5% rate)	
Path 2:	1996 attacks (24.9% rate)	
Path 3:	2008 attacks (25.1% rate)	
ATTACKS: Mean attacks/frame: 1.008		

EXPERIMENT 3 COMPLETED
Execution Time: 113.8s
Results:
Oracle: 7777.26 (theoretical maximum)
EXPNeuralUCB: 5584.49 (28.2% gap) [WINNER]
GNeuralUCB: 5417.64 (30.3% gap)
EXPUCB: 4675.52 (39.9% gap)
Winner: EXPNeuralUCB

QUANTUM EVALUATION COMPLETED!

Total experiments: 3
Total time: 184.0s
Average time per experiment: 61.3s

DETAILED EXPERIMENT ANALYSIS

Attack Configuration:
Strategy: RANDOM
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 26.2s)
+-- Oracle: 3192.62 (theoretical maximum)
+-- EXPNeuralUCB: 2097.30 (34.3% gap)
+-- GNeuralUCB: 2420.84 (24.2% gap)
+-- EXPUCB: 2024.79 (36.6% gap)
Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 43.2s)
+-- Oracle: 5567.05 (theoretical maximum)
+-- EXPNeuralUCB: 3910.27 (29.8% gap)

```
EXPNeuralUCB: 3840.18 ( 31.0% gap)
+-- GNeuralUCB: 3840.18 ( 31.0% gap)
+-- EXPUCB: 3310.90 ( 40.5% gap)
Winner: EXPNeuralUCB
```

EXPERIMENT 3 (8,000 frames, 113.8s)

```
|
+-- Oracle: 7777.26 (theoretical maximum)
+-- EXPNeuralUCB: 5584.49 ( 28.2% gap)
+-- GNeuralUCB: 5417.64 ( 30.3% gap)
+-- EXPUCB: 4675.52 ( 39.9% gap)
Winner: EXPNeuralUCB
```

ORACLE-RELATIVE GAP ANALYSIS:

LEARNING PERFORMANCE (Gap Reduction to Oracle):

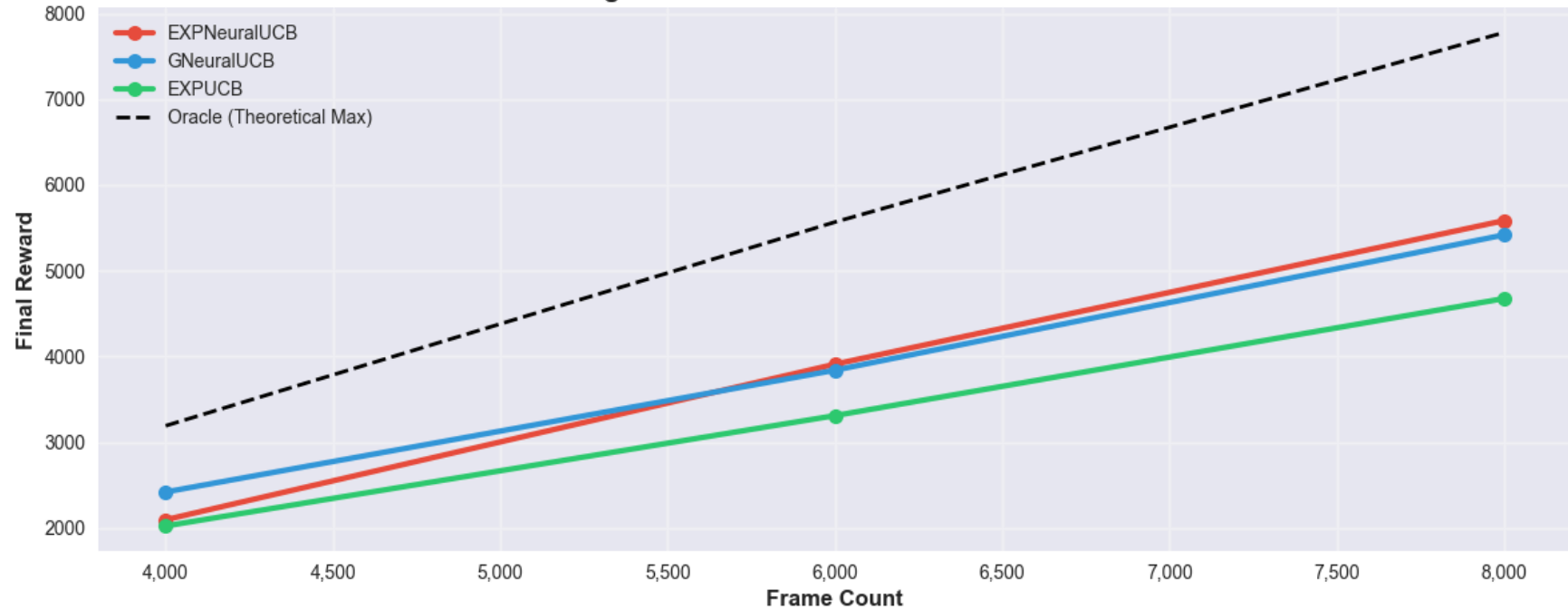
```
#1: EXPNeuralUCB -> +6.1pp improvement (gap closing)
    Gap: 34.3% -> 28.2%
#2: EXPUCB      -> +3.3pp decline (gap widening)
    Gap: 36.6% -> 39.9%
#3: GNeuralUCB  -> +6.2pp decline (gap widening)
    Gap: 24.2% -> 30.3%
```

Interpretation:

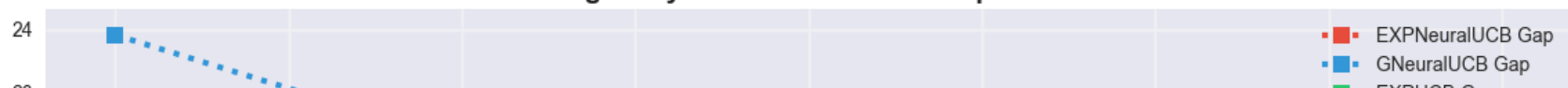
- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

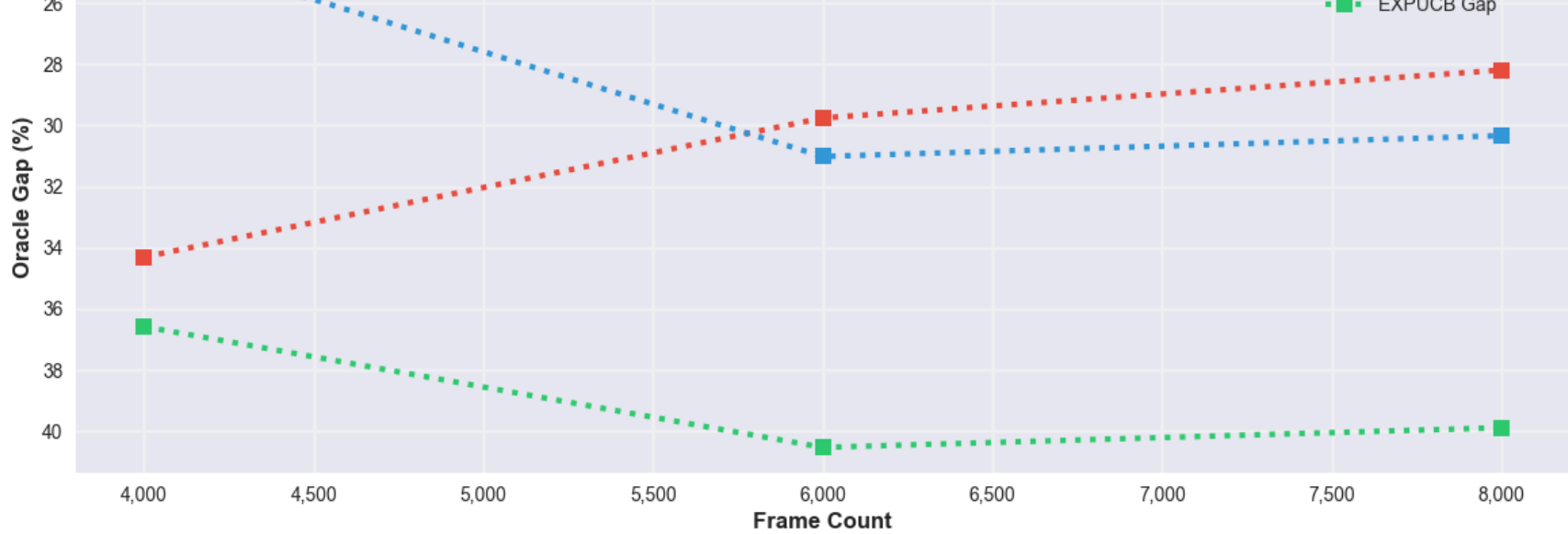
Generating performance visualization...

Algorithm Performance - RANDOM Attack



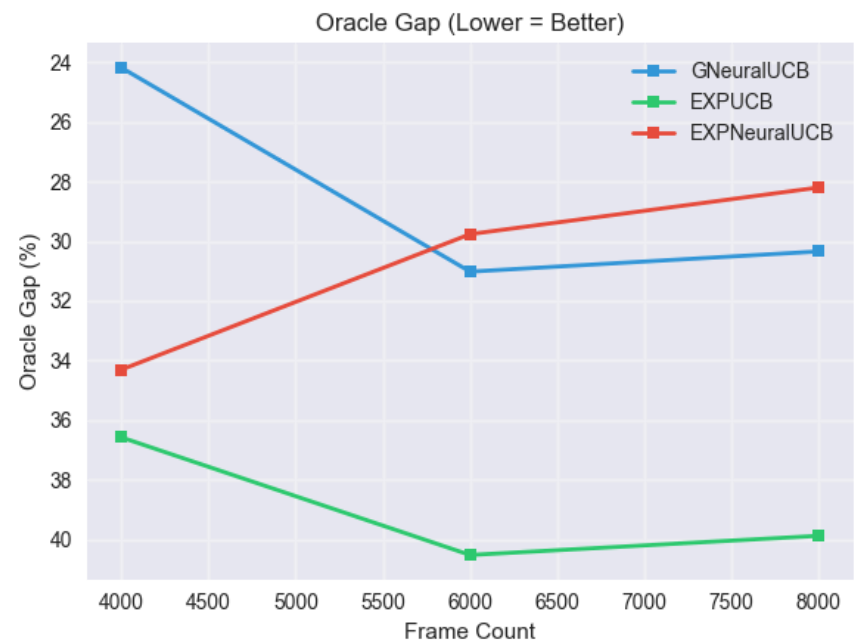
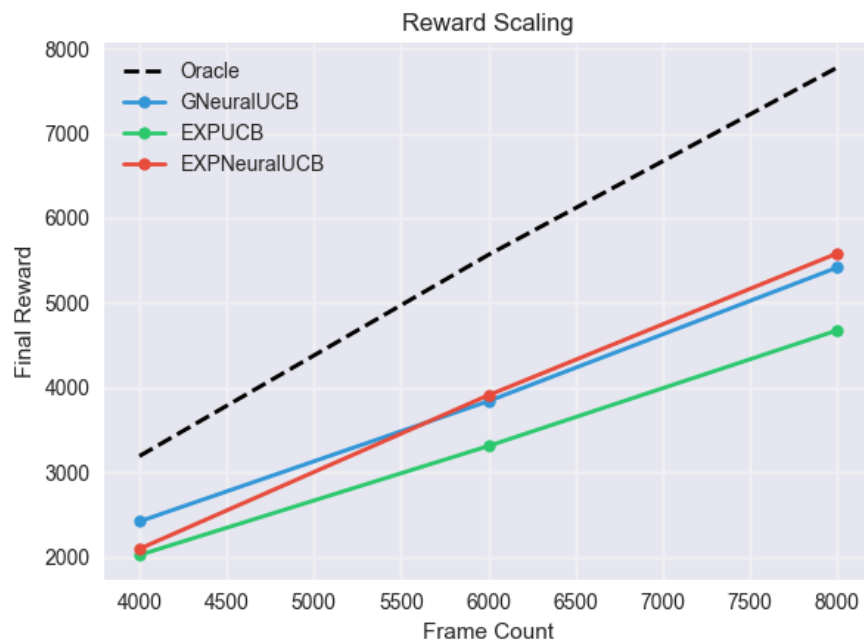
Learning Analysis: Oracle-Relative Gap Reduction





Visualization complete!

Random Environment Results



```

=====
TESTING MARKOV
=====
Running test for markov environment...

=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
Configuration:
- Base frames:      4,000
- Evaluation frames: 2,000
  
```

```
- Frame step:      2,000
- Base seed:      12345
- Attack strategy: MARKOV
- Attack intensity: 1.0
- Progress bars:   Disabled
- Analysis method: Oracle-relative gap analysis
```

STARTING QUANTUM MULTIRUN EVALUATION

Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000

PROGRESS: Experiment 1/3

EXPERIMENT 1

Frame Count: 4,000
Attack: markov (intensity=1.0)

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	4,000	

ADVERSARIAL CONFIGURATION:

Attack Strategy:	MarkovAttack	
Attack Rate:	1.00	
Simultaneous:	2	

Environment: markov (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3051.08

COMPLETED: in 0.00s, final reward: 3051.08

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]	

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties	
-----------	--------	------------	--

```
| Group Selection | Simple UCB | NOT Adversarial |
| Action Selection | Neural UCB | Robust |
| Parameters | beta=0.2 | UCB Confidence |
=====
- NEURAL Progress: 100%|██████████| 4000/4000 [00:08<00:00, 479.42it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | EXP3 | Adversarially |
| Action Selection | Linear UCB | Robust |
| Parameters | gamma=0.100, eta=0.005 | Linear |
| | beta=0.2 | Learning |
| | | EXP3 Config |
| | | Linear UCB |
=====
```

```
- EXP3 Progress: 100%|██████████| 4000/4000 [00:02<00:00, 1548.16it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

```
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | EXP3 | Adversarially |
| Action Selection | Neural UCB | Robust |
| Parameters | gamma=0.010, eta=0.050 | Nonlinear |
| | beta=0.2 | Learning |
| | | EXP3 Config |
| | | Neural UCB |
=====
```

```
- HYBRID Progress: 100%|██████████| 4000/4000 [00:12<00:00, 316.21it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

RESULTS (completed in 23.76s):

```
=====
Oracle (step-wise) : 3051.08
GNeuralUCB (batch) : 1137.52
=====
```

EXPUCB (batch) : 1453.00
EXPNeuralUCB (batch) : 1795.46
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 58.8%

ATTACK PATTERN ANALYSIS:

=====			
Strategy:	MarkovAttack		
Total Frames:	4,000		
Total Attacks:	8,000		
Attack Intensity:	0.500		

PATH-SPECIFIC ANALYSIS:			

Path 0:	2543 attacks (63.6% rate)		
Path 1:	1474 attacks (36.9% rate)		
Path 2:	2482 attacks (62.1% rate)		
Path 3:	1501 attacks (37.5% rate)		
=====			
ATTACKS: Mean attacks/frame: 2.000			
=====			

EXPERIMENT 1 COMPLETED

Execution Time: 23.8s

Results:

Oracle: 3051.08 (theoretical maximum)
EXPNeuralUCB: 1795.46 (41.2% gap) [WINNER]
GNeuralUCB: 1137.52 (62.7% gap)
EXPUCB: 1453.00 (52.4% gap)
Winner: EXPNeuralUCB

PROGRESS: Experiment 2/3

EXPERIMENT 2

Frame Count: 6,000

Attack: markov (intensity=1.0)

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====			
Network Paths:	4		
Qubit Config:	(8, 10, 8, 9)		
Frame Length:	6,000		
=====			

ADVERSARIAL CONFIGURATION:

=====			
Attack Strategy:	MarkovAttack		
Attack Rate:	1.00		
Simultaneous:	2		
=====			

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5474.92

COMPLETED: in 0.01s, final reward: 5474.92

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
| Action Selection   | Neural UCB      | Robust              |
|                    |                  | Nonlinear            |
|                    |                  | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====
```

```
- NEURAL Progress: 100%|██████████| 6000/6000 [00:14<00:00, 408.33it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
| Action Selection   | Linear UCB      | Robust              |
|                    |                  | Linear              |
|                    |                  | Learning             |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config        |
|                    | beta=0.2        | Linear UCB          |
=====
```

```
- EXP3 Progress: 100%|██████████| 6000/6000 [00:05<00:00, 1049.22it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
```

Group Selection	EXP3	Adversarially
Action Selection	Neural UCB	Robust
Parameters	gamma=0.010, eta=0.050	Nonlinear Learning
	beta=0.2	EXP3 Config Neural UCB

- HYBRID Progress: 100%[██████████] 6000/6000 [00:19<00:00, 300.72it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 40.62s):

```
=====
Oracle      (step-wise) : 5474.92
GNeuralUCB  (batch)    : 3211.32
EXPUCB      (batch)    : 2604.25
EXPNeuralUCB (batch)   : 3260.87
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 59.6%
```

ATTACK PATTERN ANALYSIS:

```
=====
| Strategy:      | MarkovAttack |
| Total Frames:  | 6,000        |
| Total Attacks: | 12,000       |
| Attack Intensity: | 0.500      |
=====
```

PATH-SPECIFIC ANALYSIS:

```
=====
| Path 0:      | 3798 attacks (63.3% rate) |
| Path 1:      | 2217 attacks (37.0% rate) |
| Path 2:      | 3789 attacks (63.1% rate) |
| Path 3:      | 2196 attacks (36.6% rate) |
=====
```

ATTACKS: Mean attacks/frame: 2.000

EXPERIMENT 2 COMPLETED

Execution Time: 40.6s

Results:

```
Oracle:      5474.92 (theoretical maximum)
EXPNeuralUCB: 3260.87 ( 40.4% gap) [WINNER]
GNeuralUCB:  3211.32 ( 41.3% gap)
EXPUCB:      2604.25 ( 52.4% gap)
Winner: EXPNeuralUCB
=====
```

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: markov (intensity=1.0)

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
| Network Paths: | 4 |
| Qubit Config:  | (8, 10, 8, 9) |
| Frame Length:  | 8,000 |
=====
```

ADVERSARIAL CONFIGURATION:

Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2

Environment: markov (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 8000 steps, final reward: 7744.38
COMPLETED: in 0.01s, final reward: 7744.38
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%[██████████] 8000/8000 [01:07<00:00, 117.77it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

- EXP3 Progress: 100%[██████████] 8000/8000 [00:10<00:00, 792.58it/s]
COMPLETED: EXPUCB (batch) completed successfully

ORACLE ANALYSIS:

=====		
Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]	
=====		

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

=====		
Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB
=====		

- HYBRID Progress: 100%[██████████] 8000/8000 [00:28<00:00, 280.10it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 106.93s):

=====

Oracle	(step-wise)	: 7744.38
GNeuralUCB	(batch)	: 4552.69
EXPUCB	(batch)	: 3605.14
EXPNeuralUCB	(batch)	: 4738.67

WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 61.2%

ATTACK PATTERN ANALYSIS:

=====		
Strategy:	MarkovAttack	
Total Frames:	8,000	
Total Attacks:	16,000	
Attack Intensity:	0.500	
=====		

PATH-SPECIFIC ANALYSIS:		

Path 0:	5076 attacks (63.4% rate)	
Path 1:	2935 attacks (36.7% rate)	
Path 2:	5036 attacks (62.9% rate)	
Path 3:	2953 attacks (36.9% rate)	

ATTACKS: Mean attacks/frame: 2.000

EXPERIMENT 3 COMPLETED

Execution Time: 106.9s

Results:

Oracle:	7744.38 (theoretical maximum)
EXPNeuralUCB:	4738.67 (38.8% gap) [WINNER]
GNeuralUCB:	4552.69 (41.2% gap)
EXPUCB:	3605.14 (53.4% gap)

Winner: EXPNeuralUCB

=====

QUANTUM EVALUATION COMPLETED:

Total experiments: 3

Total time: 172.3s

Average time per experiment: 57.4s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:

Strategy: MARKOV

Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 23.8s)

|

+-- Oracle: 3051.08 (theoretical maximum)

+-- EXPNeuralUCB: 1795.46 (41.2% gap)

+-- GNeuralUCB: 1137.52 (62.7% gap)

+-- EXPUCB: 1453.00 (52.4% gap)

Winner: EXPNeuralUCB

EXPERIMENT 2 (6,000 frames, 40.6s)

|

+-- Oracle: 5474.92 (theoretical maximum)

+-- EXPNeuralUCB: 3260.87 (40.4% gap)

+-- GNeuralUCB: 3211.32 (41.3% gap)

+-- EXPUCB: 2604.25 (52.4% gap)

Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 106.9s)

|

+-- Oracle: 7744.38 (theoretical maximum)

+-- EXPNeuralUCB: 4738.67 (38.8% gap)

+-- GNeuralUCB: 4552.69 (41.2% gap)

+-- EXPUCB: 3605.14 (53.4% gap)

Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

LEARNING PERFORMANCE (Gap Reduction to Oracle):

#1: GNeuralUCB -> +21.5pp improvement (gap closing)

Gap: 62.7% -> 41.2%

#2: EXPNeuralUCB -> +2.3pp improvement (gap closing)

Gap: 41.2% -> 38.8%

#3: EXPUCB -> +1.1pp decline (gap widening)

Gap: 52.4% -> 53.4%

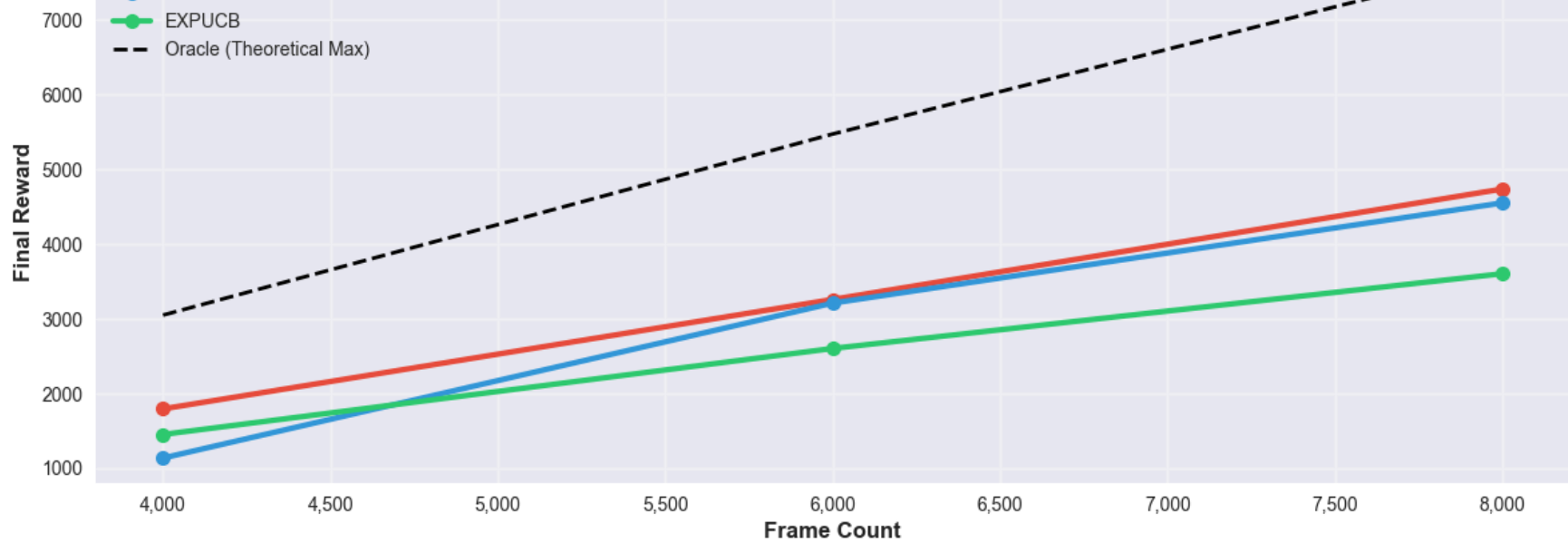
Interpretation:

- Lower gaps = Better performance relative to Oracle
 - Gap reduction = Algorithm learning and improving
 - Best learner closes the gap to Oracle over time
-

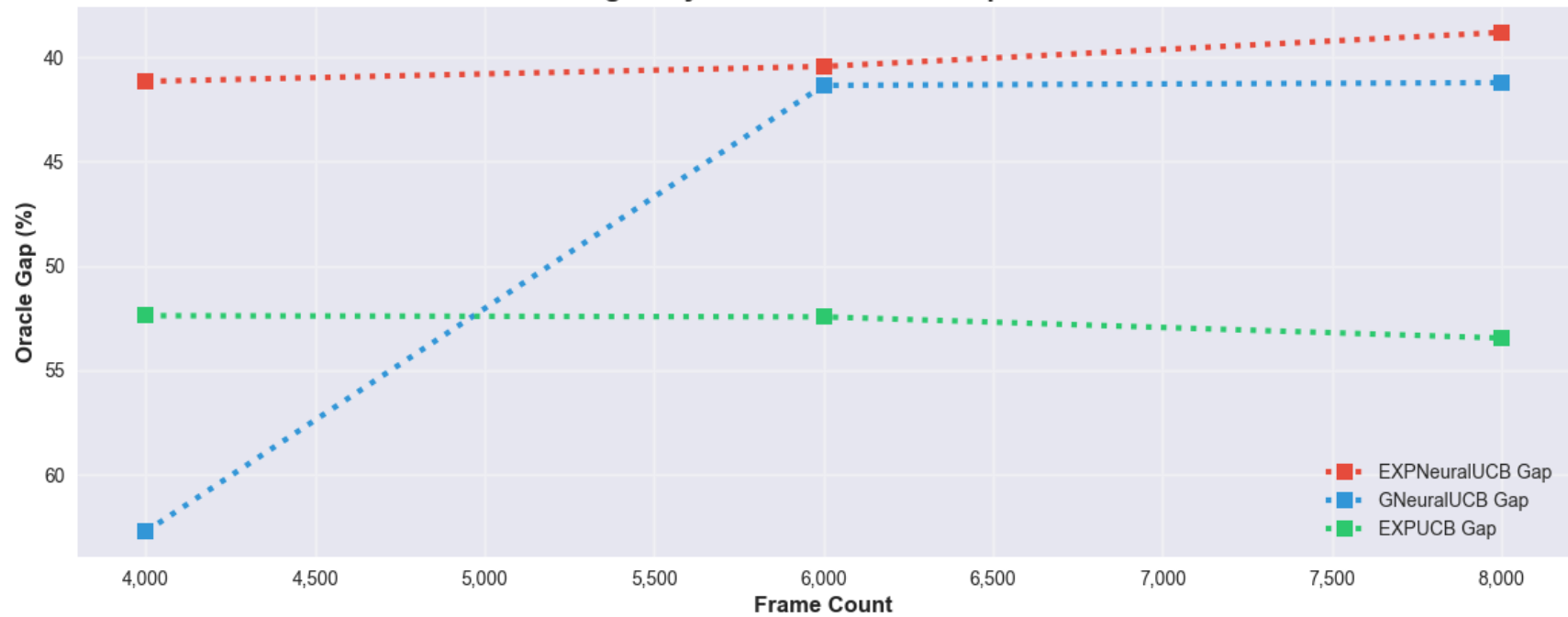
Generating performance visualization...

Algorithm Performance - MARKOV Attack



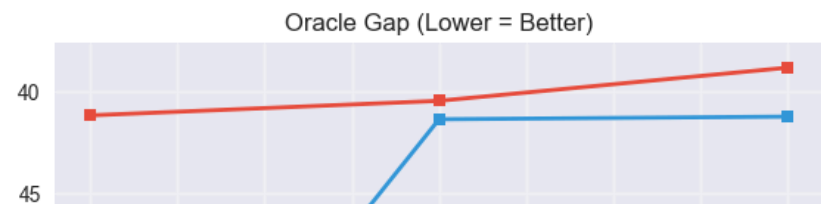


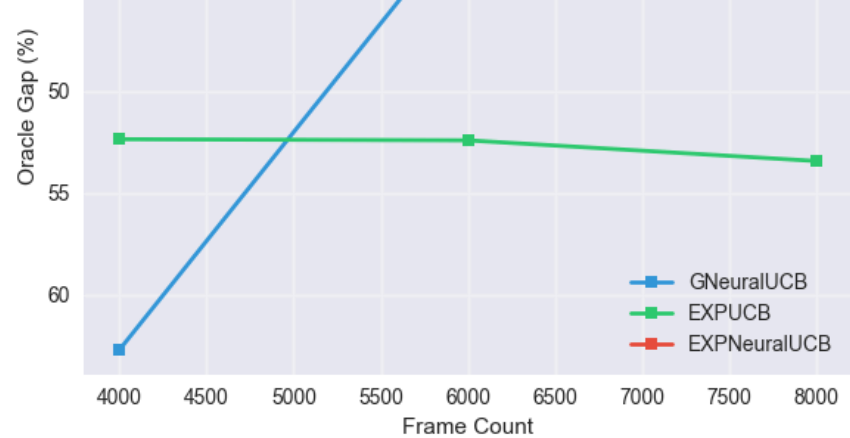
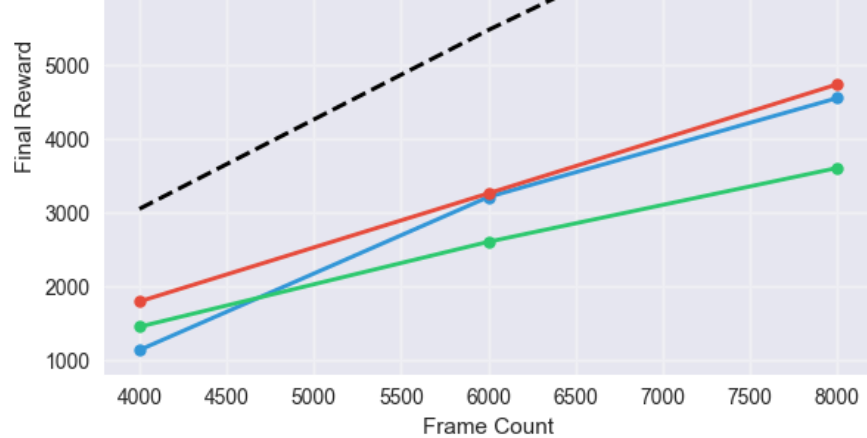
Learning Analysis: Oracle-Relative Gap Reduction



Visualization complete!

Markov Environment Results





```
=====
TESTING ADAPTIVE
=====
Running test for adaptive environment...
```

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
Configuration:
- Base frames:      4,000
- Frame step:       2,000
- Base seed:        12345
- Attack strategy:  ADAPTIVE
- Attack intensity: 1.0
- Progress bars:    Disabled
- Analysis method:  Oracle-relative gap analysis
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000
=====
```

PROGRESS: Experiment 1/3

```
=====
EXPERIMENT 1
=====
Frame Count: 4,000
Attack: adaptive (intensity=1.0)
=====
```

EXPERIMENT 1: 4000 frames

```
QUANTUM ENVIRONMENT INITIALIZED:
=====
| Network Paths: | 4 |
| Qubit Config:  | (8, 10, 8, 9) |
| Frame Length:  | 4,000 |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy: | AdaptiveAttack |
| Attack Rate:    | 1.00          |
| Memory Window:  | 50            |
=====
```

Environment: adaptive (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3240.72

COMPLETED: in 0.00s, final reward: 3240.72

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0              |
| Optimal Action:  | 4              |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
| Action Selection   | Neural UCB      | Robust              |
|                    |                  | Nonlinear           |
|                    |                  | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====
```

- NEURAL Progress: 100%[██████████] 4000/4000 [00:10<00:00, 367.61it/s]

COMPLETED: GNeuralUCB (batch) completed successfully

Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0              |
| Optimal Action:  | 4              |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
| Action Selection   | Linear UCB      | Robust              |
|                    |                  | Linear              |
|                    |                  | Learning             |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config        |
|                    | beta=0.2        | Linear UCB          |
=====
```

- EXP3 Progress: 100%[██████████] 4000/4000 [00:02<00:00, 1536.88it/s]

COMPLETED: EXPUCB (batch) completed successfully

Running EXPNeuralUCB...

```
ORACLE ANALYSIS:
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component          | Method              | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3                | Adversarially      |
|                   |                     | Robust              |
| Action Selection   | Neural UCB          | Nonlinear           |
|                   |                     | Learning            |
| Parameters         | gamma=0.010, eta=0.050 | EXP3 Config        |
|                   | beta=0.2            | Neural UCB          |
=====
- HYBRID Progress: 100%[██████████] 4000/4000 [00:13<00:00, 303.10it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

```
RESULTS (completed in 26.76s):
=====
Oracle      (step-wise) : 3240.72
GNeuralUCB  (batch)     : 2542.22
EXPUCB      (batch)     : 2093.89
EXPNeuralUCB (batch)    : 2016.44
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 62.2%
```

```
ATTACK PATTERN ANALYSIS:
=====
| Strategy:          | AdaptiveAttack      |
| Total Frames:      | 4,000               |
| Total Attacks:     | 4,000               |
| Attack Intensity:  | 0.250               |
=====
| PATH-SPECIFIC ANALYSIS: |
|-----|-----|
| Path 0:            | 841 attacks (21.0% rate) |
| Path 1:            | 1159 attacks (29.0% rate) |
| Path 2:            | 896 attacks (22.4% rate) |
| Path 3:            | 1104 attacks (27.6% rate) |
|-----|-----|
ATTACKS: Mean attacks/frame: 1.000
=====
```

```
EXPERIMENT 1 COMPLETED
Execution Time: 26.8s
Results:
Oracle:      3240.72 (theoretical maximum)
EXPNeuralUCB: 2016.44 ( 37.8% gap)
GNeuralUCB:  2542.22 ( 21.6% gap) [WINNER]
EXPUCB:      2093.89 ( 35.4% gap)
Winner: GNeuralUCB
=====
```

EXPERIMENT 2

Frame Count: 6,000
Attack: adaptive (intensity=1.0)

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000

ADVERSARIAL CONFIGURATION:

Attack Strategy:	AdaptiveAttack
Attack Rate:	1.00
Memory Window:	50

Environment: adaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 6000 steps, final reward: 5584.16
COMPLETED: in 0.01s, final reward: 5584.16
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%[██████████] 6000/6000 [00:15<00:00, 375.14it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

- EXP3 Progress: 100% ██████████ | 6000/6000 [00:05<00:00, 1045.03it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

- HYBRID Progress: 100% ██████████ | 6000/6000 [00:23<00:00, 252.92it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 45.56s):

Oracle (step-wise)	: 5584.16
GNeuralUCB (batch)	: 3903.85
EXPUCB (batch)	: 3318.91
EXPNeuralUCB (batch)	: 3980.39
WINNER:	EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 71.3%	

ATTACK PATTERN ANALYSIS:

Strategy:	AdaptiveAttack
Total Frames:	6,000
Total Attacks:	6,000
Attack Intensity:	0.250

PATH-SPECIFIC ANALYSIS:

Path 0:	2142 attacks (35.7% rate)
Path 1:	1439 attacks (24.0% rate)
Path 2:	947 attacks (15.8% rate)
Path 3:	1472 attacks (24.5% rate)

ATTACKS: Mean attacks/frame: 1.000

=====

EXPERIMENT 2 COMPLETED
Execution Time: 45.6s
Results:
Oracle: 5584.16 (theoretical maximum)
EXPNeuralUCB: 3980.39 (28.7% gap) [WINNER]
GNeuralUCB: 3903.85 (30.1% gap)
EXPUCB: 3318.91 (40.6% gap)
Winner: EXPNeuralUCB

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000
Attack: adaptive (intensity=1.0)

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	8,000	

=====

ADVERSARIAL CONFIGURATION:

=====

Attack Strategy:	AdaptiveAttack	
Attack Rate:	1.00	
Memory Window:	50	

=====

Environment: adaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 8000 steps, final reward: 7832.21
COMPLETED: in 0.01s, final reward: 7832.21
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]	

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
Action Selection	Neural UCB	Robust
		Nonlinear
		Learning

```
| Parameters | beta=0.2 | UCB Confidence |
=====
- NEURAL Progress: 100%|██████████| 8000/8000 [01:11<00:00, 112.07it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | EXP3 | Adversarially Robust |
| Action Selection | Linear UCB | Linear Learning |
| Parameters | gamma=0.100, eta=0.005 | EXP3 Config |
| | beta=0.2 | Linear UCB |
=====
```

```
- EXP3 Progress: 100%|██████████| 8000/8000 [00:10<00:00, 779.08it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

```
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | EXP3 | Adversarially Robust |
| Action Selection | Neural UCB | Nonlinear Learning |
| Parameters | gamma=0.010, eta=0.050 | EXP3 Config |
| | beta=0.2 | Neural UCB |
=====
```

```
- HYBRID Progress: 100%|██████████| 8000/8000 [00:30<00:00, 265.54it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

RESULTS (completed in 111.91s):

```
=====
Oracle (step-wise) : 7832.21
GNeuralUCB (batch) : 5597.04
EXPUCB (batch) : 4744.51
EXPNeuralUCB (batch) : 5498.66
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 70.2%
```

```
ATTACK PATTERN ANALYSIS:
=====
| Strategy:      | AdaptiveAttack      |
| Total Frames:  | 8,000               |
| Total Attacks: | 8,000               |
| Attack Intensity: | 0.250              |
=====
| PATH-SPECIFIC ANALYSIS: |
=====
| Path 0:      | 1844 attacks (23.1% rate) |
| Path 1:      | 2126 attacks (26.6% rate) |
| Path 2:      | 2217 attacks (27.7% rate) |
| Path 3:      | 1813 attacks (22.7% rate) |
=====
ATTACKS: Mean attacks/frame: 1.000
=====
```

```
EXPERIMENT 3 COMPLETED
Execution Time: 111.9s
Results:
  Oracle:      7832.21 (theoretical maximum)
  EXPNeuralUCB: 5498.66 ( 29.8% gap)
  GNeuralUCB:   5597.04 ( 28.5% gap) [WINNER]
  EXPUCB:      4744.51 ( 39.4% gap)
  Winner: GNeuralUCB
=====
```

```
=====
QUANTUM EVALUATION COMPLETED!
=====
Total experiments: 3
Total time: 185.5s
Average time per experiment: 61.8s
=====
```

```
=====
DETAILED EXPERIMENT ANALYSIS
=====
```

```
Attack Configuration:
  Strategy: ADAPTIVE
  Intensity: 1.0
```

```
Per-Experiment Results:
=====
```

```
EXPERIMENT 1 (4,000 frames, 26.8s)
|
+-- Oracle:      3240.72 (theoretical maximum)
+-- EXPNeuralUCB: 2016.44 ( 37.8% gap)
+-- GNeuralUCB:   2542.22 ( 21.6% gap)
+-- EXPUCB:      2093.89 ( 35.4% gap)
  Winner: GNeuralUCB
```

```
EXPERIMENT 2 (6,000 frames, 45.6s)
|
+-- Oracle:      5584.16 (theoretical maximum)
+-- EXPNeuralUCB: 3980.39 ( 28.7% gap)
+-- GNeuralUCB:   3903.85 ( 30.1% gap)
+-- EXPUCB:      3318.91 ( 40.6% gap)
  Winner: EXPNeuralUCB
```

```
EXPERIMENT 3 (8,000 frames, 111.9s)
```

Oracle: 7832.21 (theoretical maximum)
EXPNeuralUCB: 5498.66 (29.8% gap)
GNeuralUCB: 5597.04 (28.5% gap)
EXPUCB: 4744.51 (39.4% gap)
Winner: GNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

LEARNING PERFORMANCE (Gap Reduction to Oracle):

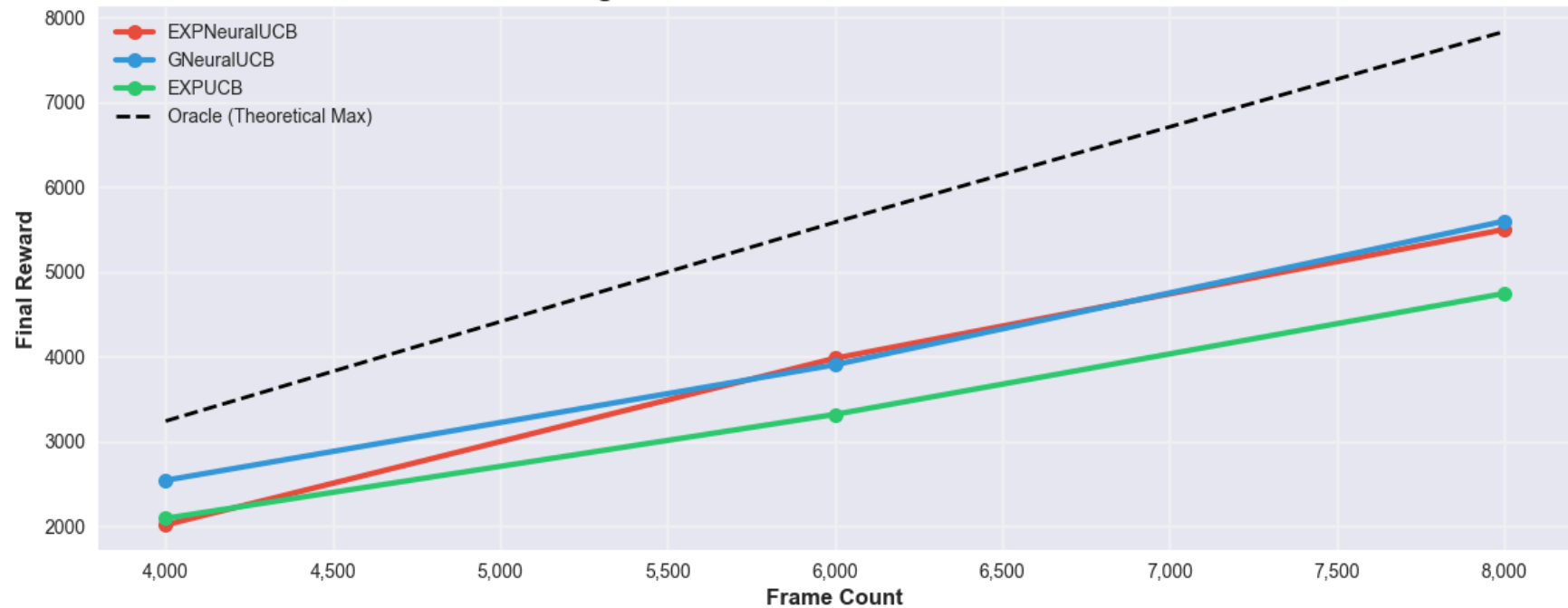
- #1: EXPNeuralUCB -> +8.0pp improvement (gap closing)
Gap: 37.8% -> 29.8%
- #2: EXPUCB -> +4.0pp decline (gap widening)
Gap: 35.4% -> 39.4%
- #3: GNeuralUCB -> +7.0pp decline (gap widening)
Gap: 21.6% -> 28.5%

Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

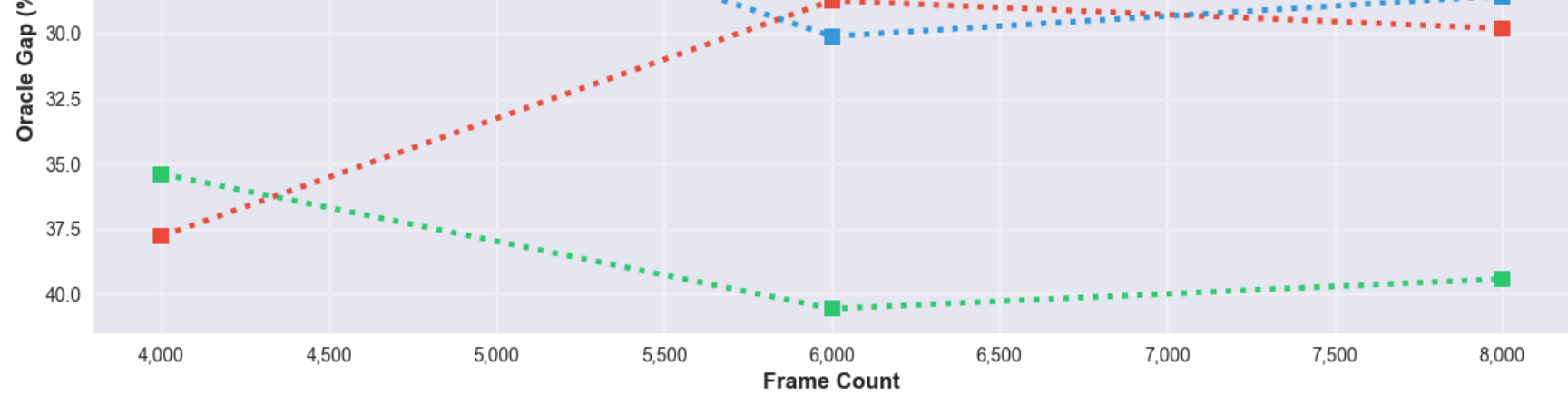
Generating performance visualization...

Algorithm Performance - ADAPTIVE Attack



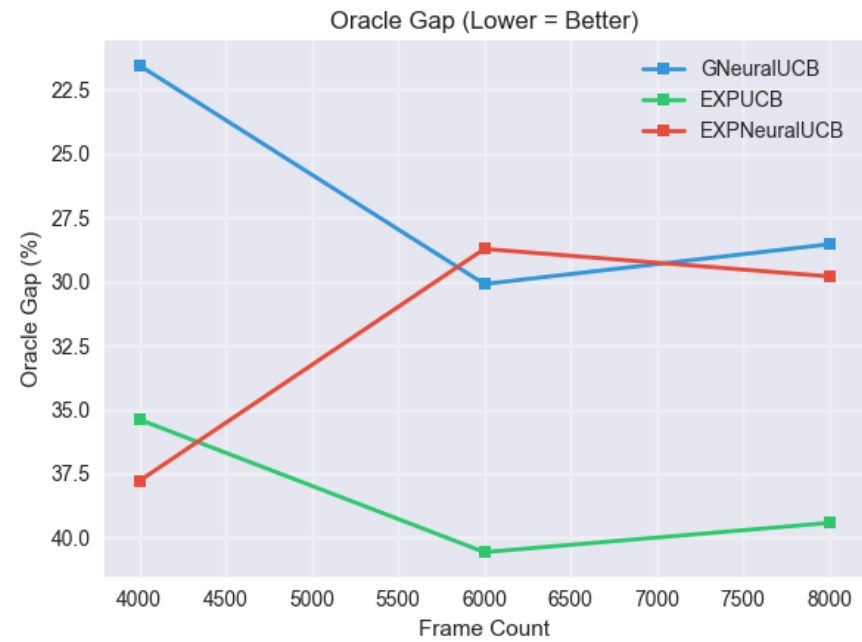
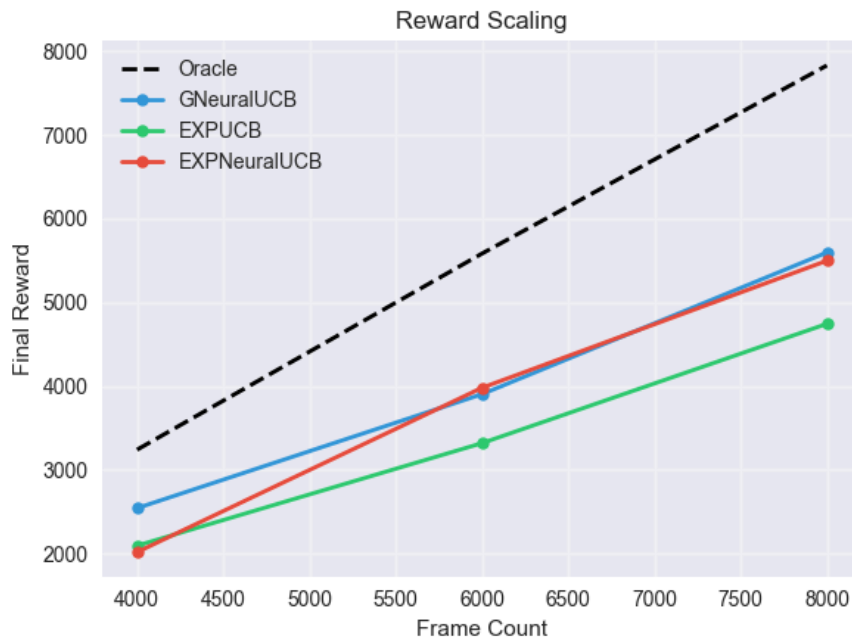
Learning Analysis: Oracle-Relative Gap Reduction



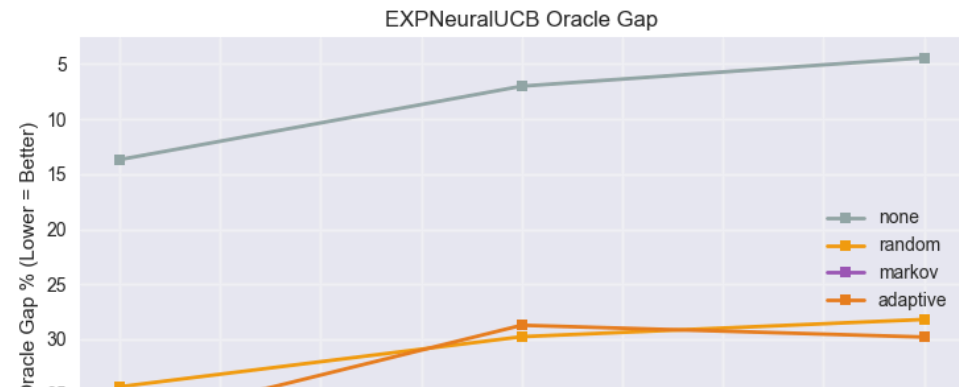
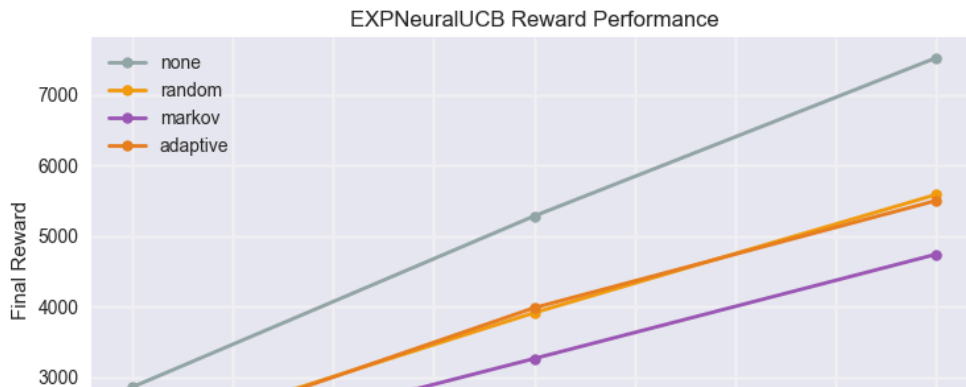


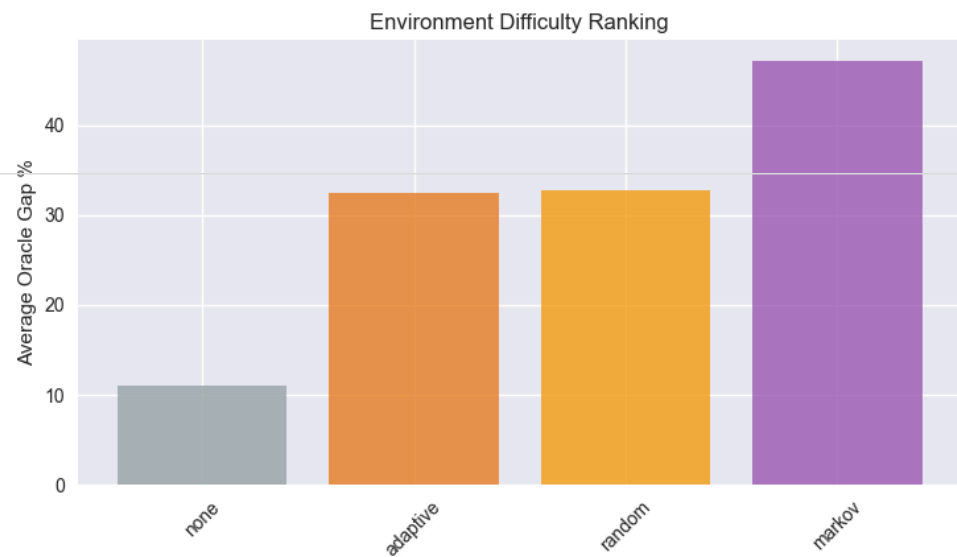
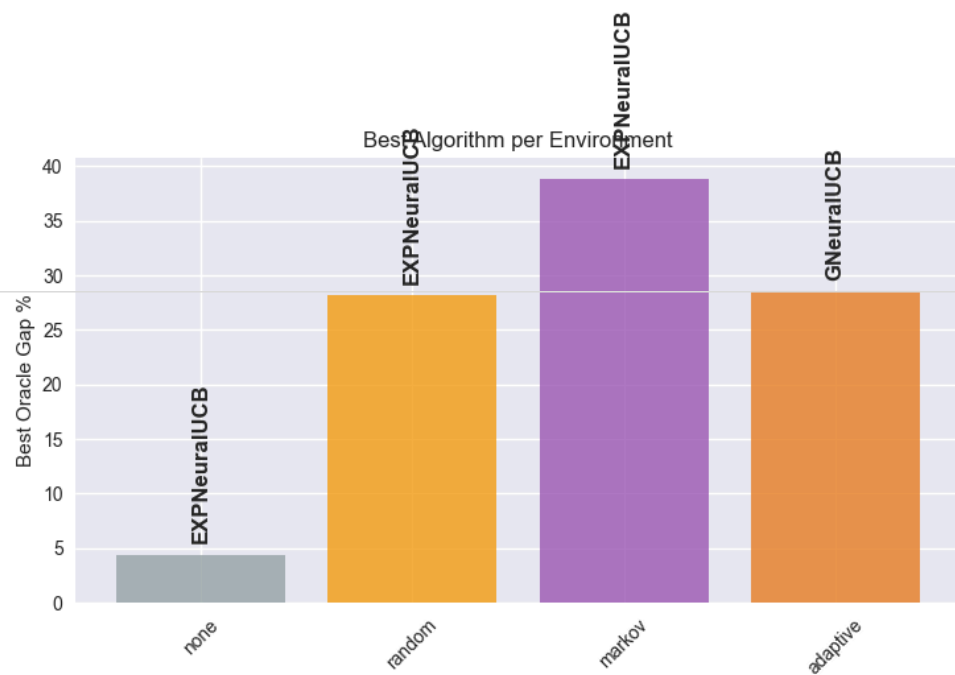
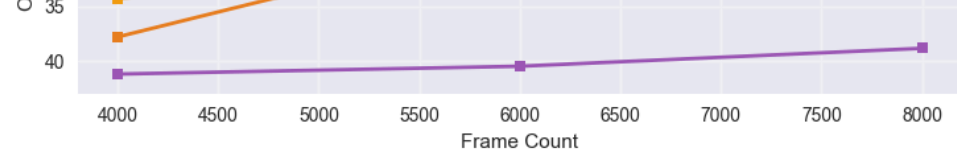
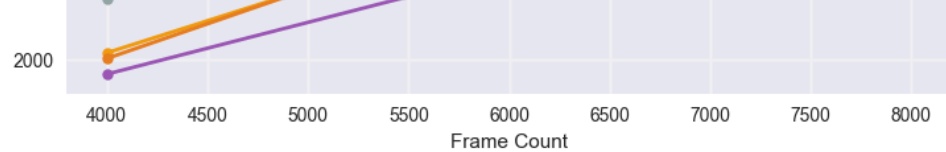
Visualization complete!

Adaptive Environment Results



ENVIRONMENT COMPARISON





```

1 # quantum_visualizer_enhanced.py
2
3 import matplotlib.pyplot as plt
4 import pandas as pd
5 import numpy as np
6 import seaborn as sns
7
8 class EnhancedQuantumVisualizer(QuantumVisualizer):
9     """Enhanced visualization extending the base QuantumVisualizer with advanced plots"""
10
11     def __init__(self):
12         super().__init__()
13         self._setup_style()
14
15     def _setup_style(self):
16         plt.style.use('seaborn-v0_8')
17         # Keep existing colors from base class
18         self.env_colors['onlineadaptive'] = '#16a085' # Add missing color
19
20     def _extract_data(self, attack_type):
21         """Extract series for plotting from env_experiments structure (with fallbacks)."""
22         env_data = self._get_env_data(attack_type)
23
24         # canonical order; include only those present
25         first_exp = next(iter(env_data.values()))
26         available_algs = set(first_exp.get('results', {}).keys())
27         algorithms = [a for a in ("EXPNeuralUCB", "GNeuralUCB", "EXPUCB") if a in available_algs]
28
29         frame_counts = []
30         oracle_rewards = []
31         alg_rewards = {alg: [] for alg in algorithms}
32         alg_gaps = {alg: [] for alg in algorithms}
33
34         for exp_id in sorted(env_data.keys()):
35             exp = env_data[exp_id]
36             res = exp.get('results', {})
37             gaps = exp.get('gaps', {})
38
39             frame_counts.append(int(exp.get('frame_count', 0)))
40             oracle_rewards.append(float(res.get('Oracle', {}).get('final_reward', 0.0)))
41
42             for alg in algorithms:
43                 alg_rewards[alg].append(float(res.get(alg, {}).get('final_reward', 0.0)))
44                 alg_gaps[alg].append(float(gaps.get(alg, np.nan)))
45
46         return frame_counts, oracle_rewards, alg_rewards, alg_gaps, algorithms
47
48     def plot_performance_dashboard(self, attack_types):
49         """Enhanced dashboard visualization (cross-environment)."""
50         missing = [env for env in attack_types if env not in self.evaluators]
51         if missing:
52             print(f"Missing environments: {missing}")
53             return
54
55         fig, axes = plt.subplots(2, 3, figsize=(18, 12))
56         fig.suptitle(
57             'QUANTUM ROUTING ALGORITHM PERFORMANCE DASHBOARD\n'
58             'Lower Oracle Gap = Better Performance | Higher Reward = Better Performance',
59             fontsize=16, fontweight='bold'
60         )
61
62         # Collect all data
63         all_gaps = []

```

```

64 all_rewards = []
65
66 for env in attack_types:
67     try:
68         frame_counts, oracle_rewards, alg_rewards, alg_gaps, algorithms = self._extract_data(env)
69     except Exception:
70         continue
71
72     for alg in algorithms:
73         for frame, gap, reward in zip(frame_counts, alg_gaps[alg], alg_rewards[alg]):
74             all_gaps.append({
75                 'environment': env,
76                 'algorithm': alg,
77                 'frame_count': frame,
78                 'gap_percent': gap,
79                 'final_reward': reward
80             })
81             all_rewards.append({
82                 'environment': env,
83                 'algorithm': alg,
84                 'frame_count': frame,
85                 'final_reward': reward
86             })
87
88 gap_df = pd.DataFrame(all_gaps)
89 reward_df = pd.DataFrame(all_rewards)
90
91 # 1. Oracle Gap Heatmap
92 if not gap_df.empty:
93     gap_matrix = gap_df.pivot_table(index='environment', columns='algorithm',
94                                     values='gap_percent', aggfunc='mean')
95     sns.heatmap(gap_matrix, annot=True, fmt='.1f', cmap='RdYlGn_r',
96                ax=axes[0,0], cbar_kws={'label': 'Oracle Gap % (Lower=Better)'})
97     axes[0,0].set_title('Oracle Gap by Environment\n(GREEN = Better)')
98 else:
99     axes[0,0].set_visible(False)
100
101 # 2. Final Reward Heatmap
102 if not reward_df.empty:
103     reward_matrix = reward_df.pivot_table(index='environment', columns='algorithm',
104                                           values='final_reward', aggfunc='mean')
105     sns.heatmap(reward_matrix, annot=True, fmt='.1f', cmap='RdYlGn',
106                ax=axes[0,1], cbar_kws={'label': 'Final Reward (Higher=Better)'})
107     axes[0,1].set_title('Final Reward by Environment\n(GREEN = Better)')
108 else:
109     axes[0,1].set_visible(False)
110
111 # 3. Algorithm Ranking (by avg gap)
112 if not gap_df.empty:
113     avg_gaps = gap_df.groupby('algorithm')['gap_percent'].mean().sort_values()
114     colors = ['gold', 'silver', '#CD7F32', 'gray'][:len(avg_gaps)]
115     axes[0,2].bar(range(len(avg_gaps)), avg_gaps.values, color=colors)
116     axes[0,2].set_xticks(range(len(avg_gaps)))
117     axes[0,2].set_xticklabels(avg_gaps.index, rotation=45)
118     axes[0,2].set_title('Overall Algorithm Ranking\n(Lower = Better)')
119     axes[0,2].set_ylabel('Average Oracle Gap %')
120 else:
121     axes[0,2].set_visible(False)
122
123 # 4. Performance vs Frame Count
124 for env in attack_types:
125     try:
126         frame_counts, _, alg_rewards, alg_gaps, algorithms = self._extract_data(env)
127     except Exception:
128         continue

```

```

129         for alg in algorithms:
130             axes[1,0].plot(frame_counts, alg_rewards[alg], 'o-',
131                             label=f"{alg} ({env})",
132                             color=self.colors.get(alg, 'gray'), alpha=0.7)
133
134         axes[1,0].set_title('Reward Scaling with Frame Count')
135         axes[1,0].set_xlabel('Frame Count')
136         axes[1,0].set_ylabel('Final Reward')
137         axes[1,0].legend(bbox_to_anchor=(1.05, 1), loc='upper left')
138         axes[1,0].grid(True, alpha=0.3)
139
140     # 5. Gap vs Frame Count
141     for env in attack_types:
142         try:
143             frame_counts, _, alg_rewards, alg_gaps, algorithms = self._extract_data(env)
144         except Exception:
145             continue
146         for alg in algorithms:
147             axes[1,1].plot(frame_counts, alg_gaps[alg], 's-',
148                             label=f"{alg} ({env})",
149                             color=self.colors.get(alg, 'gray'), alpha=0.7)
150
151         axes[1,1].set_title('Oracle Gap vs Frame Count')
152         axes[1,1].set_xlabel('Frame Count')
153         axes[1,1].set_ylabel('Oracle Gap %')
154         axes[1,1].legend(bbox_to_anchor=(1.05, 1), loc='upper left')
155         axes[1,1].grid(True, alpha=0.3)
156         axes[1,1].invert_yaxis()
157
158     # 6. Environment Difficulty
159     if not gap_df.empty:
160         env_difficulty = gap_df.groupby('environment')['gap_percent'].mean().sort_values()
161         colors = [self.env_colors.get(env, 'gray') for env in env_difficulty.index]
162         axes[1,2].bar(range(len(env_difficulty)), env_difficulty.values, color=colors)
163         axes[1,2].set_xticks(range(len(env_difficulty)))
164         axes[1,2].set_xticklabels(env_difficulty.index, rotation=45)
165         axes[1,2].set_title('Environment Difficulty\n(Lower = Easier)')
166         axes[1,2].set_ylabel('Average Oracle Gap %')
167     else:
168         axes[1,2].set_visible(False)
169
170     plt.tight_layout()
171     plt.savefig('performance_dashboard.png', dpi=300, bbox_inches='tight')
172     plt.show()
173
174     def plot_algorithm_comparison(self, attack_types):
175         """Algorithm comparison across environments."""
176         fig, axes = plt.subplots(2, 2, figsize=(15, 12))
177         fig.suptitle('ALGORITHM COMPARISON ANALYSIS', fontsize=16, fontweight='bold')
178
179         all_data = []
180         for env in attack_types:
181             try:
182                 frame_counts, _, alg_rewards, alg_gaps, algorithms = self._extract_data(env)
183             except Exception:
184                 continue
185             for alg in algorithms:
186                 for frame, gap, reward in zip(frame_counts, alg_gaps[alg], alg_rewards[alg]):
187                     all_data.append({
188                         'environment': env,
189                         'algorithm': alg,
190                         'frame_count': frame,
191                         'gap_percent': gap,
192                         'final_reward': reward
193                     })

```

```

194
195     df = pd.DataFrame(all_data)
196     if df.empty:
197         plt.close(fig)
198         print("No data to compare.")
199         return
200
201     sns.boxplot(data=df, x='algorithm', y='gap_percent', ax=axes[0,0])
202     axes[0,0].set_title('Performance Distribution (Lower = Better)')
203     axes[0,0].set_ylabel('Oracle Gap %')
204     axes[0,0].tick_params(axis='x', rotation=45)
205
206     sns.boxplot(data=df, x='environment', y='gap_percent', ax=axes[0,1])
207     axes[0,1].set_title('Environment Difficulty')
208     axes[0,1].set_ylabel('Oracle Gap %')
209     axes[0,1].tick_params(axis='x', rotation=45)
210
211     pivot = df.pivot_table(index='algorithm', columns='environment',
212                             values='gap_percent', aggfunc='mean')
213     sns.heatmap(pivot, annot=True, fmt='.1f', cmap='RdYlGn_r', ax=axes[1,0])
214     axes[1,0].set_title('Algorithm vs Environment\n(Green = Better)')
215
216     # Scaling improvement
217     scaling = []
218     for alg in df['algorithm'].unique():
219         for env in df['environment'].unique():
220             sub = df[(df['algorithm'] == alg) & (df['environment'] == env)]
221             if sub['frame_count'].nunique() >= 2:
222                 min_f = sub['frame_count'].min()
223                 max_f = sub['frame_count'].max()
224                 min_gap = sub[sub['frame_count'] == min_f]['gap_percent'].mean()
225                 max_gap = sub[sub['frame_count'] == max_f]['gap_percent'].mean()
226                 scaling.append({'algorithm': alg, 'environment': env, 'improvement': (min_gap - max_gap)})
227
228     if scaling:
229         sc_df = pd.DataFrame(scaling)
230         sns.barplot(data=sc_df, x='algorithm', y='improvement', hue='environment', ax=axes[1,1])
231         axes[1,1].set_title('Scaling Improvement (Higher = Better Learning)')
232         axes[1,1].set_ylabel('Gap Improvement (pp)')
233         axes[1,1].tick_params(axis='x', rotation=45)
234     else:
235         axes[1,1].set_visible(False)
236
237     plt.tight_layout()
238     plt.savefig('algorithm_comparison.png', dpi=300, bbox_inches='tight')
239     plt.show()
240
241 # Test functions using the enhanced class
242 def test_enhanced_visualizations():
243     viz = EnhancedQuantumVisualizer()
244     environments = ["none", "random", "markov", "adaptive"]
245
246     for env in environments:
247         print(f"\n{' '*50}\nTESTING {env.upper()}\n{' '*50}")
248         viz.run_test(env)
249         viz.plot_results(env) # Inherited from base class
250
251     print("\n" + "="*60 + "\nCREATING ENHANCED VISUALIZATIONS\n" + "="*60)
252     viz.plot_performance_dashboard(environments)
253     viz.plot_algorithm_comparison(environments)
254     viz.compare_environments(environments) # Inherited from base class
255     return viz
256
257 if __name__ == "__main__":
258     print("TESTING ENHANCED QUANTUM VISUALIZER:")

```



TESTING ENHANCED QUANTUM VISUALIZER:

=====

TESTING NONE

=====

Running test for none environment...

=====

QUANTUM MULTIRUN EVALUATOR INITIALIZED

=====

Configuration:

- Base frames: 4,000
- Frame step: 2,000
- Base seed: 12345
- Attack strategy: NONE
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis

=====

=====

STARTING QUANTUM MULTIRUN EVALUATION

=====

Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000

=====

PROGRESS: Experiment 1/3

=====

EXPERIMENT 1

=====

Frame Count: 4,000
Attack: none (intensity=1.0)

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	4,000	

=====

ADVERSARIAL CONFIGURATION:

=====

Attack Strategy:	NoAttack	
------------------	----------	--

=====

Environment: none (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 4000 steps, final reward: 3307.29

COMPLETED: in 0.00s, final reward: 3307.29

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0	
---------------	---	--

```
Optimal Path: 0
Optimal Action: 4
Path Performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

```
- NEURAL Progress: 100%|██████████| 4000/4000 [00:13<00:00, 300.33it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

ORACLE ANALYSIS:

```
Optimal Path: 0
Optimal Action: 4
Path Performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

```
- EXP3 Progress: 100%|██████████| 4000/4000 [00:02<00:00, 1477.43it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
Optimal Path: 0
Optimal Action: 4
Path Performance: [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear

```

| Parameters | gamma=0.010, eta=0.050 | EXP3 Config |
|            | beta=0.2                | Neural UCB  |
=====
- HYBRID Progress: 100%|██████████| 4000/4000 [00:14<00:00, 267.67it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully

```

RESULTS (completed in 31.06s):

```

=====
Oracle      (step-wise) : 3307.29
GNeuralUCB  (batch)     : 3235.40
EXPUCB      (batch)     : 2794.69
EXPNeuralUCB (batch)    : 2854.95
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 86.3%

```

ATTACK PATTERN ANALYSIS:

```

=====
| Strategy:      | NoAttack          |
| Total Frames:  | 4,000             |
| Total Attacks: | 0                 |
| Attack Intensity: | 0.000           |
=====
| PATH-SPECIFIC ANALYSIS: |
=====
| Path 0:        | 0 attacks ( 0.0% rate) |
| Path 1:        | 0 attacks ( 0.0% rate) |
| Path 2:        | 0 attacks ( 0.0% rate) |
| Path 3:        | 0 attacks ( 0.0% rate) |
=====
ATTACKS: Mean attacks/frame: 0.000
=====

```

EXPERIMENT 1 COMPLETED

Execution Time: 31.1s

Results:

```

Oracle:      3307.29 (theoretical maximum)
EXPNeuralUCB: 2854.95 ( 13.7% gap)
GNeuralUCB:  3235.40 (  2.2% gap) [WINNER]
EXPUCB:      2794.69 ( 15.5% gap)
Winner: GNeuralUCB

```

PROGRESS: Experiment 2/3

=====

EXPERIMENT 2

=====

Frame Count: 6,000

Attack: none (intensity=1.0)

=====

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```

=====
| Network Paths:  | 4                 |
| Qubit Config:   | (8, 10, 8, 9)    |
| Frame Length:   | 6,000             |
=====

```

ADVERSARIAL CONFIGURATION:

```

=====
| Attack Strategy: | NoAttack          |

```

```
Environment: none (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 6000 steps, final reward: 5676.68
COMPLETED: in 0.01s, final reward: 5676.68
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...
```

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial
Action Selection	Neural UCB	Robust
Parameters	beta=0.2	Nonlinear
		Learning
		UCB Confidence

```
- NEURAL Progress: 100%[██████████] 6000/6000 [00:22<00:00, 269.66it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...
```

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially
Action Selection	Linear UCB	Robust
Parameters	gamma=0.100, eta=0.005	Linear
	beta=0.2	Learning
		EXP3 Config
		Linear UCB

```
- EXP3 Progress: 100%[██████████] 6000/6000 [00:05<00:00, 1050.10it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4

| Path Performance: | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

- HYBRID Progress: 100%|██████████| 6000/6000 [00:17<00:00, 336.93it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 45.87s):

Oracle (step-wise) : 5676.68
GNeuralUCB (batch) : 5179.98
EXPUCB (batch) : 4552.40
EXPNeuralUCB (batch) : 5279.10
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 93.0%

ATTACK PATTERN ANALYSIS:

Strategy:	NoAttack
Total Frames:	6,000
Total Attacks:	0
Attack Intensity:	0.000

PATH-SPECIFIC ANALYSIS:

Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)

ATTACKS: Mean attacks/frame: 0.000

EXPERIMENT 2 COMPLETED

Execution Time: 45.9s

Results:

Oracle: 5676.68 (theoretical maximum)
EXPNeuralUCB: 5279.10 (7.0% gap) [WINNER]
GNeuralUCB: 5179.98 (8.7% gap)
EXPUCB: 4552.40 (19.8% gap)
Winner: EXPNeuralUCB

PROGRESS: Experiment 3/3

EXPERIMENT 3

Frame Count: 8,000
Attack: none (intensity=1.0)

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	8,000

ADVERSARIAL CONFIGURATION:

Attack Strategy:	NoAttack
------------------	----------

Environment: none (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 8000 steps, final reward: 7868.91
COMPLETED: in 0.00s, final reward: 7868.91
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%[██████████] 8000/8000 [00:53<00:00, 148.29it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust

Action Selection	Linear UCB	Linear
Parameters	gamma=0.100, eta=0.005	EXP3 Config
	beta=0.2	Linear UCB

```
=====
- EXP3 Progress: 100%|██████████| 8000/8000 [00:07<00:00, 1133.72it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050	EXP3 Config
	beta=0.2	Neural UCB

```
=====
- HYBRID Progress: 100%|██████████| 8000/8000 [00:23<00:00, 344.68it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

RESULTS (completed in 84.30s):

```
=====
Oracle      (step-wise) : 7868.91
GNeuralUCB  (batch)     : 7266.99
EXPUCB      (batch)     : 6307.33
EXPNeuralUCB (batch)    : 7520.98
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 95.6%
```

ATTACK PATTERN ANALYSIS:

Strategy:	NoAttack
Total Frames:	8,000
Total Attacks:	0
Attack Intensity:	0.000

PATH-SPECIFIC ANALYSIS:

Path 0:	0 attacks (0.0% rate)
Path 1:	0 attacks (0.0% rate)
Path 2:	0 attacks (0.0% rate)
Path 3:	0 attacks (0.0% rate)

```
=====
ATTACKS: Mean attacks/frame: 0.000
=====
```

EXPERIMENT 3 COMPLETED

Execution Time: 84.3s

Results:

```
Oracle: 7868.91 (theoretical maximum)
EXPNeuralUCB: 7520.98 ( 4.4% gap) [WINNER]
```

GNeuralUCB: 7266.99 (7.6% gap)
EXPUCB: 6307.33 (19.8% gap)
Winner: EXPNeuralUCB

QUANTUM EVALUATION COMPLETED!

Total experiments: 3
Total time: 162.3s
Average time per experiment: 54.1s

DETAILED EXPERIMENT ANALYSIS

Attack Configuration:
Strategy: NONE
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 31.1s)

|
+-- Oracle: 3307.29 (theoretical maximum)
+-- EXPNeuralUCB: 2854.95 (13.7% gap)
+-- GNeuralUCB: 3235.40 (2.2% gap)
+-- EXPUCB: 2794.69 (15.5% gap)
Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 45.9s)

|
+-- Oracle: 5676.68 (theoretical maximum)
+-- EXPNeuralUCB: 5279.10 (7.0% gap)
+-- GNeuralUCB: 5179.98 (8.7% gap)
+-- EXPUCB: 4552.40 (19.8% gap)
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 84.3s)

|
+-- Oracle: 7868.91 (theoretical maximum)
+-- EXPNeuralUCB: 7520.98 (4.4% gap)
+-- GNeuralUCB: 7266.99 (7.6% gap)
+-- EXPUCB: 6307.33 (19.8% gap)
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

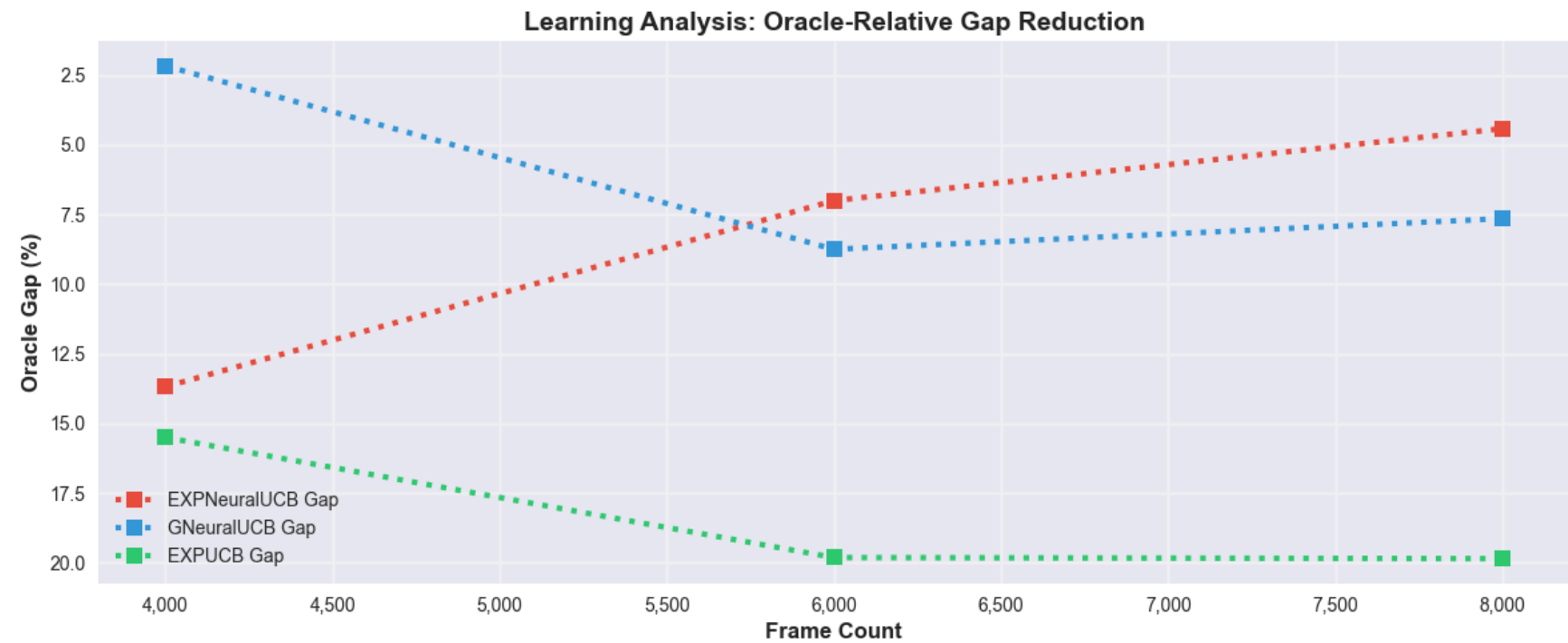
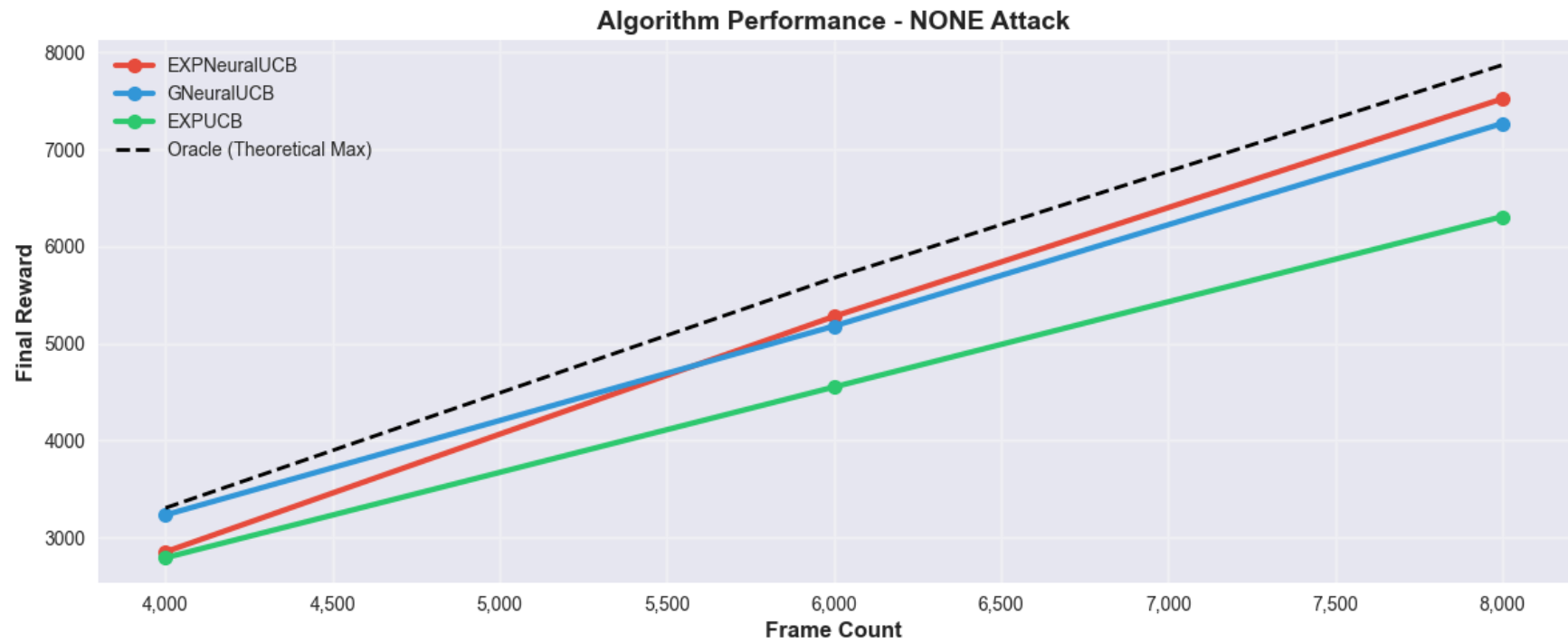
LEARNING PERFORMANCE (Gap Reduction to Oracle):

#1: EXPNeuralUCB -> +9.3pp improvement (gap closing)
Gap: 13.7% -> 4.4%
#2: EXPUCB -> +4.3pp decline (gap widening)
Gap: 15.5% -> 19.8%
#3: GNeuralUCB -> +5.5pp decline (gap widening)
Gap: 2.2% -> 7.6%

Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

Generating performance visualization...

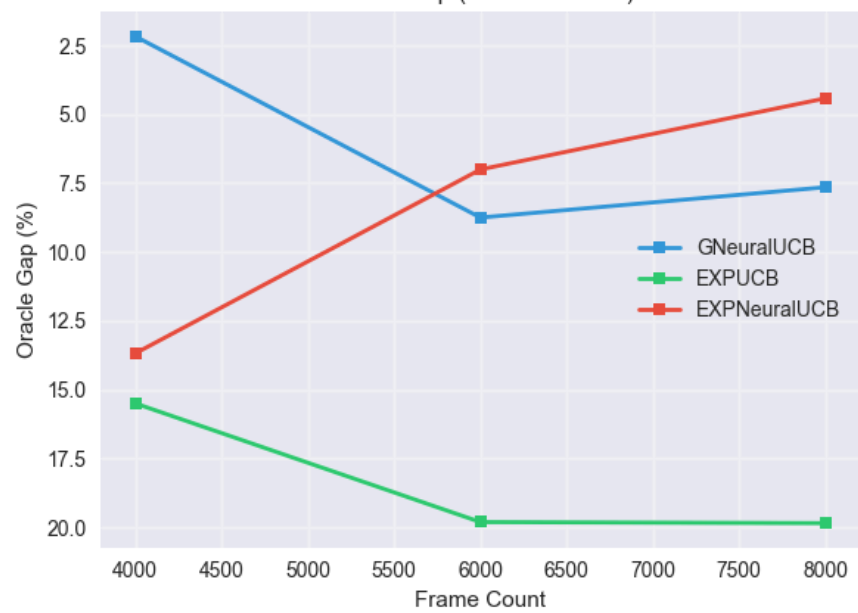
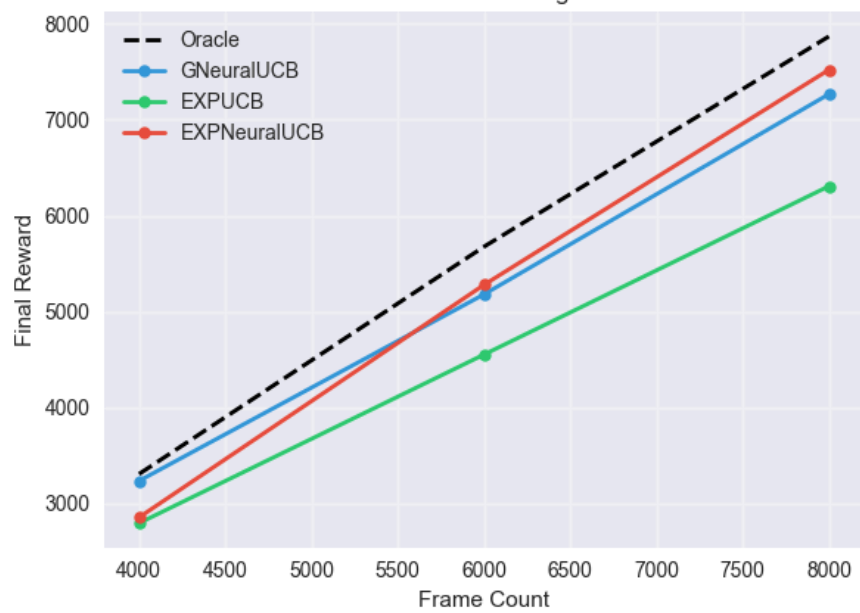


Visualization complete!

None Environment Results

Reward Scaling

Oracle Gap (Lower = Better)



```
=====
TESTING RANDOM
=====
Running test for random environment...
```

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
Configuration:
- Base frames:      4,000
- Frame step:       2,000
- Base seed:        12345
- Attack strategy:  RANDOM
- Attack intensity: 1.0
- Progress bars:    Disabled
- Analysis method:  Oracle-relative gap analysis
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000
=====
```

PROGRESS: Experiment 1/3

```
=====
EXPERIMENT 1
=====
Frame Count: 4,000
Attack: random (intensity=1.0)
=====
```

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

```
=====
QUANTUM ENVIRONMENT INITIALIZED:
=====
| Network Paths:      | 4          |
| Qubit Config:       | (8, 10, 8, 9) |
| Frame Length:       | 4,000       |
=====
```

ADVERSARIAL CONFIGURATION:

```
=====
| Attack Strategy:    | RandomAttack  |
| Attack Rate:        | 0.25          |
=====
```

```
Environment: random (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 4000 steps, final reward: 3192.62
COMPLETED: in 0.00s, final reward: 3192.62
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:       | 0          |
| Optimal Action:     | 4          |
| Path Performance:   | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB      | NOT Adversarial    |
|                   |                 | Robust              |
| Action Selection   | Neural UCB      | Nonlinear           |
|                   |                 | Learning             |
| Parameters         | beta=0.2        | UCB Confidence      |
=====
```

```
- NEURAL Progress: 100%[██████████] 4000/4000 [00:06<00:00, 576.69it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:       | 0          |
| Optimal Action:     | 4          |
| Path Performance:   | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                   |                 | Robust              |
| Action Selection   | Linear UCB      | Linear              |
|                   |                 | Learning             |
=====
```

```
Parameters | gamma=0.100, eta=0.005 | EXP3 Config |
beta=0.2 | Linear UCB |
=====
- EXP3 Progress: 100%|██████████| 4000/4000 [00:01<00:00, 2188.33it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...

ORACLE ANALYSIS:
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

```
=====
- HYBRID Progress: 100%|██████████| 4000/4000 [00:09<00:00, 433.70it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

RESULTS (completed in 18.03s):

```
=====
Oracle      (step-wise) : 3192.62
GNeuralUCB  (batch)    : 2420.84
EXPUCB      (batch)    : 2024.79
EXPNeuralUCB (batch)   : 2097.30
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 65.7%
```

ATTACK PATTERN ANALYSIS:

```
=====
| Strategy: | RandomAttack |
| Total Frames: | 4,000 |
| Total Attacks: | 4,024 |
| Attack Intensity: | 0.252 |
=====
```

PATH-SPECIFIC ANALYSIS:

```
=====
| Path 0: | 986 attacks (24.6% rate) |
| Path 1: | 1041 attacks (26.0% rate) |
| Path 2: | 1003 attacks (25.1% rate) |
| Path 3: | 994 attacks (24.9% rate) |
=====
```

ATTACKS: Mean attacks/frame: 1.006

EXPERIMENT 1 COMPLETED

Execution Time: 18.0s

Results:

```
Oracle: 3192.62 (theoretical maximum)
EXPNeuralUCB: 2097.30 ( 34.3% gap)
GNeuralUCB: 2420.84 ( 24.2% gap) [WINNER]
EXPUCB: 2024.79 ( 36.6% gap)
```

PROGRESS: Experiment 2/3

EXPERIMENT 2

Frame Count: 6,000
Attack: random (intensity=1.0)

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000

ADVERSARIAL CONFIGURATION:

Attack Strategy:	RandomAttack
Attack Rate:	0.25

Environment: random (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 6000 steps, final reward: 5567.05
COMPLETED: in 0.00s, final reward: 5567.05
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100%[██████████] 6000/6000 [00:10<00:00, 549.82it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4

| Path Performance: | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

- EXP3 Progress: 100%|██████████| 6000/6000 [00:04<00:00, 1471.45it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

- HYBRID Progress: 100%|██████████| 6000/6000 [00:14<00:00, 413.28it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 29.56s):

Oracle (step-wise) : 5567.05
GNeuralUCB (batch) : 3840.18
EXPUCB (batch) : 3310.90
EXPNeuralUCB (batch) : 3910.27
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 70.2%

ATTACK PATTERN ANALYSIS:

Strategy:	RandomAttack
Total Frames:	6,000
Total Attacks:	6,001
Attack Intensity:	0.250

PATH-SPECIFIC ANALYSIS:

Path 0:	1490 attacks (24.8% rate)
---------	---------------------------

Path 1:	1511 attacks (25.2% rate)
Path 2:	1476 attacks (24.6% rate)
Path 3:	1524 attacks (25.4% rate)

=====

ATTACKS: Mean attacks/frame: 1.000

=====

EXPERIMENT 2 COMPLETED

Execution Time: 29.6s

Results:

Oracle: 5567.05 (theoretical maximum)
EXPNeuralUCB: 3910.27 (29.8% gap) [WINNER]
GNeuralUCB: 3840.18 (31.0% gap)
EXPUCB: 3310.90 (40.5% gap)
Winner: EXPNeuralUCB

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: random (intensity=1.0)

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	8,000	

=====

ADVERSARIAL CONFIGURATION:

=====

Attack Strategy:	RandomAttack	
Attack Rate:	0.25	

=====

Environment: random (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 8000 steps, final reward: 7777.26

COMPLETED: in 0.01s, final reward: 7777.26

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

=====

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]	

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties	
-----	-----	-----	

Group Selection	Simple UCB	NOT Adversarial
Action Selection	Neural UCB	Robust
Parameters	beta=0.2	Nonlinear
		Learning
		UCB Confidence

```

=====
- NEURAL Progress: 100%|██████████| 8000/8000 [00:50<00:00, 157.45it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...

```

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially
Action Selection	Linear UCB	Robust
Parameters	gamma=0.100, eta=0.005	Linear
	beta=0.2	Learning
		EXP3 Config
		Linear UCB

```

=====
- EXP3 Progress: 100%|██████████| 8000/8000 [00:07<00:00, 1117.11it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...

```

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially
Action Selection	Neural UCB	Robust
Parameters	gamma=0.010, eta=0.050	Nonlinear
	beta=0.2	Learning
		EXP3 Config
		Neural UCB

```

=====
- HYBRID Progress: 100%|██████████| 8000/8000 [00:21<00:00, 378.38it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully

```

RESULTS (completed in 79.18s):

```

=====
Oracle      (step-wise) : 7777.26
GNeuralUCB  (batch)    : 5417.64
EXPUCB      (batch)    : 4675.52

```

EXPNeuralUCB (batch) : 5584.49
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 71.8%

ATTACK PATTERN ANALYSIS:

Strategy:	RandomAttack	
Total Frames:	8,000	
Total Attacks:	8,063	
Attack Intensity:	0.252	
PATH-SPECIFIC ANALYSIS:		
Path 0:	2018 attacks (25.2% rate)	
Path 1:	2041 attacks (25.5% rate)	
Path 2:	1996 attacks (24.9% rate)	
Path 3:	2008 attacks (25.1% rate)	
ATTACKS: Mean attacks/frame: 1.008		

EXPERIMENT 3 COMPLETED
Execution Time: 79.2s
Results:
Oracle: 7777.26 (theoretical maximum)
EXPNeuralUCB: 5584.49 (28.2% gap) [WINNER]
GNeuralUCB: 5417.64 (30.3% gap)
EXPUCB: 4675.52 (39.9% gap)
Winner: EXPNeuralUCB

QUANTUM EVALUATION COMPLETED!

Total experiments: 3
Total time: 127.5s
Average time per experiment: 42.5s

DETAILED EXPERIMENT ANALYSIS

Attack Configuration:
Strategy: RANDOM
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 18.0s)
+-- Oracle: 3192.62 (theoretical maximum)
+-- EXPNeuralUCB: 2097.30 (34.3% gap)
+-- GNeuralUCB: 2420.84 (24.2% gap)
+-- EXPUCB: 2024.79 (36.6% gap)
Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 29.6s)
+-- Oracle: 5567.05 (theoretical maximum)
+-- EXPNeuralUCB: 3910.27 (29.8% gap)
+-- GNeuralUCB: 3840.18 (31.0% gap)
+-- EXPUCB: 3310.90 (40.5% gap)

EXPNeuralUCB: 5584.49 (28.2% gap)
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 79.2s)

|
+-- Oracle: 7777.26 (theoretical maximum)
+-- EXPNeuralUCB: 5584.49 (28.2% gap)
+-- GNeuralUCB: 5417.64 (30.3% gap)
+-- EXPUCB: 4675.52 (39.9% gap)
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

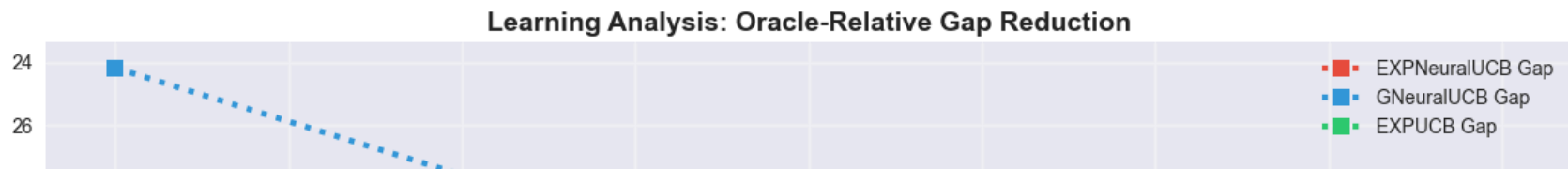
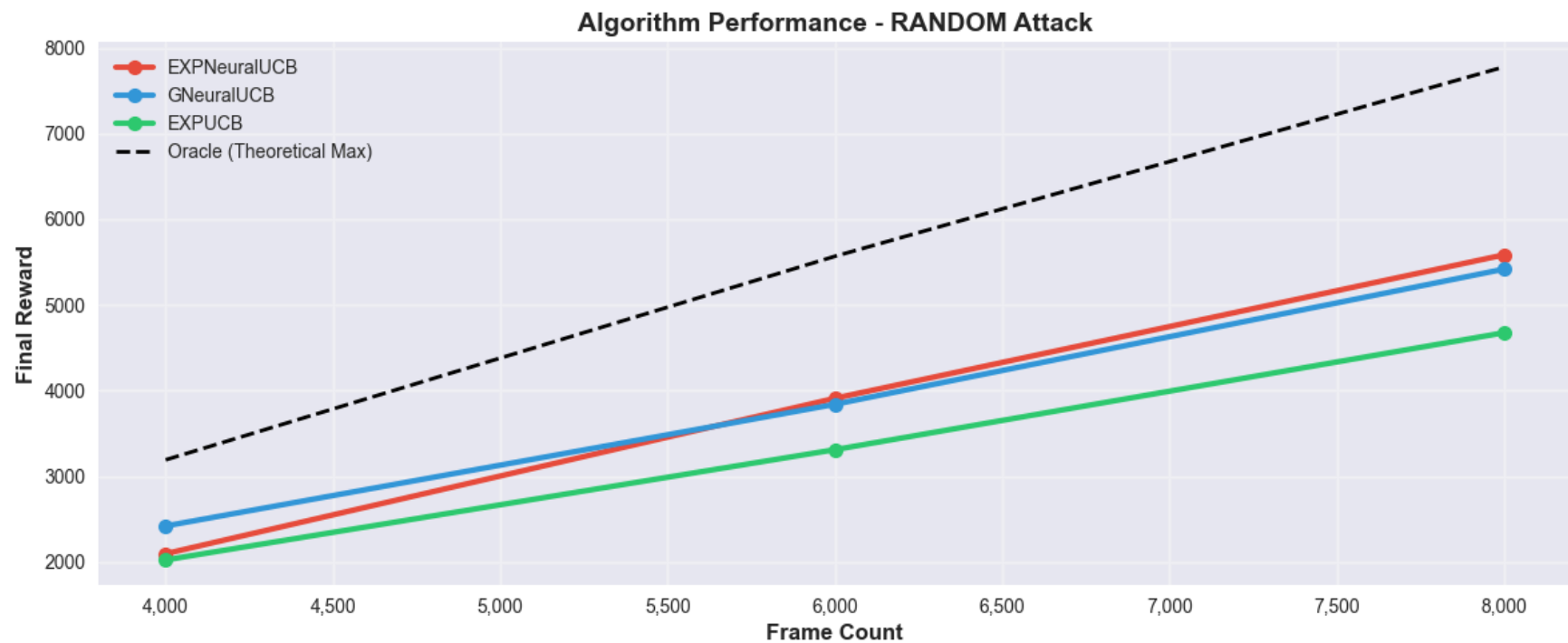
LEARNING PERFORMANCE (Gap Reduction to Oracle):

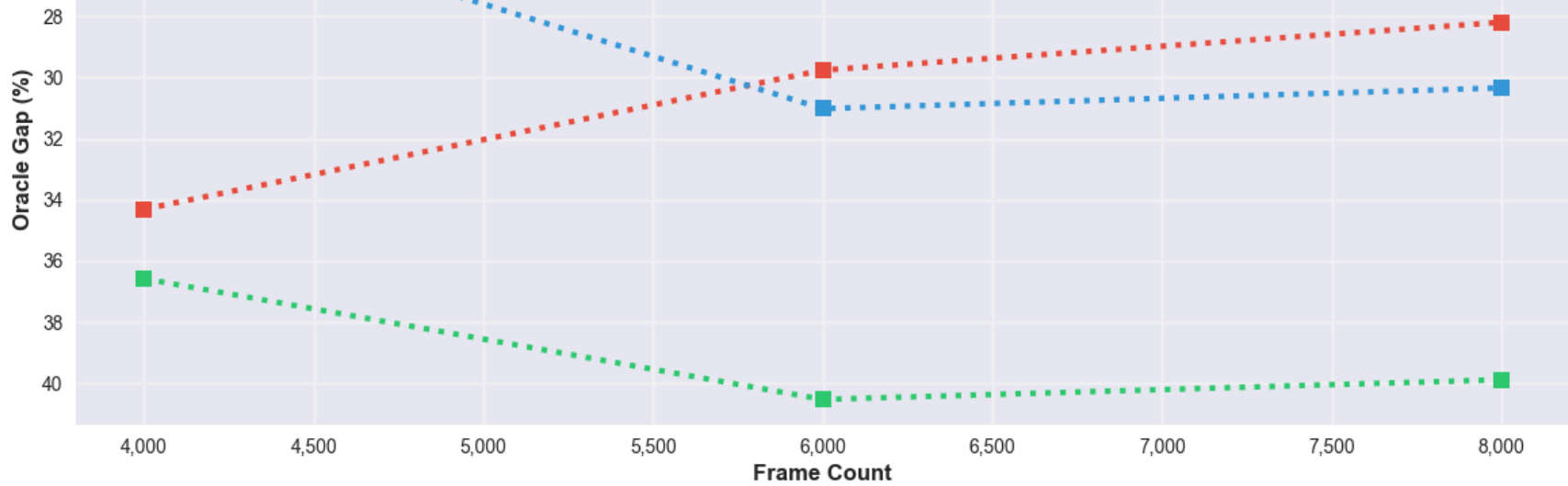
#1: EXPNeuralUCB -> +6.1pp improvement (gap closing)
Gap: 34.3% -> 28.2%
#2: EXPUCB -> +3.3pp decline (gap widening)
Gap: 36.6% -> 39.9%
#3: GNeuralUCB -> +6.2pp decline (gap widening)
Gap: 24.2% -> 30.3%

Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

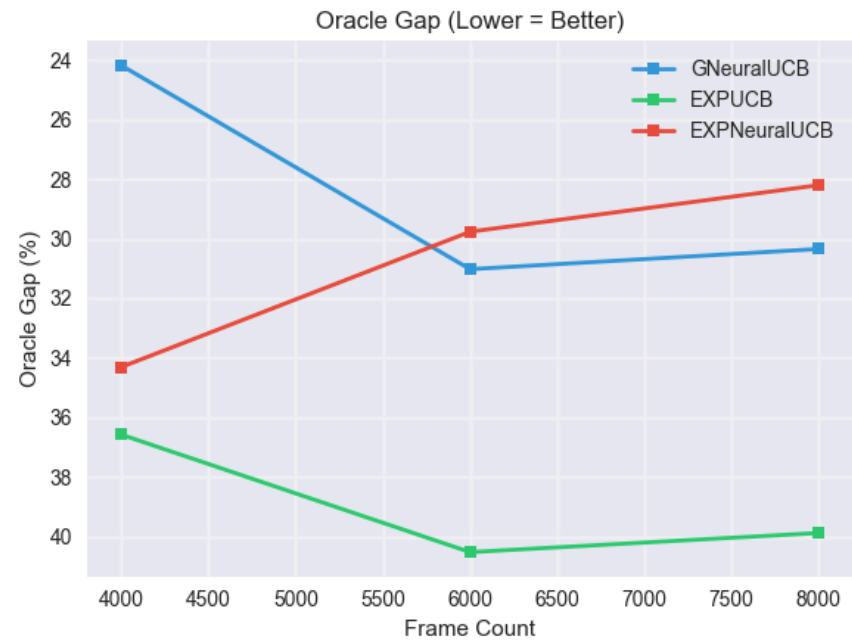
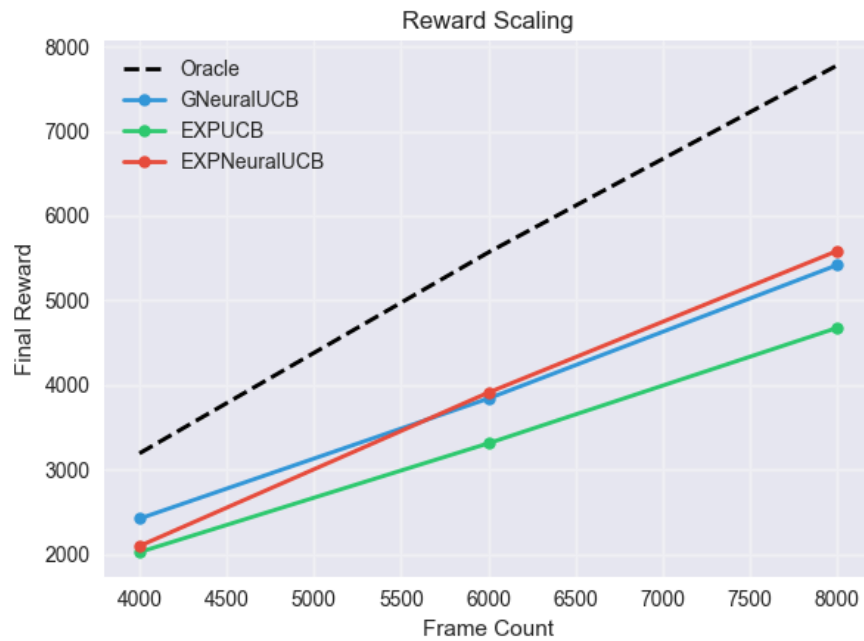
Generating performance visualization...





Visualization complete!

Random Environment Results



=====

TESTING MARKOV

=====

Running test for markov environment...

=====

QUANTUM MULTIRUN EVALUATOR INITIALIZED

=====

Configuration:

- Base frames: 4,000

- Frame step: 2,000

- Base seed: 12345

- Environment: MARKOV

```
- Attack strategy: MARKOV
- Attack intensity: 1.0
- Progress bars: Disabled
- Analysis method: Oracle-relative gap analysis
```

STARTING QUANTUM MULTIRUN EVALUATION

Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000

PROGRESS: Experiment 1/3

EXPERIMENT 1

Frame Count: 4,000
Attack: markov (intensity=1.0)

EXPERIMENT 1: 4000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	4,000

ADVERSARIAL CONFIGURATION:

Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2

Environment: markov (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 4000 steps, final reward: 3051.08
COMPLETED: in 0.00s, final reward: 3051.08
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial

```
| Action Selection | Neural UCB | Nonlinear |
| Parameters      | beta=0.2   | UCB Confidence |
=====
- NEURAL Progress: 100%|██████████| 4000/4000 [00:04<00:00, 822.58it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0 |
| Optimal Action:  | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component      | Method          | Properties          |
|-----|-----|-----|
| Group Selection | EXP3            | Adversarially      |
| Action Selection | Linear UCB      | Linear             |
| Parameters      | gamma=0.100, eta=0.005 | EXP3 Config        |
|                 | beta=0.2        | Linear UCB         |
=====
```

```
- EXP3 Progress: 100%|██████████| 4000/4000 [00:01<00:00, 2172.04it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:    | 0 |
| Optimal Action:  | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

```
=====
| Component      | Method          | Properties          |
|-----|-----|-----|
| Group Selection | EXP3            | Adversarially      |
| Action Selection | Neural UCB      | Nonlinear           |
| Parameters      | gamma=0.010, eta=0.050 | EXP3 Config        |
|                 | beta=0.2        | Neural UCB         |
=====
```

```
- HYBRID Progress: 100%|██████████| 4000/4000 [00:08<00:00, 458.70it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

RESULTS (completed in 15.55s):

```
=====
Oracle      (step-wise) : 3051.08
GNeuralUCB  (batch)      : 1137.52
EXPUCB      (batch)      : 1453.00
EXPNeuralUCB (batch)     : 1795.46
```

WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 58.8%

ATTACK PATTERN ANALYSIS:

Strategy:	MarkovAttack	
Total Frames:	4,000	
Total Attacks:	8,000	
Attack Intensity:	0.500	

PATH-SPECIFIC ANALYSIS:	
Path 0:	2543 attacks (63.6% rate)
Path 1:	1474 attacks (36.9% rate)
Path 2:	2482 attacks (62.1% rate)
Path 3:	1501 attacks (37.5% rate)

ATTACKS: Mean attacks/frame: 2.000

EXPERIMENT 1 COMPLETED

Execution Time: 15.5s

Results:

Oracle: 3051.08 (theoretical maximum)
EXPNeuralUCB: 1795.46 (41.2% gap) [WINNER]
GNeuralUCB: 1137.52 (62.7% gap)
EXPUCB: 1453.00 (52.4% gap)
Winner: EXPNeuralUCB

PROGRESS: Experiment 2/3

EXPERIMENT 2

Frame Count: 6,000

Attack: markov (intensity=1.0)

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	6,000

ADVERSARIAL CONFIGURATION:

Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2

Environment: markov (intensity=1.0)

Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']

Running Oracle...

Context strategy: context-free

take_action params: 0, update params: 3

COMPLETED: 6000 steps, final reward: 5474.92

COMPLETED: in 0.00s, final reward: 5474.92

COMPLETED: Oracle (step-wise) completed successfully

Running GNeuralUCB...

ORACLE ANALYSIS:

```

=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====

```

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

```

=====
| Component          | Method                | Properties          |
|-----|-----|-----|
| Group Selection    | Simple UCB            | NOT Adversarial    |
|                   |                       | Robust              |
| Action Selection   | Neural UCB            | Nonlinear           |
|                   |                       | Learning             |
| Parameters         | beta=0.2              | UCB Confidence      |
=====

```

```

- NEURAL Progress: 100%|██████████| 6000/6000 [00:10<00:00, 586.52it/s]
  COMPLETED: GNeuralUCB (batch) completed successfully
  Running EXPUCB...

```

ORACLE ANALYSIS:

```

=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====

```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```

=====
| Component          | Method                | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3                  | Adversarially      |
|                   |                       | Robust              |
| Action Selection   | Linear UCB            | Linear              |
|                   |                       | Learning             |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config         |
|                   | beta=0.2              | Linear UCB          |
=====

```

```

- EXP3 Progress: 100%|██████████| 6000/6000 [00:04<00:00, 1450.28it/s]
  COMPLETED: EXPUCB (batch) completed successfully
  Running EXPNeuralUCB...

```

ORACLE ANALYSIS:

```

=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]
=====

```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

```

=====
| Component          | Method                | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3                  | Adversarially      |

```

Action Selection	Neural UCB	Robust
		Nonlinear
Parameters	gamma=0.010, eta=0.050	Learning
	beta=0.2	EXP3 Config
		Neural UCB

=====

- HYBRID Progress: 100%|██████████| 6000/6000 [00:13<00:00, 434.65it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 28.34s):

=====

Oracle	(step-wise)	: 5474.92
GNeuralUCB	(batch)	: 3211.32
EXPUCB	(batch)	: 2604.25
EXPNeuralUCB	(batch)	: 3260.87

WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 59.6%

ATTACK PATTERN ANALYSIS:

=====

Strategy:	MarkovAttack	
Total Frames:	6,000	
Total Attacks:	12,000	
Attack Intensity:	0.500	

=====

PATH-SPECIFIC ANALYSIS:

=====

Path 0:	3798 attacks (63.3% rate)
Path 1:	2217 attacks (37.0% rate)
Path 2:	3789 attacks (63.1% rate)
Path 3:	2196 attacks (36.6% rate)

=====

ATTACKS: Mean attacks/frame: 2.000

EXPERIMENT 2 COMPLETED

Execution Time: 28.3s

Results:

Oracle:	5474.92 (theoretical maximum)
EXPNeuralUCB:	3260.87 (40.4% gap) [WINNER]
GNeuralUCB:	3211.32 (41.3% gap)
EXPUCB:	2604.25 (52.4% gap)

Winner: EXPNeuralUCB

PROGRESS: Experiment 3/3

=====

EXPERIMENT 3

=====

Frame Count: 8,000

Attack: markov (intensity=1.0)

=====

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:

=====

Network Paths:	4
Qubit Config:	(8, 10, 8, 9)
Frame Length:	8,000

=====

ADVERSARIAL CONFIGURATION:

Attack Strategy:	MarkovAttack
Attack Rate:	1.00
Simultaneous:	2

Environment: markov (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 8000 steps, final reward: 7744.38
COMPLETED: in 0.00s, final reward: 7744.38
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'neural' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	Simple UCB	NOT Adversarial Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	beta=0.2	UCB Confidence

- NEURAL Progress: 100% [██████████] 8000/8000 [00:48<00:00, 163.66it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

- EXP3 Progress: 100% [██████████] 8000/8000 [00:06<00:00, 1151.01it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

```
ORACLE ANALYSIS:
=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component      | Method          | Properties          |
|-----|-----|-----|
| Group Selection | EXP3            | Adversarially      |
|                 |                 | Robust              |
| Action Selection | Neural UCB      | Nonlinear           |
|                 |                 | Learning            |
| Parameters      | gamma=0.010, eta=0.050 | EXP3 Config        |
|                 | beta=0.2        | Neural UCB          |
=====
- HYBRID Progress: 100%[██████████] 8000/8000 [00:21<00:00, 373.39it/s]
  COMPLETED: EXPNeuralUCB (batch) completed successfully
```

```
RESULTS (completed in 77.48s):
=====
Oracle      (step-wise) : 7744.38
GNeuralUCB  (batch)      : 4552.69
EXPUCB      (batch)      : 3605.14
EXPNeuralUCB (batch)     : 4738.67
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 61.2%
```

```
ATTACK PATTERN ANALYSIS:
=====
| Strategy:      | MarkovAttack    |
| Total Frames:  | 8,000           |
| Total Attacks: | 16,000          |
| Attack Intensity: | 0.500         |
=====
| PATH-SPECIFIC ANALYSIS: |
=====
| Path 0:      | 5076 attacks (63.4% rate) |
| Path 1:      | 2935 attacks (36.7% rate) |
| Path 2:      | 5036 attacks (62.9% rate) |
| Path 3:      | 2953 attacks (36.9% rate) |
=====
ATTACKS: Mean attacks/frame: 2.000
=====
```

```
EXPERIMENT 3 COMPLETED
Execution Time: 77.5s
Results:
Oracle:      7744.38 (theoretical maximum)
EXPNeuralUCB: 4738.67 ( 38.8% gap) [WINNER]
GNeuralUCB:  4552.69 ( 41.2% gap)
EXPUCB:      3605.14 ( 53.4% gap)
Winner: EXPNeuralUCB
=====
```

```
=====
QUANTUM EVALUATION COMPLETED!
=====
Total experiments: 3
```

Total experiments: 3
Total time: 122.2s
Average time per experiment: 40.7s
=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:
Strategy: MARKOV
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 15.5s)

|
+-- Oracle: 3051.08 (theoretical maximum)
+-- EXPNeuralUCB: 1795.46 (41.2% gap)
+-- GNeuralUCB: 1137.52 (62.7% gap)
+-- EXPUCB: 1453.00 (52.4% gap)
Winner: EXPNeuralUCB

EXPERIMENT 2 (6,000 frames, 28.3s)

|
+-- Oracle: 5474.92 (theoretical maximum)
+-- EXPNeuralUCB: 3260.87 (40.4% gap)
+-- GNeuralUCB: 3211.32 (41.3% gap)
+-- EXPUCB: 2604.25 (52.4% gap)
Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 77.5s)

|
+-- Oracle: 7744.38 (theoretical maximum)
+-- EXPNeuralUCB: 4738.67 (38.8% gap)
+-- GNeuralUCB: 4552.69 (41.2% gap)
+-- EXPUCB: 3605.14 (53.4% gap)
Winner: EXPNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

LEARNING PERFORMANCE (Gap Reduction to Oracle):

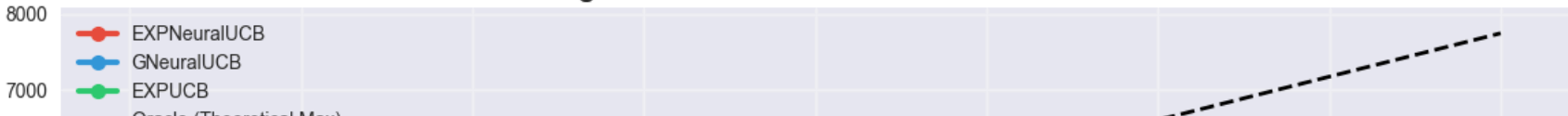
- #1: GNeuralUCB -> +21.5pp improvement (gap closing)
Gap: 62.7% -> 41.2%
- #2: EXPNeuralUCB -> +2.3pp improvement (gap closing)
Gap: 41.2% -> 38.8%
- #3: EXPUCB -> +1.1pp decline (gap widening)
Gap: 52.4% -> 53.4%

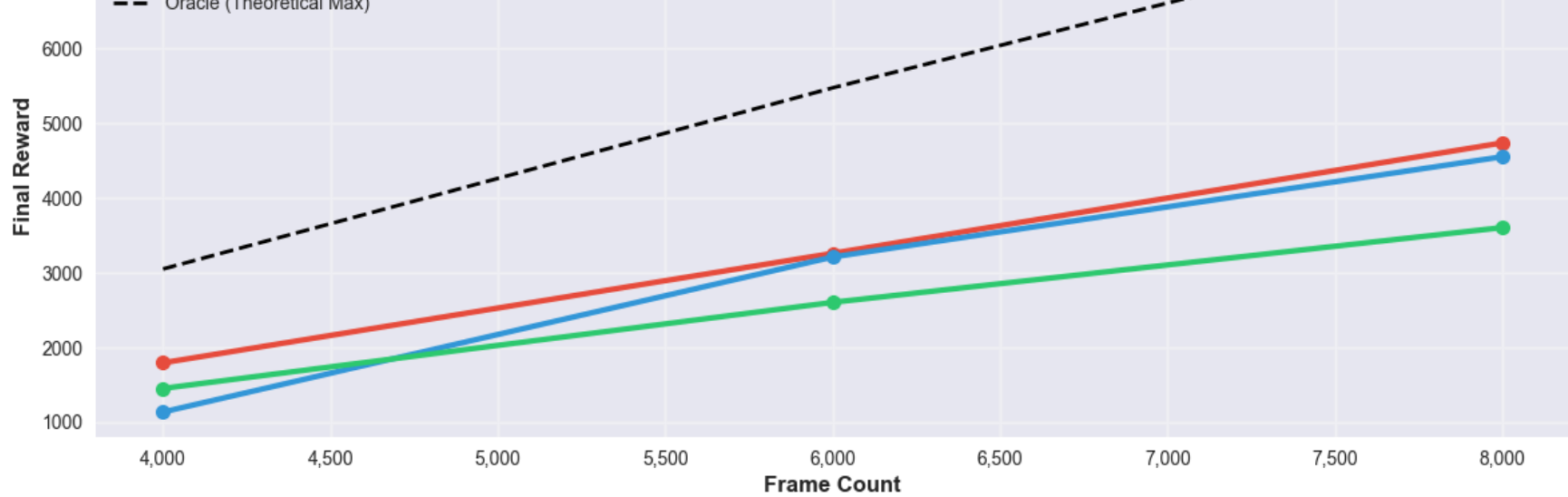
Interpretation:

- Lower gaps = Better performance relative to Oracle
 - Gap reduction = Algorithm learning and improving
 - Best learner closes the gap to Oracle over time
-

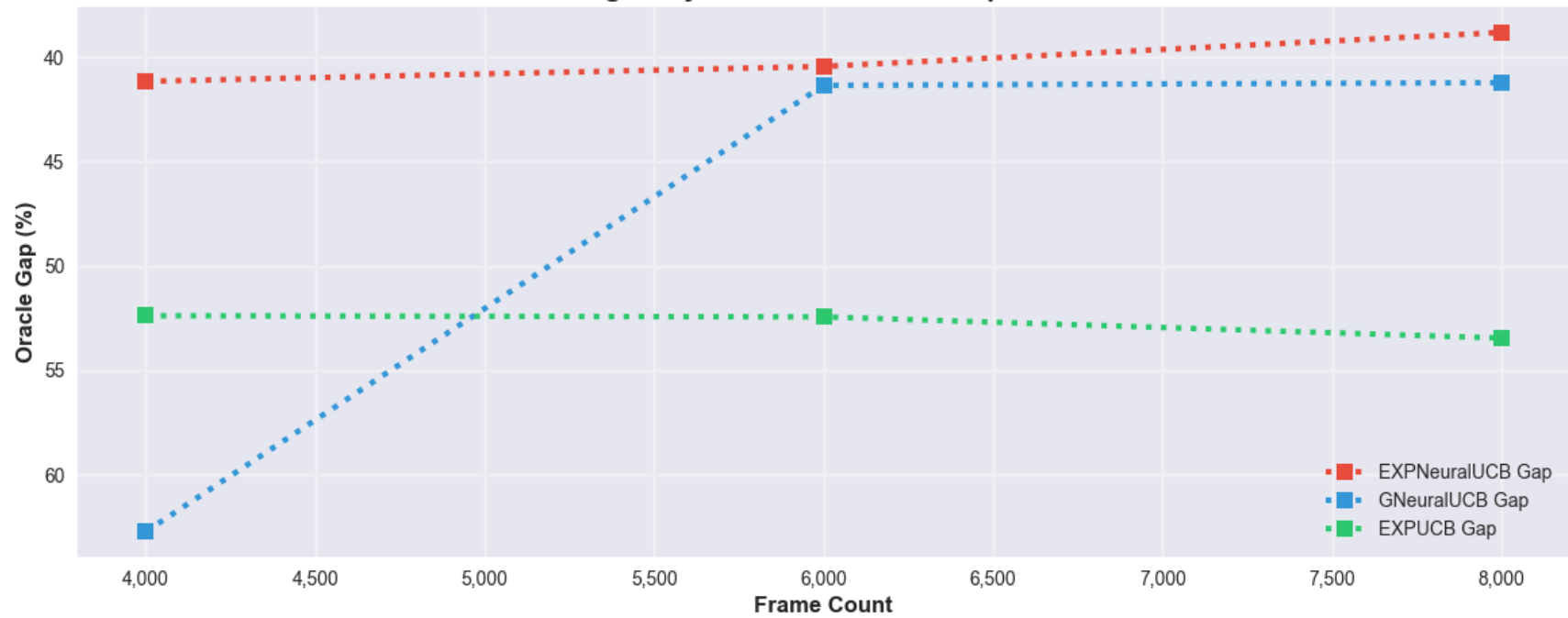
Generating performance visualization...

Algorithm Performance - MARKOV Attack



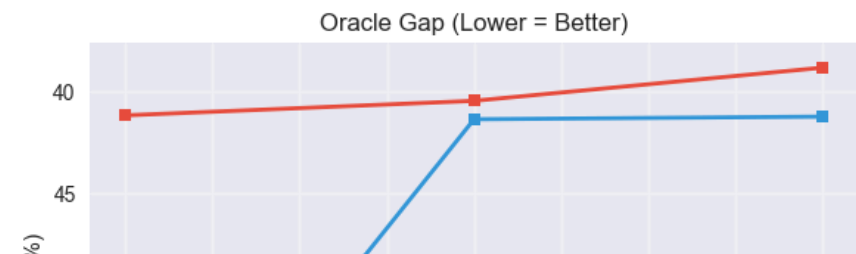


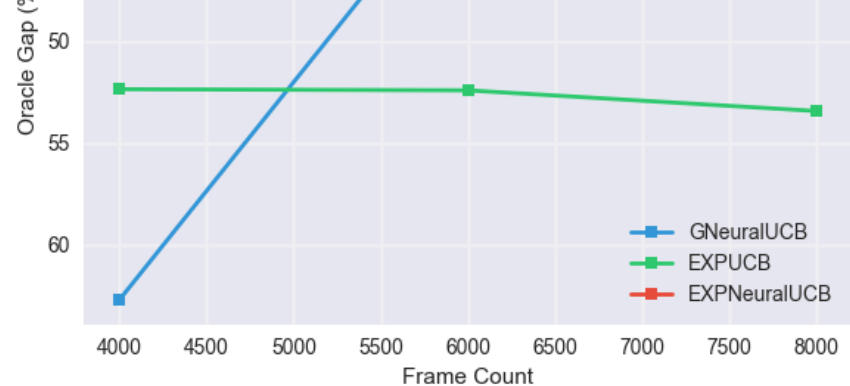
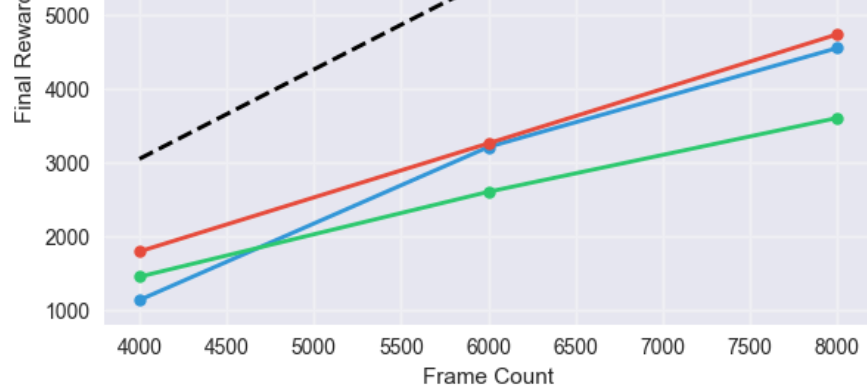
Learning Analysis: Oracle-Relative Gap Reduction



Visualization complete!

Markov Environment Results





```
=====
TESTING ADAPTIVE
=====
```

```
Running test for adaptive environment...
```

```
=====
QUANTUM MULTIRUN EVALUATOR INITIALIZED
=====
```

```
Configuration:
- Base frames:      4,000
- Frame step:       2,000
- Base seed:        12345
- Attack strategy:  ADAPTIVE
- Attack intensity: 1.0
- Progress bars:    Disabled
- Analysis method:  Oracle-relative gap analysis
=====
```

```
=====
STARTING QUANTUM MULTIRUN EVALUATION
=====
```

```
Planning 3 experiments
Frame counts: 4,000, 6,000, 8,000
=====
```

```
PROGRESS: Experiment 1/3
```

```
=====
EXPERIMENT 1
=====
```

```
Frame Count: 4,000
Attack: adaptive (intensity=1.0)
=====
```

```
EXPERIMENT 1: 4000 frames
```

```
QUANTUM ENVIRONMENT INITIALIZED:
=====
```

```
| Network Paths: | 4 |
| Qubit Config:  | (8, 10, 8, 9) |
| Frame Length:  | 4,000 |
=====
```

```
ADVERSARIAL CONFIGURATION:
=====
```

```
Attack Strategy: | AdaptiveAttack |
Attack Rate: | 1.00 |
Memory Window: | 50 |
=====
Environment: adaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 4000 steps, final reward: 3240.72
COMPLETED: in 0.00s, final reward: 3240.72
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'neural' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | Simple UCB | NOT Adversarial |
| Action Selection | Neural UCB | Robust |
| Parameters | beta=0.2 | Nonlinear |
| | | Learning |
| | | UCB Confidence |
=====
- NEURAL Progress: 100%|██████████| 4000/4000 [00:07<00:00, 562.82it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path: | 0 |
| Optimal Action: | 4 |
| Path Performance: | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)] |
=====
```

EXPNeuralUCB initialized in 'exp3' mode

```
=====
ALGORITHM CONFIGURATION
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | EXP3 | Adversarially |
| Action Selection | Linear UCB | Robust |
| Parameters | gamma=0.100, eta=0.005 | Linear |
| | beta=0.2 | Learning |
| | | EXP3 Config |
| | | Linear UCB |
=====
- EXP3 Progress: 100%|██████████| 4000/4000 [00:01<00:00, 2177.65it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...
```

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0 |
| Optimal Action:    | 4 |
| Path Performance:  | [np.float64(0.8268235367879645), np.float64(0.7476684769175549), np.float64(0.6599253551504429), np.float64(0.5816045675472745)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

=====
ALGORITHM CONFIGURATION
=====

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

=====
- HYBRID Progress: 100%|██████████| 4000/4000 [00:08<00:00, 449.15it/s]
 COMPLETED: EXPNeuralUCB (batch) completed successfully
=====

RESULTS (completed in 17.91s):

=====
Oracle (step-wise) : 3240.72
GNeuralUCB (batch) : 2542.22
EXPUCB (batch) : 2093.89
EXPNeuralUCB (batch) : 2016.44
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 62.2%
=====

ATTACK PATTERN ANALYSIS:

=====

Strategy:	AdaptiveAttack
Total Frames:	4,000
Total Attacks:	4,000
Attack Intensity:	0.250

=====

=====

PATH-SPECIFIC ANALYSIS:	
Path 0:	841 attacks (21.0% rate)
Path 1:	1159 attacks (29.0% rate)
Path 2:	896 attacks (22.4% rate)
Path 3:	1104 attacks (27.6% rate)

=====

ATTACKS: Mean attacks/frame: 1.000
=====

EXPERIMENT 1 COMPLETED
Execution Time: 17.9s
Results:
Oracle: 3240.72 (theoretical maximum)
EXPNeuralUCB: 2016.44 (37.8% gap)
GNeuralUCB: 2542.22 (21.6% gap) [WINNER]
EXPUCB: 2093.89 (35.4% gap)
Winner: GNeuralUCB
=====

PROGRESS: Experiment 2/3

=====
EXPERIMENT 2
=====

Frame Count: 6,000
Attack: adaptive (intensity=1.0)
=====

EXPERIMENT 2: 6000 frames

QUANTUM ENVIRONMENT INITIALIZED:
=====

Network Paths:	4	
Qubit Config:	(8, 10, 8, 9)	
Frame Length:	6,000	

=====

ADVERSARIAL CONFIGURATION:
=====

Attack Strategy:	AdaptiveAttack	
Attack Rate:	1.00	
Memory Window:	50	

=====

Environment: adaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
Running Oracle...
Context strategy: context-free
take_action params: 0, update params: 3
COMPLETED: 6000 steps, final reward: 5584.16
COMPLETED: in 0.00s, final reward: 5584.16
COMPLETED: Oracle (step-wise) completed successfully
Running GNeuralUCB...

ORACLE ANALYSIS:
=====

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]	

=====

EXPNeuralUCB initialized in 'neural' mode

=====

ALGORITHM CONFIGURATION

=====

Component	Method	Properties	
Group Selection	Simple UCB	NOT Adversarial	
Action Selection	Neural UCB	Robust	
Parameters	beta=0.2	Nonlinear	
		Learning	
		UCB Confidence	

=====

- NEURAL Progress: 100%|██████████| 6000/6000 [00:11<00:00, 537.91it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:
=====

Optimal Path:	0	
Optimal Action:	4	
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]	

=====

EXPNeuralUCB initialized in 'exp3' mode

=====

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Linear UCB	Linear Learning
Parameters	gamma=0.100, eta=0.005 beta=0.2	EXP3 Config Linear UCB

- EXP3 Progress: 100% | 6000/6000 [00:04<00:00, 1477.07it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

Optimal Path:	0
Optimal Action:	4
Path Performance:	[np.float64(0.9461134921196183), np.float64(0.902918807873247), np.float64(0.8603089615724164), np.float64(0.8116651900474303)]

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

Component	Method	Properties
Group Selection	EXP3	Adversarially Robust
Action Selection	Neural UCB	Nonlinear Learning
Parameters	gamma=0.010, eta=0.050 beta=0.2	EXP3 Config Neural UCB

- HYBRID Progress: 100% | 6000/6000 [00:14<00:00, 404.77it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 30.11s):

Oracle (step-wise) : 5584.16
GNeuralUCB (batch) : 3903.85
EXPUCB (batch) : 3318.91
EXPNeuralUCB (batch) : 3980.39
WINNER: EXPNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 71.3%

ATTACK PATTERN ANALYSIS:

Strategy:	AdaptiveAttack
Total Frames:	6,000
Total Attacks:	6,000
Attack Intensity:	0.250

PATH-SPECIFIC ANALYSIS:

Path 0:	2142 attacks (35.7% rate)
Path 1:	1439 attacks (24.0% rate)
Path 2:	947 attacks (15.8% rate)
Path 3:	1472 attacks (24.5% rate)

ATTACKS: Mean attacks/frame: 1.000

```
EXPERIMENT 2 COMPLETED
Execution Time: 30.1s
Results:
  Oracle:      5584.16 (theoretical maximum)
  EXPNeuralUCB: 3980.39 ( 28.7% gap) [WINNER]
  GNeuralUCB:  3903.85 ( 30.1% gap)
  EXPUCB:      3318.91 ( 40.6% gap)
  Winner: EXPNeuralUCB

-----

PROGRESS: Experiment 3/3

=====
EXPERIMENT 3
=====
Frame Count: 8,000
Attack: adaptive (intensity=1.0)
-----

EXPERIMENT 3: 8000 frames

QUANTUM ENVIRONMENT INITIALIZED:
=====
| Network Paths:  | 4 |
| Qubit Config:  | (8, 10, 8, 9) |
| Frame Length:  | 8,000 |
=====

ADVERSARIAL CONFIGURATION:
=====
| Attack Strategy: | AdaptiveAttack |
| Attack Rate:     | 1.00 |
| Memory Window:   | 50 |
=====
Environment: adaptive (intensity=1.0)
Running algorithms: ['Oracle', 'GNeuralUCB', 'EXPUCB', 'EXPNeuralUCB']
  Running Oracle...
    Context strategy: context-free
    take_action params: 0, update params: 3
    COMPLETED: 8000 steps, final reward: 7832.21
    COMPLETED: in 0.00s, final reward: 7832.21
    COMPLETED: Oracle (step-wise) completed successfully
  Running GNeuralUCB...

ORACLE ANALYSIS:
=====
| Optimal Path:    | 0 |
| Optimal Action:  | 4 |
| Path Performance: | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)] |
=====

EXPNeuralUCB initialized in 'neural' mode

=====
ALGORITHM CONFIGURATION
=====
| Component | Method | Properties |
|-----|-----|-----|
| Group Selection | Simple UCB | NOT Adversarial |
| Action Selection | Neural UCB | Robust |
| Parameters | beta=0.2 | Nonlinear |
| | | Learning |
| | UCB Confidence |
```

- NEURAL Progress: 100%[██████████] 8000/8000 [00:51<00:00, 156.64it/s]
COMPLETED: GNeuralUCB (batch) completed successfully
Running EXPUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'exp3' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                   |                 | Robust             |
| Action Selection   | Linear UCB      | Linear             |
|                   |                 | Learning           |
| Parameters         | gamma=0.100, eta=0.005 | EXP3 Config      |
|                   | beta=0.2        | Linear UCB         |
=====
```

- EXP3 Progress: 100%[██████████] 8000/8000 [00:07<00:00, 1035.12it/s]
COMPLETED: EXPUCB (batch) completed successfully
Running EXPNeuralUCB...

ORACLE ANALYSIS:

```
=====
| Optimal Path:      | 0          |
| Optimal Action:    | 4          |
| Path Performance:  | [np.float64(0.9836141108302526), np.float64(0.9637113766945926), np.float64(0.9435345403529187), np.float64(0.9202691292754374)]
=====
```

EXPNeuralUCB initialized in 'hybrid' mode

ALGORITHM CONFIGURATION

```
=====
| Component          | Method          | Properties          |
|-----|-----|-----|
| Group Selection    | EXP3            | Adversarially      |
|                   |                 | Robust             |
| Action Selection   | Neural UCB      | Nonlinear          |
|                   |                 | Learning           |
| Parameters         | gamma=0.010, eta=0.050 | EXP3 Config      |
|                   | beta=0.2        | Neural UCB         |
=====
```

- HYBRID Progress: 100%[██████████] 8000/8000 [00:21<00:00, 366.98it/s]
COMPLETED: EXPNeuralUCB (batch) completed successfully

RESULTS (completed in 80.70s):

```
=====
Oracle      (step-wise) : 7832.21
GNeuralUCB  (batch)     : 5597.04
EXPUCB      (batch)     : 4744.51
EXPNeuralUCB (batch)    : 5498.66
WINNER: GNeuralUCB
EFFICIENCY: EXPNeuralUCB Oracle Efficiency: 70.2%
=====
```

ATTACK PATTERN ANALYSIS:

Strategy:	AdaptiveAttack	
Total Frames:	8,000	
Total Attacks:	8,000	
Attack Intensity:	0.250	

PATH-SPECIFIC ANALYSIS:		

Path 0:	1844 attacks (23.1% rate)	
Path 1:	2126 attacks (26.6% rate)	
Path 2:	2217 attacks (27.7% rate)	
Path 3:	1813 attacks (22.7% rate)	
=====		
ATTACKS: Mean attacks/frame: 1.000		
=====		

EXPERIMENT 3 COMPLETED
Execution Time: 80.7s
Results:
Oracle: 7832.21 (theoretical maximum)
EXPNeuralUCB: 5498.66 (29.8% gap)
GNeuralUCB: 5597.04 (28.5% gap) [WINNER]
EXPUCB: 4744.51 (39.4% gap)
Winner: GNeuralUCB

=====

QUANTUM EVALUATION COMPLETED!

=====

Total experiments: 3
Total time: 129.6s
Average time per experiment: 43.2s

=====

=====

DETAILED EXPERIMENT ANALYSIS

=====

Attack Configuration:
Strategy: ADAPTIVE
Intensity: 1.0

Per-Experiment Results:

EXPERIMENT 1 (4,000 frames, 17.9s)

|

+-- Oracle: 3240.72 (theoretical maximum)

+-- EXPNeuralUCB: 2016.44 (37.8% gap)

+-- GNeuralUCB: 2542.22 (21.6% gap)

+-- EXPUCB: 2093.89 (35.4% gap)

Winner: GNeuralUCB

EXPERIMENT 2 (6,000 frames, 30.1s)

|

+-- Oracle: 5584.16 (theoretical maximum)

+-- EXPNeuralUCB: 3980.39 (28.7% gap)

+-- GNeuralUCB: 3903.85 (30.1% gap)

+-- EXPUCB: 3318.91 (40.6% gap)

Winner: EXPNeuralUCB

EXPERIMENT 3 (8,000 frames, 80.7s)

|

+-- Oracle: 7832.21 (theoretical maximum)

+-- EXPNeuralUCB: 5498.66 (29.8% gap)

+-- GNeuralUCB: 5597.04 (28.5% gap) [WINNER]

+-- EXPUCB: 4744.51 (39.4% gap)

Winner: GNeuralUCB

EXPNeuralUCB: 5498.66 (29.8% gap)
GNeuralUCB: 5597.04 (28.5% gap)
EXPUCB: 4744.51 (39.4% gap)
Winner: GNeuralUCB

ORACLE-RELATIVE GAP ANALYSIS:

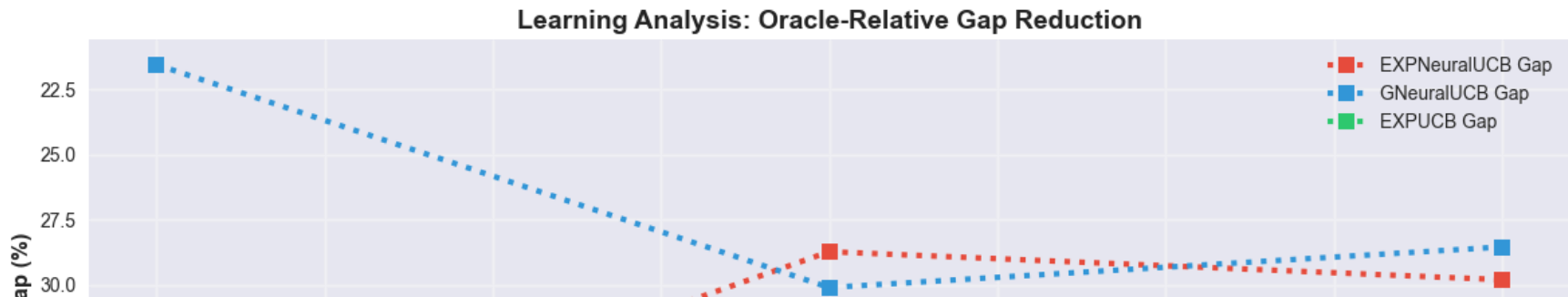
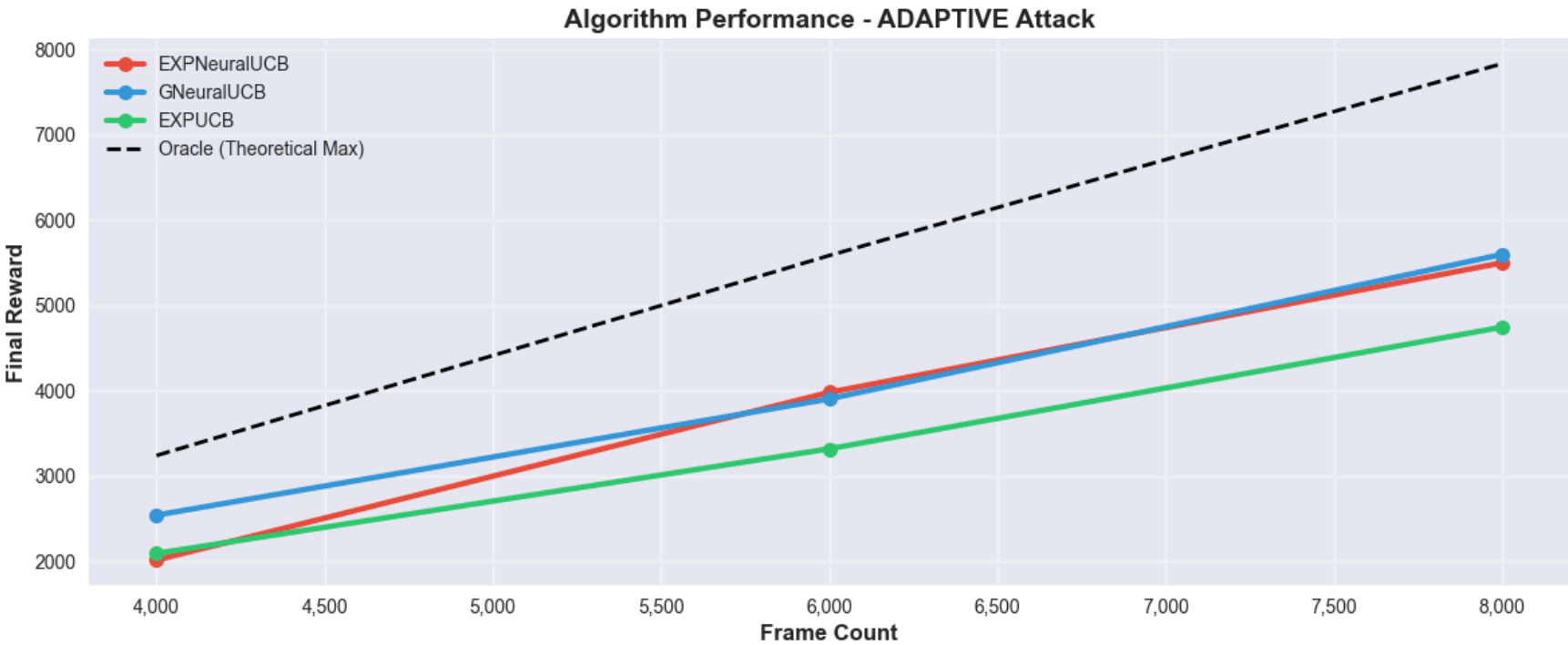
LEARNING PERFORMANCE (Gap Reduction to Oracle):

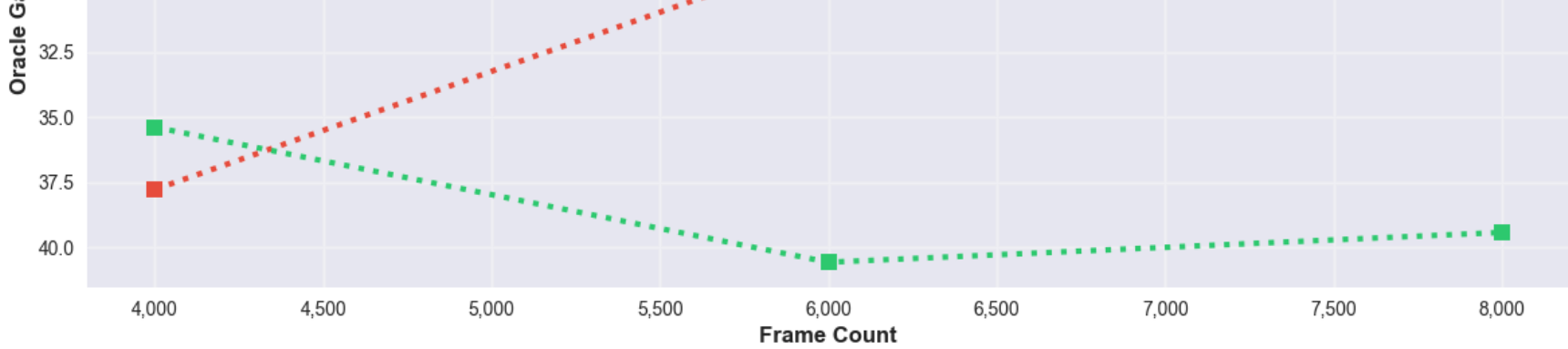
- #1: EXPNeuralUCB -> +8.0pp improvement (gap closing)
Gap: 37.8% -> 29.8%
- #2: EXPUCB -> +4.0pp decline (gap widening)
Gap: 35.4% -> 39.4%
- #3: GNeuralUCB -> +7.0pp decline (gap widening)
Gap: 21.6% -> 28.5%

Interpretation:

- Lower gaps = Better performance relative to Oracle
- Gap reduction = Algorithm learning and improving
- Best learner closes the gap to Oracle over time

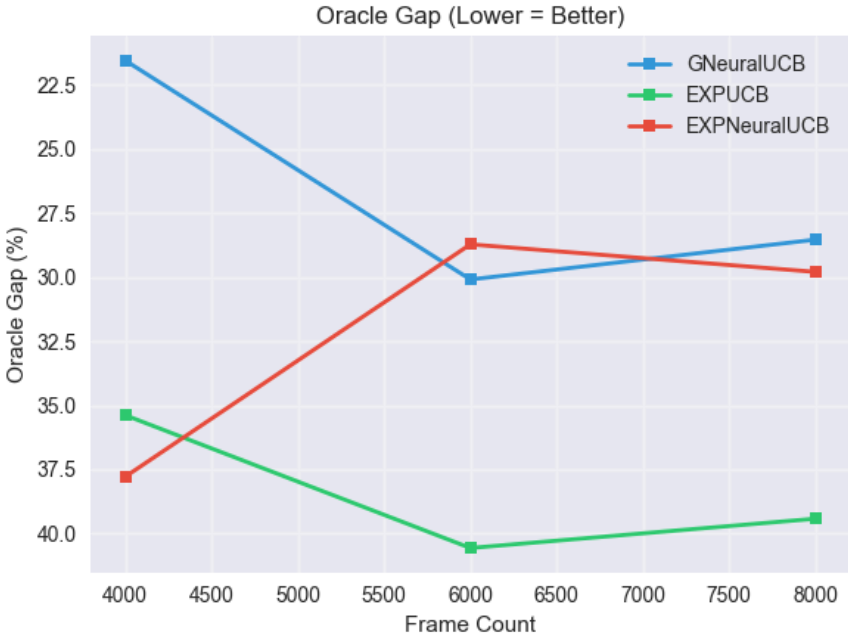
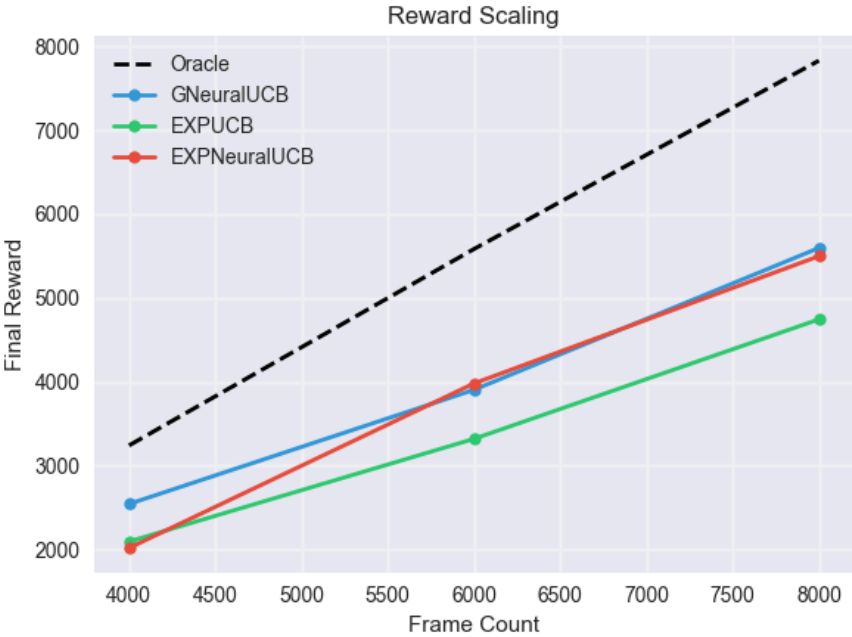
Generating performance visualization...





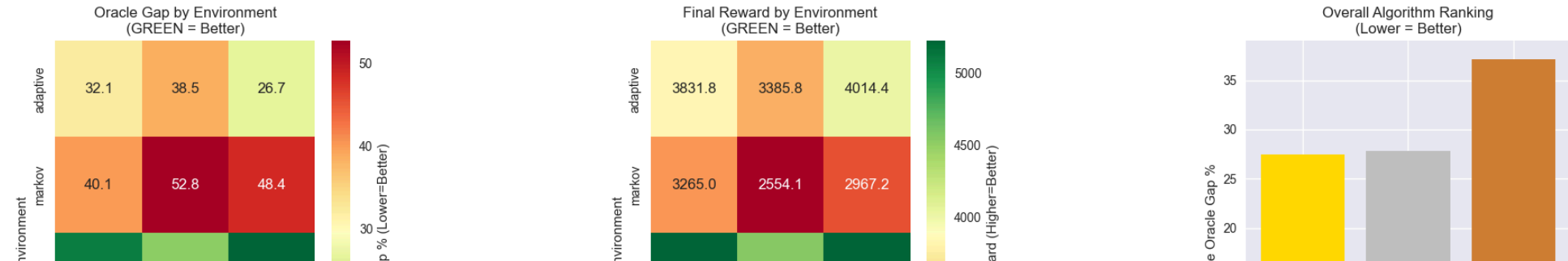
Visualization complete!

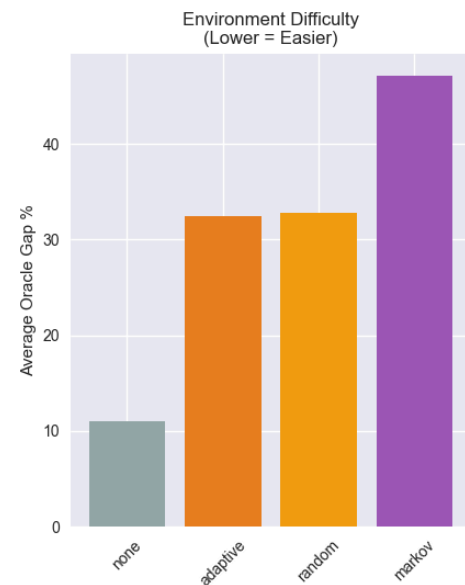
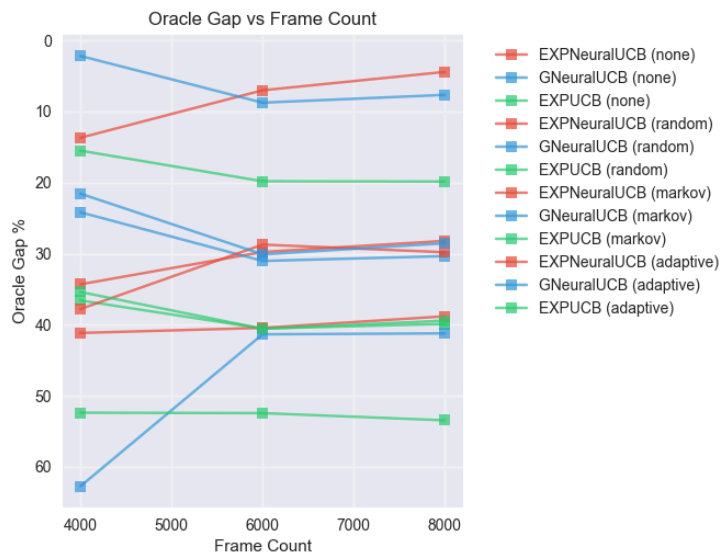
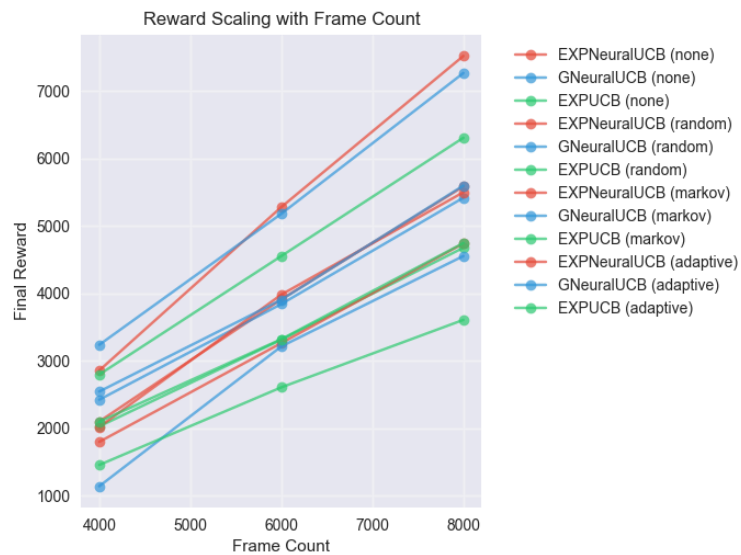
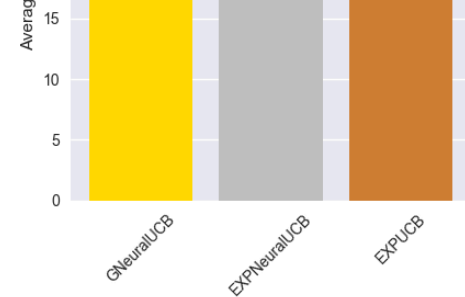
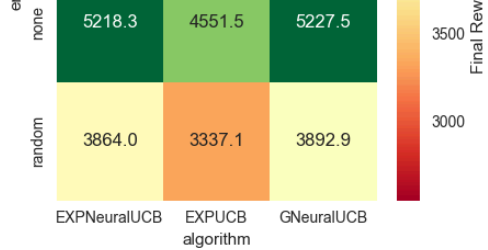
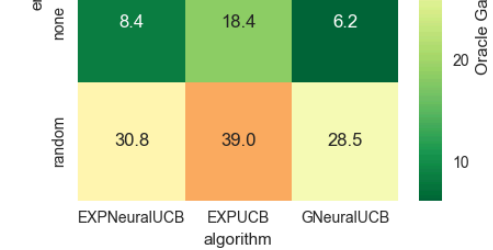
Adaptive Environment Results



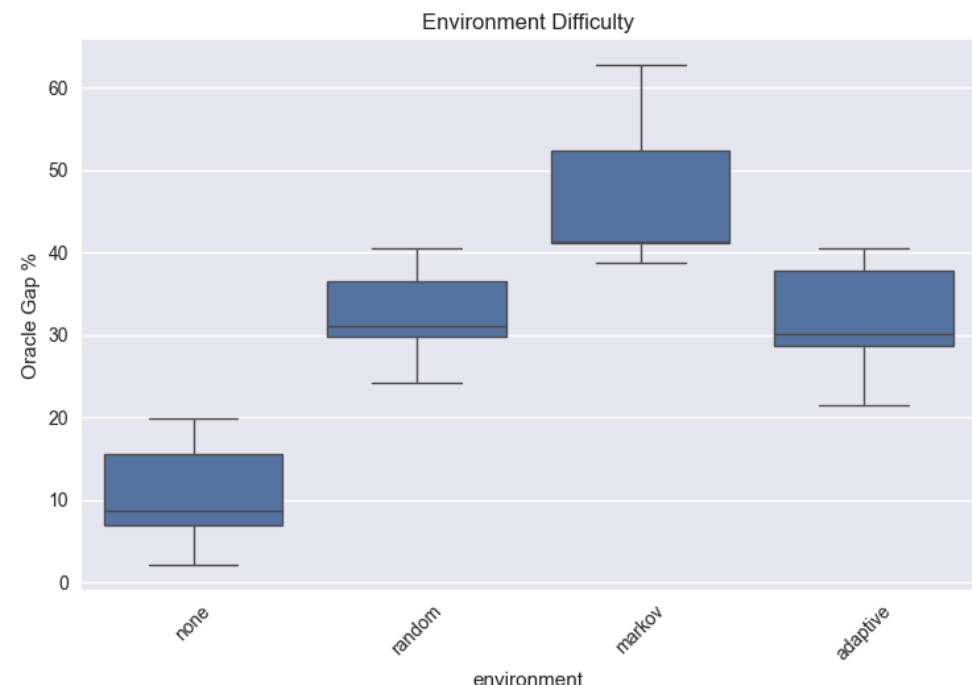
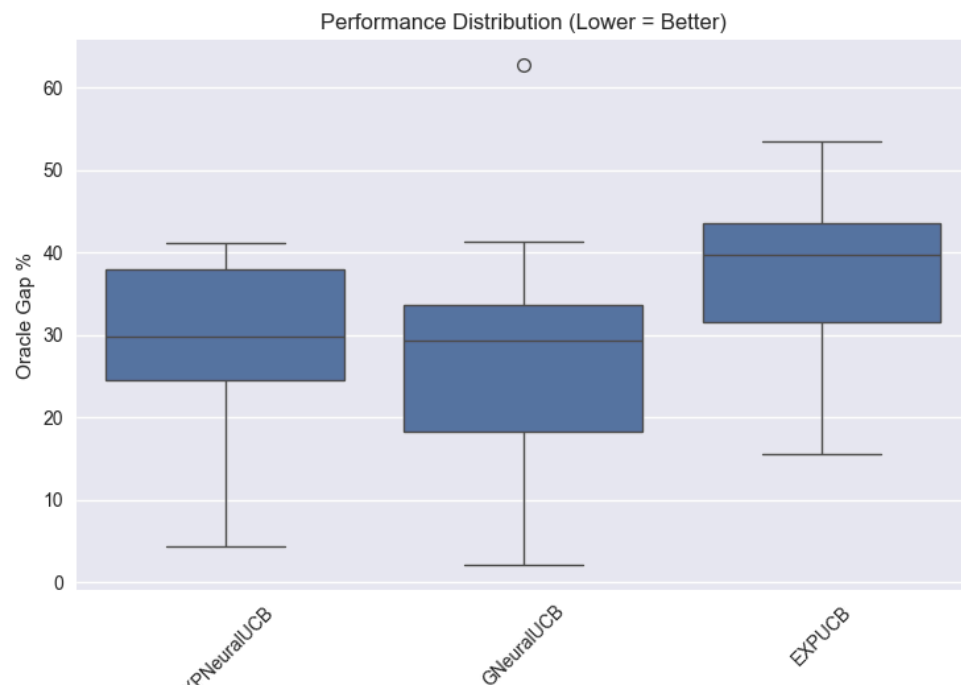
CREATING ENHANCED VISUALIZATIONS

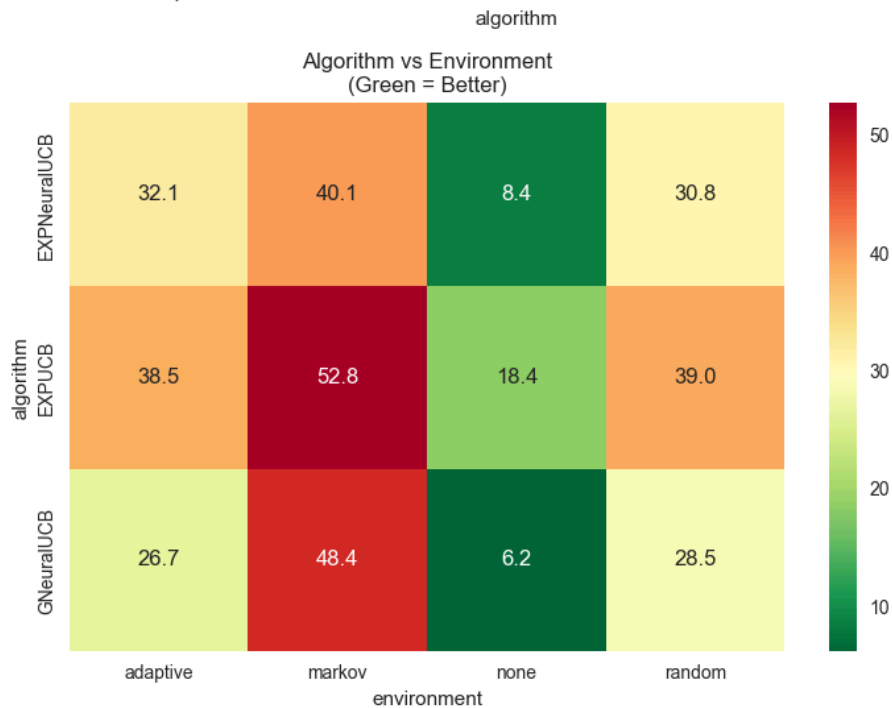
QUANTUM ROUTING ALGORITHM PERFORMANCE DASHBOARD
Lower Oracle Gap = Better Performance | Higher Reward = Better Performance



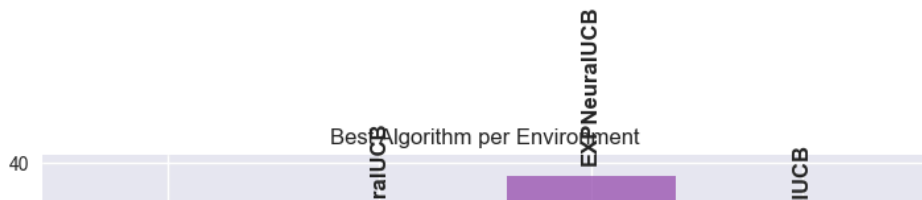
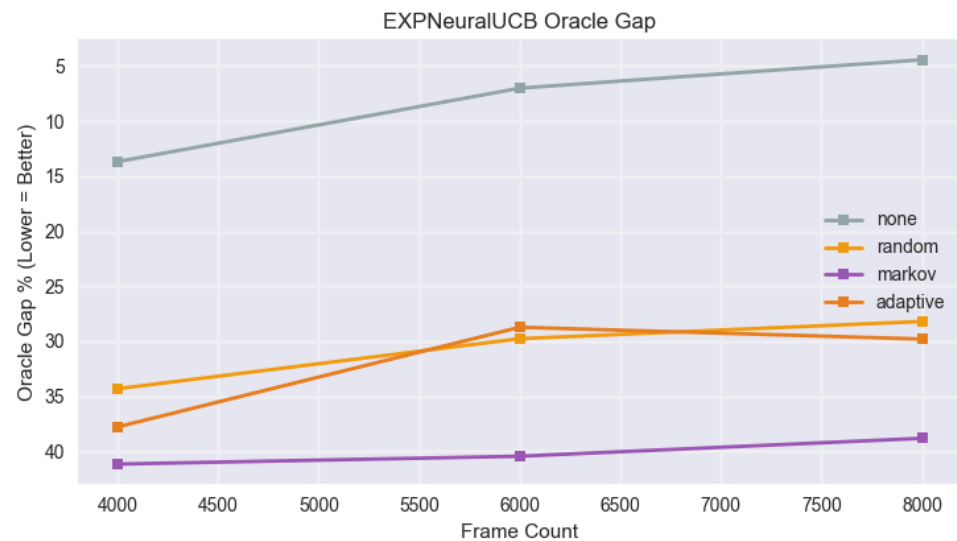
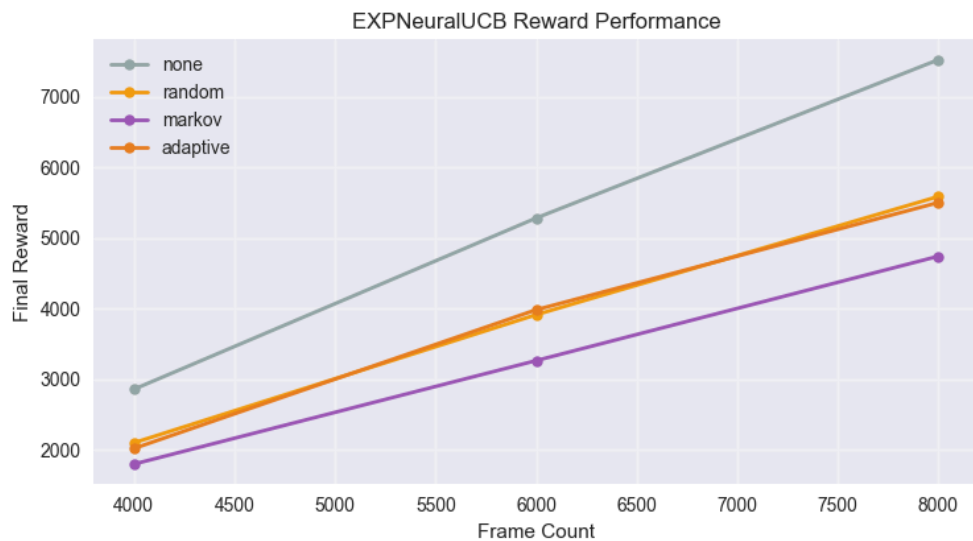


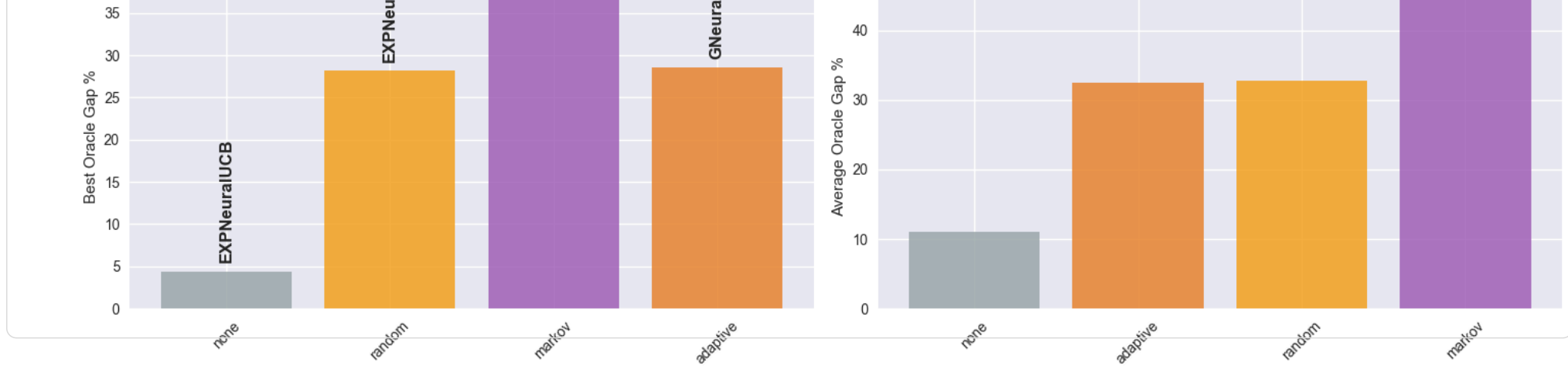
ALGORITHM COMPARISON ANALYSIS





ENVIRONMENT COMPARISON





1 Start coding or [generate](#) with AI.

1 Start coding or [generate](#) with AI.

1 Start coding or [generate](#) with AI.

Quantum Routing Test: NeuralUCB vs LinUCB Comparison

```
1 #-*- coding: utf-8 -*-
2 """
3 Quantum Routing Test: NeuralUCB vs LinUCB Comparison
4 Replicates the exact test from quantum.ipynb
5 """
6
7 import matplotlib.pyplot as plt
8 from tqdm import trange
9 import random
10 import torch
11 from quantum_tools import *
12 from quantum_models import *
13
14 # Set random seeds for reproducibility
15 np.random.seed(42)
16 torch.manual_seed(42)
17 random.seed(42)
18
19 def get_reward_2hop(X_n):
20     """
21     2-hop quantum path reward function
22     pe_a_1, pe_a_2: single-qubit single-attempt entanglement success rates
23     """
24     A = 4000 # Number of attempts per time slot
25     reward = []
26
27     # Single-hop single-qubit single-attempt entanglement success rates
28     pe_a_1 = 1.5e-4
29     pe_a_2 = 1.5e-4
30
31     # Single-hop single-qubit success rates after A attempts
32     p_a_1 = 1 - (1 - pe_a_1)**A
33     p_a_2 = 1 - (1 - pe_a_2)**A
34
35     # Calculate success rate for each qubit allocation
36     for j in range(len(X_n)):
37         P_a_1 = 1 - (1 - p_a_1)**X_n[j][0] # First hop success
38         P_a_2 = 1 - (1 - p_a_2)**X_n[j][1] # Second hop success
39         P_a = P_a_1 * P_a_2 # Overall path success rate
40         reward.append(P_a)
41
42     return reward
43
44 def get_reward_3hop(X_n):
45     """
46     3-hop quantum path reward function
47     """
48     A = 4000
49     reward = []
50
51     # Single-hop single-qubit single-attempt entanglement success rates
```

```

52 pe_c_1 = 2e-4
53 pe_c_2 = 2e-4
54 pe_c_3 = 2e-4
55
56 # Single-hop single-qubit success rates after A attempts
57 p_c_1 = 1 - (1 - pe_c_1)**A
58 p_c_2 = 1 - (1 - pe_c_2)**A
59 p_c_3 = 1 - (1 - pe_c_3)**A
60
61 # Calculate success rate for each qubit allocation
62 for j in range(len(X_n)):
63     P_c_1 = 1 - (1 - p_c_1)**X_n[j][0] # First hop success
64     P_c_2 = 1 - (1 - p_c_2)**X_n[j][1] # Second hop success
65     P_c_3 = 1 - (1 - p_c_3)**X_n[j][2] # Third hop success
66     P_c = P_c_1 * P_c_2 * P_c_3 # Overall path success rate
67     reward.append(P_c)
68
69 return reward
70
71 def run_quantum_test_2hop():
72     """Test 1: 2-hop quantum path (NeuralUCB vs LinUCB)"""
73     print("="*60)
74     print("QUANTUM TEST 1: 2-HOP PATH ROUTING")
75     print("="*60)
76
77     # Generate 2D context (qubit allocation for 2-hop path)
78     qubit = 8
79     X_n = []
80     for i in range(qubit+1):
81         new_d = np.array([i, qubit-i])
82         X_n.append(new_d)
83     X_n = np.array(X_n)
84
85     # Calculate rewards
86     context = X_n
87     reward = get_reward_2hop(X_n)
88
89     print(f"Context shape: {X_n.shape}")
90     print(f"Number of allocation strategies: {len(X_n)}")
91     print(f"Max reward: {max(reward):.4f} at allocation {X_n[reward.index(max(reward))]}")
92     print(f"Rewards: {[f'{r:.4f}' for r in reward]}")
93
94     # Test NeuralUCB
95     print("\nTesting NeuralUCB...")
96     neuralucb = NeuralUCB(2, len(X_n), beta=1, lamb=1)
97     NeuralUCBregret_list = []
98     NeuralUCBregret = 0
99
100    for i in trange(10000, desc="NeuralUCB Progress"):
101        arm = neuralucb.take_action(context)
102        X_t = np.random.choice([0,1], p=[1-reward[arm], reward[arm]])
103        NeuralUCBregret = NeuralUCBregret + max(reward) - reward[arm]
104        NeuralUCBregret_list.append(NeuralUCBregret)
105        neuralucb.update(context, arm, X_t)
106
107    print(f"NeuralUCB Final Regret: {NeuralUCBregret:.2f}")
108
109    # Test LinUCB
110    print("\nTesting LinUCB...")
111    linucb = LinUCB(2, len(X_n), beta=0.2, lamb=1)
112    LinUCBregret_list = []
113    LinUCBregret = 0
114    LinUCBreward_list = []
115    LinUCBreward = 0
116

```

```

117     for i in trange(10000, desc="LinUCB Progress"):
118         arm = linucb.take_action(context)
119         X_t = np.random.choice([0,1], p=[1-reward[arm], reward[arm]])
120         LinUCBregret = LinUCBregret + max(reward) - reward[arm]
121         LinUCBregret_list.append(LinUCBregret)
122         linucb.update(context, arm, X_t)
123         LinUCBreward += reward[arm]
124         LinUCBreward_list.append(LinUCBreward)
125
126     print(f"LinUCB Final Regret: {LinUCBregret:.2f}")
127
128     # Plot results
129     plt.figure(figsize=(15, 5))
130
131     # Plot 1: NeuralUCB regret evolution
132     plt.subplot(1, 3, 1)
133     t = np.arange(0, 2000, 1)
134     plt.plot(t, NeuralUCBregret_list[0:2000], color='red', label='Neural_UCB', linestyle="--", linewidth=2)
135     plt.legend()
136     plt.grid()
137     plt.xlabel('Frame Number')
138     plt.ylabel('Cumulative Regret')
139     plt.title('NeuralUCB Regret (2-hop)')
140
141     # Plot 2: LinUCB reward evolution
142     plt.subplot(1, 3, 2)
143     t = np.arange(0, 10000, 1)
144     plt.plot(t, LinUCBreward_list[0:10000], color='blue', label='Lin_UCB', linestyle="--", linewidth=2)
145     plt.legend()
146     plt.grid()
147     plt.xlabel('Frame Number')
148     plt.ylabel('Cumulative Reward')
149     plt.title('LinUCB Reward (2-hop)')
150
151     # Plot 3: Direct comparison
152     plt.subplot(1, 3, 3)
153     t = np.arange(0, 10000, 1)
154     plt.plot(t, NeuralUCBregret_list[0:10000], color='red', label='Neural_UCB', linestyle="--", linewidth=2)
155     plt.plot(t, LinUCBregret_list[0:10000], color='green', label='LinUCB', linestyle="--", linewidth=2)
156     plt.legend()
157     plt.grid()
158     plt.xlabel('Frame Number')
159     plt.ylabel('Cumulative Regret')
160     plt.title('2-hop Comparison')
161
162     plt.tight_layout()
163     plt.show()
164
165     return {
166         'neural_regret': NeuralUCBregret_list,
167         'lin_regret': LinUCBregret_list,
168         'neural_final': NeuralUCBregret,
169         'lin_final': LinUCBregret,
170         'rewards': reward,
171         'context': X_n
172     }
173
174 def run_quantum_test_3hop():
175     """Test 2: 3-hop quantum path (NeuralUCB vs LinUCB)"""
176     print("="*60)
177     print("QUANTUM TEST 2: 3-HOP PATH ROUTING")
178     print("="*60)
179
180     # Generate 3D context (qubit allocation for 3-hop path)
181     qubit = 8

```

```

182 X_n = []
183 for i in range(qubit+1):
184     for j in range(qubit+1-i):
185         new_d = np.array([i, j, qubit-i-j])
186         X_n.append(new_d)
187 X_n = np.array(X_n)
188
189 # Calculate rewards
190 context = X_n
191 reward = get_reward_3hop(X_n)
192
193 print(f"Context shape: {X_n.shape}")
194 print(f"Number of allocation strategies: {len(X_n)}")
195 print(f"Max reward: {max(reward):.4f} at allocation {X_n[reward.index(max(reward))]}")
196 print(f"Max reward ID: {reward.index(max(reward))}")
197
198 # Test NeuralUCB
199 print("\nTesting NeuralUCB...")
200 neuralucb = NeuralUCB(3, len(X_n), beta=1, lamb=1)
201 NeuralUCBregret_list = []
202 NeuralUCBregret = 0
203
204 for i in trange(10000, desc="NeuralUCB Progress"):
205     arm = neuralucb.take_action(context)
206     X_t = np.random.choice([0,1], p=[1-reward[arm], reward[arm]])
207     NeuralUCBregret = NeuralUCBregret + max(reward) - reward[arm]
208     NeuralUCBregret_list.append(NeuralUCBregret)
209     neuralucb.update(context, arm, X_t)
210
211 print(f"NeuralUCB Final Regret: {NeuralUCBregret:.2f}")
212
213 # Test LinUCB
214 print("\nTesting LinUCB...")
215 linucb = LinUCB(3, len(X_n), beta=0.2, lamb=1)
216 LinUCBregret_list = []
217 LinUCBregret = 0
218
219 for i in trange(10000, desc="LinUCB Progress"):
220     arm = linucb.take_action(context)
221     X_t = np.random.choice([0,1], p=[1-reward[arm], reward[arm]])
222     LinUCBregret = LinUCBregret + max(reward) - reward[arm]
223     LinUCBregret_list.append(LinUCBregret)
224     linucb.update(context, arm, X_t)
225
226 print(f"LinUCB Final Regret: {LinUCBregret:.2f}")
227
228 # Plot results
229 plt.figure(figsize=(15, 5))
230
231 # Plot 1: NeuralUCB regret evolution
232 plt.subplot(1, 3, 1)
233 t = np.arange(0, 2000, 1)
234 plt.plot(t, NeuralUCBregret_list[0:2000], color='red', label='Neural_UCB', linestyle="---", linewidth=2)
235 plt.legend()
236 plt.grid()
237 plt.xlabel('Frame Number')
238 plt.ylabel('Cumulative Regret')
239 plt.title('NeuralUCB Regret (3-hop)')
240
241 # Plot 2: Full comparison
242 plt.subplot(1, 3, 2)
243 t = np.arange(0, 10000, 1)
244 plt.plot(t, NeuralUCBregret_list[0:10000], color='red', label='Neural_UCB', linestyle="---", linewidth=2)
245 plt.plot(t, LinUCBregret_list[0:10000], color='green', label='LinUCB', linestyle="---", linewidth=2)
246 plt.legend()

```

```

247 plt.grid()
248 plt.xlabel('Frame Number')
249 plt.ylabel('Cumulative Regret')
250 plt.title('3-hop Comparison')
251
252 # Plot 3: Performance summary
253 plt.subplot(1, 3, 3)
254 algorithms = ['NeuralUCB', 'LinUCB']
255 final_regrets = [NeuralUCBregret, LinUCBregret]
256 colors = ['red', 'green']
257 bars = plt.bar(algorithms, final_regrets, color=colors, alpha=0.7)
258 plt.ylabel('Final Cumulative Regret')
259 plt.title('3-hop Final Performance')
260 plt.grid(axis='y')
261
262 # Add value labels on bars
263 for bar, value in zip(bars, final_regrets):
264     plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + max(final_regrets)*0.01,
265             f'{value:.0f}', ha='center', va='bottom', fontweight='bold')
266
267 plt.tight_layout()
268

```